

A Derived Approach to Classical Observables in Lagrangian Field Theory

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Abstract

I study how the derived critical locus of a function f on a manifold gives us a differential graded algebra with a Poisson structure, that has an interpretation as functions and symmetries. I introduce diffeological spaces and study their properties to show that they do not admit a direct adaptation of the previous construction on manifolds. Several generalizations of the algebra structure will be proposed that can be used to model Lagrangian field theories. In particular, I prove that these admit natural homotopy Poisson algebra structures. I study the relation of commutative differential graded algebras with L_∞ -algebras through the Chevalley-Eilenberg-complex and present a diffeological construction that is closely related to the previous results. In the end, the notion of “observables” in symplectic geometry is reconciled with the view of “observables” applied previously in this thesis.

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0.1 Introduction

In classical physical field theory the term “observables” refers intuitively to quantities such as energy or momentum, that can be measured in reality. Mathematically they can be understood as functions defined on the subset of the space of fields containing only those fields that are “on-shell” in the given theory (i.e. satisfy the given field equations). A problem that arises when formalizing this point of view is, that neither the subspace we consider nor the space of fields itself are manifolds in general. The former space might contain singularities, while the latter is usually infinite dimensional and hence cannot be treated naively as a manifold either.

In any field theory, however, we have notions of measurements, symmetries and higher symmetries. They resemble the symmetries we can construct on a subspace of a manifold when assuming the subspace is cut out by a certain form of differential equations. Together, these notions form a differentially graded complex of modules, defined up to quasi-isomorphism, that is called the complex of classical observables. The complex plays an important role in quantum field theory and gives rise to an interesting application of factorization algebras in quantization (cf. [5]).

Therefore, Costello and Gwilliam (cf. [5]) introduce the classical observables as the derived critical locus of the action functional, where they assume that the space of fields is a manifold. They then formulate conjectures about the structure of the observables in the general case. But by the categorical nature of the construction of the derived critical locus, the definition can also be applied to diffeological spaces. These spaces are a generalization of manifolds, that have already seen use in other areas of mathematical physics to cover a wider generality of field theories.

To eliminate the necessity to directly compute the observables, Costello and Gwilliam use a different definition of a “classical field theory”, that is implicitly containing the observables (cf. [5]). Their approach is related to more standard notions of “classical field theories” through a theorem by Lurie. It concerns the equivalence of formal moduli problems and differential graded Lie-algebras via the consideration of deformation problems (cf. [11, 14]).

Let us also note that, to physicists, usually not all possible observables given through functions on the “on-shell” fields are of interest, but the measurements that have a fundamental physical meaning. In the case where the field theory admits a notion of space and time, this approach leads to a different concept of observables arising through techniques of multi-symplectic geometry (cf. [6]). They correspond to integrals of local quantities that are time-independent, such as energy, momentum, or other conserved currents. Thus they can be related to the previously discussed notion of observables (cf. [6], [18]).

This thesis aims to contribute to the studies of the complex of classical

observables and its relation to the notions described above. Let us briefly review the main subjects and results here.

Section 1 acts as an introduction to the methods to compute the derived critical locus of a function on a manifold. This gives rise to our motivating example (cf. Example 1.1.2): Assume that the space of fields $\mathfrak{F}$ of a certain field theory is a manifold. Then the derived critical locus of the action functional S is equipped with the structure sheaf induced by a commutative differential algebra structure on $\mathfrak{X}(\mathfrak{F})$, where the differential (of degree -1) is given by $\mathrm{vd}S$.

In section 2, our aim will be to understand what the derived critical locus of a function on the space of fields $\mathfrak{F}$ might be, where we view $\mathfrak{F}$ as a diffeological space. Therefore, we first need to review the concept of diffeological spaces and its core constructions, such as tangent and cotangent bundle of $\mathfrak{F}$.

Even statements about manifolds that are considered basic, such as the vector bundle structure of the tangent space, do no longer hold for general diffeological spaces. This is why we need to consider some specific proofs about properties of $\mathfrak{F}$. My main contribution to these discussions is the fact, that the smooth dual to $T_\varphi^*\mathfrak{F}$ (which is defined as the smooth dual to $T_\varphi\mathfrak{F}$), is again given by $T_\varphi\mathfrak{F}$, even though the space is infinite dimensional (cf. Theorem 2.2.2).

This allows us to finally formulate the object that we intended to study: The sheaf of functions on the derived critical locus of an action function on $\mathfrak{F}$. This is an object that, to my knowledge, previously has not been considered in a diffeological context. However, we will see that there are relevant differences in the form of this complex when compared to the computations on manifolds (cf. Theorem 2.3.8). This leads to the conclusion that there are several possible generalisations of the complex of observables through diffeological constructions. When also considering field theories where the Lagrangian does not integrate to a well-defined function on $\mathfrak{F}$, we get even more possible models of the observables.

Regarding these options from the viewpoint of [5], it would be a desired property for the complex of observables to admit a homotopy Poisson algebra structure. However, the bracket on $\mathfrak{X}^1(M)$ for a manifold M is defined as a bracket on derivations and can only be generalized to a bracket on derivations of diffeological spaces at first. I show in this thesis, that this bracket can be restricted to the space of vector fields $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ (cf. Theorem 2.5.1).

In section 3, we establish a relation between the constructions of [5] and the objects studied previously in the second chapter. To do this, we will review the key concepts of formal moduli problems and L_∞ -algebras. A theorem by Lurie will be presented about an equivalence between these concepts together with a sketch of a categorical construction he employed.

We can use this set-up to motivate why the definition of a classical field theory as an L_∞ -algebra is, in a sense, natural. I then present an L_∞ -

algebra (with possible variations) together with a diffeological version of the Chevalley-Eilenberg-complex, that is closely related to one of the previous generalizations of the complex of observables. I further show that while that L_∞ -algebra depends on additional structure on the bundle, the resulting complex does not (cf. Theorem 3.2.9 and Proposition 3.2.11).

In chapter 4, this thesis concludes with an additional review of Hamiltonian observables, which is a notion from symplectic geometry. We will recapitulate the necessary underlying concepts of symplectic forms and the variational bicomplex to define the Hamiltonian observables. Then it will be highlighted, how these objects can be understood as functions contained in $C^\infty(\mathfrak{F})$.

0.2 Acknowledgements

Looking back a year ago, when I started to write this thesis, I was expecting many things. I was looking forward to learning a lot – which I did. I was prepared for it to be intense and challenging work – which it was. And I was already expecting for the topic to be too vast to be covered just in one thesis – which I took for all the more reason to try. But the global crisis that is the current COVID pandemic was something that I absolutely couldn't have imagined back then.

So suddenly all the libraries were closed and meetings and university lectures were only held online, which created a range of unexpected problems. For example, even though I have a desk at home, I share the workspace we have with my partner, who was having lectures and meetings as well. But what was I missing most these last month were the casual conversations with other students, to broaden my horizon, to recapitulate the current status of my thesis or simply to take a small break from work.

It is thanks to a few people, that I have still been able to overcome the challenge of the situation. On a personal level, it is certainly due mostly to my partner Jakob that I was able to keep up both concentration and morale throughout the past months. Overall, the support of my parents and friends, with whom I could share struggles and successes has been wonderful and helped me to keep going.

On a professional level, I was supported among others by Prof. Dr. Catharina Stroppel, who was kind enough to very spontaneously agree to be my second advisor for this thesis.

But without a doubt the person who has contributed most to helping me finish this work is Dr. Christian Blohmann, my thesis advisor. From the start, he encouraged me to believe in my abilities and take on the fairly ambitious topic of my master's thesis. Our meetings were always valuable contributions to my thought process and I often came home with a new idea or a name of a paper for research, but his support even extended beyond the mathematical problems.

Together with Lory Kadiyan, he also organized a seminar for all of his students, which brought me in touch with interesting areas of mathematics and, using the online medium to our advantage, connected us students with working mathematicians living far away.

I would like to thank all of the persons mentioned above, especially Christian, to get me safely through all the troubles that I have met in the past year. This thesis is not only my achievement, as it would not have been possible without the incredible support I have experienced and I am very grateful for all of it.

0.3 Notational remarks

In this thesis I will mostly use notational conventions that are not unusual throughout the literature that I’ve considered or the lectures I’ve taken. However I felt that some notational conventions I adapted could benefit from clarification.

$\Gamma(f)$	denotes the graph of a function
$\Gamma^\infty(M, F)$	denotes the smooth sections of a fiber bundle
$C^\infty(M)$	denotes the ring of smooth functions on a manifold, sometimes considered as a sheaf on M
$\mathcal{O}(M)$	denotes the sheaf of dg-modules over $C^\infty(U)$, where $U \subset M$ open, that replaces the ring of functions in the category of dg-manifolds

As there are quite a few concepts known as “observables” I have also taken the freedom of giving them more precise names, such as “perturbative observables” in the case of the notion of Costello and Gwilliam (cf. [5]) or “Hamiltonian observables” in the case of the notion from symplectic geometry (cf. [6]).

1 The Derived Critical Locus

1.1 The Derived Critical Locus of a Function on a Manifold

The critical locus of a function f on a manifold M is usually understood as just the set $\{m \in M \mid df(m) = 0\}$. This does not necessarily describe a manifold because the vanishing set of df might have singularities, but as the subset of a manifold, we can at least transfer the topology and even some of the smooth structure. For example we could say that a function on the critical locus is smooth if and only if it can be extended to a smooth function on M .

In some cases, especially in the case of field theory that we are interested in, we no longer have a unique ring of functions on the manifold, but rather a complex $\mathcal{O}(M)$, that is only defined up to quasi-isomorphism. In physics, this is often interpreted to describe “measurements” in degree 0, “symmetries” in degree 1 (such that one is not able to distinguish measurements that only differ by the action of a symmetry) and “higher symmetries” in higher degrees.

When considering the critical locus of a function from this viewpoint, we can use a different setting to work with, namely the category of derived smooth manifolds. It is not necessary for this work to be familiar with the concept of derived smooth manifolds, as the construction we are interested in can be described very explicitly.

Assume we are given a manifold M with its ring of functions $C^\infty(M)$. By considering the dg-algebra $\mathcal{O}(M)$ concentrated in degree 0, where it is given by $C^\infty(M)$, we obtain $(M, \mathcal{O}(M))$ as a dg-manifold with grading only in positive degree. Here we are using the definition of dg-manifold as given in [15], as the terminology throughout the literature is unfortunately not uniform. By the Dold-Kan-correspondence, we can understand any such dg-manifold as a derived manifold, as described in [13]. The theorem also allows us in general to consider the spaces we work with as spaces equipped on each open $U \subset M$ with a commutative differential graded $C^\infty(U)$ -algebra (instead of a simplicial ring with certain properties).

Let $f \in C^\infty(M)$.

Definition 1.1.1. The derived critical locus of f is the pull-back D in the following diagram of derived smooth manifolds:

$$\begin{array}{ccc} D & \longrightarrow & (M, \mathcal{O}(M)) \\ \downarrow & & \downarrow df \\ (M, \mathcal{O}(M)) & \xrightarrow{0} & (T^*M, \mathcal{O}(T^*M)). \end{array}$$

Both for reasons of computation and application, I will always consider D as a dg-manifold $(|D|, \mathcal{O}(D))$ here, where $|D|$ is a topological space and

$\mathcal{O}(D)$ a chain complex. The underlying space $|D|$ is a manifold, which is not an invariant under equivalences of dg-manifolds. We only know that it admits the vanishing locus of df as a subset that can be reconstructed via $\mathcal{O}(D)$.

The commutative differential graded algebra of functions $\mathcal{O}(D)$ is, at least from the physics perspective, of high interest. For example, it inherits a key role in computing the observables of a classical field theory in the BV formalism.

Example 1.1.2. ([5], pp. 45 f.) Let M be a manifold and $\pi : F \rightarrow M$ a fiber bundle (also called the configuration bundle in the context of a field theory). Let $\mathfrak{F} = \Gamma^\infty(M, F)$ be the space of smooth sections (also called space of fields).

Assume that $\mathfrak{F}$ is a manifold and $S : \mathfrak{F} \rightarrow \mathbb{R}$ is a smooth function (we think of it as an action functional, that is everywhere well defined). Then the classical observables on M are denoted with $Obs^{cl}(M)$ and given by the chain complex of functions $\mathcal{O}(D)$ on the derived critical locus of S .

In this case the observables can be explicitly computed as a dg-algebra induced by $\mathfrak{X}^*(\mathfrak{F}) = \Gamma^\infty(\mathfrak{F}, \wedge^* T\mathfrak{F})$. In general, however, the assumption that $\mathfrak{F}$ is finite is far too strict to apply it to any non-trivial field theory.

It is one of the aims of this thesis to consider the case of more complicated field theories, where we cannot assume that $\mathfrak{F}$ is a manifold. To do so, let us first analyze the methods of the finite dimensional case, so we can generalize them later.

By the construction of the category of derived smooth manifolds, $\mathcal{O}(D)$ is the homotopy push-out given by the following diagram.

$$\begin{array}{ccc} (\cdots \leftarrow 0 \leftarrow C^\infty(T^*M) \leftarrow 0 \leftarrow \cdots) & \xrightarrow{df^*} & (\cdots \leftarrow 0 \leftarrow C^\infty(M) \leftarrow 0 \leftarrow \cdots) \\ \downarrow 0^* & & \downarrow \\ (\cdots \leftarrow 0 \leftarrow C^\infty(M) \leftarrow 0 \leftarrow \cdots) & \longrightarrow & \mathcal{O}(D) \end{array}$$

This tells us, that $\mathcal{O}(D) = C^\infty(M) \otimes_{C^\infty(T^*M)}^{\mathbb{L}} C^\infty(M)$ (cf. [5], p. 46).

To compute this derived tensor product, we need to interpret $C^\infty(M)$ as a $C^\infty(T^*M)$ -module through both of the given morphisms respectively. Then we need to replace one of the modules by a flat resolution. The $C^\infty(T^*M)$ -module structure on $C^\infty(M)$ is given each by

$$C^\infty(M) = \frac{C^\infty(T^*M)}{\{f : f|_{\Gamma(df)} = 0\}}$$

and

$$C^\infty(M) = \frac{C^\infty(T^*M)}{\{f : f|_{\Gamma(0)} = 0\}},$$

where we use the notation of Γ for the graph of the respective functions.

The actual computation of $\mathcal{O}(D)$ is easier if we compute it for the opens of M , and then apply gluing to reconstruct the sheaf. As any sheaf is uniquely determined by their stalks, it is even possible for us to just restrict ourselves to considerations of these. Costello and Gwilliam [5] already mention that in this context, we can use a form of Koszul resolution as the flat resolution. We will carry this out in the following.

For the sake of simplicity, let us from now on define that $\dim(M) = n$.

Lemma 1.1.3. *Let $m \in M$ be a point with coordinate functions x^i for $1 \leq i \leq n$ on a neighborhood U of m and let y^i denote the remaining coordinates on T^*M given by the basis dx^i of the fibers. Then the ideal $I = \{g \in C^\infty(\pi_{(T^*M \rightarrow M)}^{-1}(U)) : g|_{\Gamma(0)} = 0\}$ is generated by $g^j : (x^i, y^i) \mapsto y^j$ for $1 \leq j \leq n$.*

To see that the given I is indeed an ideal, we should specify, that we regard $C^\infty(-)$ always as a ring by point-wise multiplication. That I is an ideal follows easily then.

Proof. That the g^j are contained in I is easy to see, as they recover the coordinates that vanish on the given section.

Assume now that we have given some $f \in I$. Then consider the function $\tilde{f}((x^i, y^i)) = f((x^i, y^i)) - f((x^i, 0, y^2, y^3, \dots))$, such that at any point p with $g^1(p) = y^1 = 0$ we have $\tilde{f}(p) = 0$. As g^1 is fiber-wise linear, we now get a well-defined function $f' \in C^\infty(T^*M)$ that satisfies

$$f'((x^i, y^i)) \cdot g^1 = \tilde{f}((x^i, y^i)),$$

where at points $p \in T^*M$ with $g^1(p) = 0$ we have $f' = \frac{\partial}{\partial y^1} \tilde{f}(p)$. So $\tilde{f}$ is contained in the ideal generated by the coordinate functions

What remains to show now is that the function

$$(x^i, y^i) \mapsto f((x^i, 0, y^2, \dots))$$

is generated by the coordinate functions. But we can iterate the same argument as before to linearize along the second and all consecutive fiber coordinates to write

$$f = f' \cdot g^1 + f'' \cdot g^2 + \dots + f^{(n)} \cdot g^n + f((x^i, 0, \dots, 0)).$$

But by assumption, the last term vanishes on all fibers of U , such that f is indeed contained in the ideal generated by the coordinate functions g^j . Hence we can conclude that I is indeed generated by these functions. $\square$

Proposition 1.1.4. *At any given point $m \in M$ the chain complex*

$$\begin{array}{ccccccc} \longleftarrow & C^\infty(T^*M)_m^n & \longleftarrow & \wedge^2(C^\infty(T^*M)_m^n) & \longleftarrow & \wedge^3(C^\infty(T^*M)_m^n) & \longleftarrow \dots \\ & & & & & & \searrow \\ \dots & \longleftarrow & 0 & \longleftarrow & \left(\frac{C^\infty(T^*M)}{\{f: f|_{\Gamma(0)=0}\}} \right)_m & \longleftarrow & C^\infty(T^*M)_m \longleftarrow \end{array}$$

is exact. Here, the map to the lowest degree where the module is non-zero is given by projection and the others are generated as a differential on the remaining complex by

$$\begin{aligned} C^\infty(T^*M)_m^n &\rightarrow C^\infty(T^*M)_m \\ \{f_j\}_{1 \leq j \leq n} &\mapsto \sum_{j=1}^n f_j \cdot g^j, \end{aligned}$$

with the g^j being the coordinate functions as given in Lemma 1.1.3.

To make the descriptions in the proof easier, we will consider the complex graded such that the first non-zero module is contained in degree -1.

Proof. The projection map onto degree -1 of the chain complex is by definition surjective with the explicitly given kernel. That this agrees with the image of the map to degree 0 is a direct consequence of Lemma 1.1.3.

The exactness in all remaining degrees can be shown by using a similar argument for each: The $C^\infty(T^*M)_m$ -modules from which the remaining complex arises have a natural choice of basis, given through the free module structure. We have actually already used it to define the differential. So any element of these modules can be denoted as an array of functions in $C^\infty(T^*M)_m$.

We will now first show in degree 1, how we can inductively decompose a closed element into an exact element and another closed element. This new closed element furthermore has the property that its first non-zero component is at a strictly higher indexed position when compared to the previous one. As there only can be finitely many iterations until we reach 0, this implies exactness in degree 1 and the method can be generalized to higher degrees afterwards.

So let us commence with the proof. Assume that for the module in degree 1 we have given $\{f_j\}_{1 \leq i \leq n} \in \ker(d_0)$, i.e. with $\sum_{j=1}^n f_j \cdot g^j = 0$. If we evaluate the functions at any point $p_{t,q}$ in the fiber of $q \in U$ that is given by the coordinates $(x^i, t, 0, 0, \dots)$, we find that

$$0 = \sum_{j=1}^n f_j(p_{t,q}) \cdot g^j(p_{t,q}) = t \cdot f_1(p_{t,q}),$$

as the g^j recover the given coordinates. Hence we can conclude that $f_1(p_{t,q}) = 0$ for all $t \in \mathbb{R} \setminus \{0\}$ and by continuity also for $t = 0$.

If our sum had only the one non-zero summand $f_1 \cdot g^1$ (so $f_j = 0$ for all $j \neq 1$), we would find by the same argument that also for all choices of $y^2, \dots, y^n$ we would get $f((x^i, t, y^2, \dots)) = 0$ for all $t \in \mathbb{R}$. This would directly imply that $f_1 = 0$ and hence $\{f_j\}_{1 \leq j \leq n} = 0$, which is trivially in the image of d_1 .

If we have multiple non-zero f_i , we can use the method that we have already applied in Lemma 1.1.3 to write

$$f_1 = h_{1,2} \cdot g^2 + h_{1,3} \cdot g^3 + \dots + h_{1,n} \cdot g^n + f_1((x^i, y^1, 0, 0, \dots)).$$

Note that as we have proven the latter function to be 0, we do not need g^1 in the sum expansion of f_1 . For notational purposes, we thus define $h_{1,1} = 0$.

If we now consider the element

$$(\{h_{1,j}\}_{1 \leq j \leq n}) \wedge (\{\delta_{1,j}\}_{1 \leq j \leq n}) \in \wedge^2 C^\infty(T^*M)_m^n,$$

we can see that under the given d_1 it maps to $\{\tilde{f}_j\}_{1 \leq j \leq n}$ with $\tilde{f}_1 = f_1$ and $\tilde{f}_j = h_{1,j} \cdot g^1$ for $j \neq 1$. So by subtracting $\{\tilde{f}_j\}$, which is in the image of d_1 by construction, from $\{f_j\}$, we obtain a new set of functions $\{\bar{f}_j\}$. They satisfy that $\bar{f}_1 = 0$ and it is equivalent that $\{\bar{f}_j\} \in \text{Im}(d_1)$ or that $\{f_j\} \in \text{Im}(d_1)$.

Also, if already a given k has the property that $f_l = 0$ for all $1 \leq l < k$, we could find an expansion of f_k with all $h_{k,l} = 0$ and $h_{k,k} = 0$. This is because, analogous to the argument that shows we are done if f_1 is the only non-zero component, we can prove that in this case $f_k((x^i, y^1, y^2, \dots, y^k, \dots, 0)) = 0$. We can then construct a sum expansion

$$f_k = h_{k,(k+1)} \cdot g^{(k+1)} + \dots + h_{k,n} \cdot g^n + f_1((x^i, y^1, y^2, \dots, y^k, 0, \dots, 0)),$$

in which we do not need to linearize with respect to g^l for $1 \leq l < k$ or with respect to g^k . So neither would appear as a summand and we can freely define $h_{k,l}$ and $h_{k,k}$ to be 0.

A short computation verifies, that the image $\{\tilde{f}_j\}_{1 \leq j \leq n}$ agrees with $\{f_j\}_{1 \leq j \leq n}$ on the first k components. Hence we could again reduce the question if $\{f_j\} \in \text{Im}(d_1)$ to the question if $\{\tilde{f}_j\} \in \text{Im}(d_1)$, where the new element of $C^\infty(T^*M)_m^n$ has the first non-zero component at a strictly higher indexed position.

By the argument that was already mentioned this is sufficient to show that $\ker(d_0) = \text{Im}(d_1)$, such that the complex is exact in degree 1.

In higher degrees, the argument is very similar, just with more iterations. As the wedge product is taken over $C^\infty(T^*M)$, we have that we can write every element

$$f \in \wedge^k C^\infty(T^*M)_m^n$$

as

$$\sum_{1 \leq i_1 < \dots < i_{(k-1)} \leq n} \left(\{f_i^{(i_1, \dots, i_{(k-1)})}\}_{1 \leq i \leq n} \right) \wedge (\{\delta_{i_1, i}\}_{1 \leq i \leq n}) \wedge \dots \wedge \left(\{\delta_{i_{(k-1)}, i}\}_{1 \leq i \leq n} \right),$$

where $f_i^{(i_1, \dots, i_{k-1})} = 0$ if $i = i_j$ for some $1 \leq j \leq (k-1)$. Without loss of generality, we may even assume that $f_i^{(i_1, \dots, i_{k-1})} = 0$ if $i \leq i_{(k-1)}$, as we could write the component

$$\left(\{f_i^{(i_1, \dots, i_{k-1})} \cdot \delta_{i,j}\}_{1 \leq j \leq n} \right) \wedge \left(\{\delta_{i_{(k-1)},j}\}_{1 \leq j \leq n} \right)$$

as

$$\left(\{\delta_{i,j}\}_{1 \leq j \leq n} \right) \wedge \left(\{f_i^{(i_1, \dots, i_{k-1})} \cdot \delta_{i_{(k-1)},j}\}_{1 \leq j \leq n} \right)$$

instead.

If we now consider $f \in \ker(d_k)$ and $(i_1, \dots, i_{(k-1)}) = (1, \dots, (k-1))$ first, we find that

$$d_0(\{f_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n}) = 0$$

through a similar decomposition of the elements in the target (the module in degree $(k-1)$).

Assume the first non-zero entry of $\{f_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n}$ is indexed by l . By the properties of the wedge product, we can assume $l \geq k$. With the same method we have already seen in the case of degree 1, we can then pick an element of the form

$$(\{h_j\}_{1 \leq j \leq n}) \wedge (\{\delta_{l,j}\}_{1 \leq j \leq n}) \in \wedge^2 C^\infty(T^*M)_m^n,$$

where $h_j = 0$ for $j \leq l$. We choose it such that

$$d_1((\{h_j\}_{1 \leq j \leq n}) \wedge (\{\delta_{l,j}\}_{1 \leq j \leq n})) = \{\tilde{f}_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n},$$

where the term $\{\tilde{f}_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n}$ is required to agree with $\{f_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n}$ in the first non-zero component.

Let us now evaluate the differential on

$$(\{h_j\}_{1 \leq j \leq n}) \wedge (\{\delta_{l,j}\}_{1 \leq j \leq n}) \wedge (\{\delta_{i_1,j}\}_{1 \leq j \leq n}) \wedge \dots \wedge (\{\delta_{i_{(k-1)},j}\}_{1 \leq j \leq n}),$$

and apply our decomposition. We then see that the values of the components $a_j^{(1, \dots, (k-1))}$ that appear in the image only arise from

$$d_1((\{h_j\}_{1 \leq j \leq n}) \wedge (\{\delta_{l,j}\}_{1 \leq j \leq n})),$$

as we have that $l \geq k$ and $h_j \neq 0$ only if $j > k$. Hence the components are given by $\{\tilde{f}_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n}$. Especially, they agree on the first non-zero component with the first non-zero component of $\{f_j^{(1, \dots, (k-1))}\}_{1 \leq j \leq n}$. Therefore, we have found an exact element, that we can subtract from our given closed term f to obtain a term, where the first non-zero element has a strictly higher index than before.

Similarly to the argument for exactness in degree 1, we can iterate this procedure for all $1 \leq j \leq n$ for a fixed index $(i_1, \dots, i_{(k-1)})$ and then iterate

for the lexicographical order of indices. After a finite number of iterations, we have each subtracted an exact term and reached 0. This leads to the conclusion, that any closed term we begin with is indeed exact. But this was the claim we intended to show. $\square$

With this, we can now commence to compute the dg-algebra $C^\infty(D)_m$ on the derived critical locus.

Theorem 1.1.5. *The dg-algebra $C^\infty(D)_m$ on the derived critical locus of $f \in C^\infty(M)$ is given at the stalk of $m \in M$ by*

$$\begin{array}{c} \Gamma^\infty(M, \wedge^2 TM)_m \longleftarrow \Gamma^\infty(M, \wedge^3 TM)_m \longleftarrow \dots \\ \searrow \\ \dots \longleftarrow 0 \longleftarrow C^\infty(M)_m \longleftarrow \Gamma^\infty(M, TM)_m \longleftarrow \end{array}.$$

Proof. To compute the dg-algebra on the derived critical locus on the stalks, we must compute the derived tensor product, as described in [5] or [8]. As previously described in this section, this can be done by tensoring a flat resolution of $\left(\frac{C^\infty(T^*M)}{\{f:f|_{\Gamma(0)}=0\}}\right)_m$ with $\left(\frac{C^\infty(T^*M)}{\{f:f|_{\Gamma(dS)}=0\}}\right)_m$ ([comp. 21]). By Proposition 1.1.4 we know that

$$\begin{array}{c} \wedge^2(C^\infty(T^*M)_m^n) \longleftarrow \wedge^3(C^\infty(T^*M)_m^n) \longleftarrow \dots \\ \searrow \\ \dots \longleftarrow 0 \longleftarrow C^\infty(T^*M)_m \longleftarrow C^\infty(T^*M)_m^n \longleftarrow \end{array}$$

is a resolution of $\left(\frac{C^\infty(T^*M)}{\{f:f|_{\Gamma(0)}=0\}}\right)_m$ and by construction it is free and hence flat.

It remains to tensor it with

$$\left(\frac{C^\infty(T^*M)}{\{f:f|_{\Gamma(dS)}=0\}}\right)_m = C^\infty(M)_m,$$

where we then obtain the complex

$$\begin{array}{c} \wedge^2(C^\infty(M)_m^n) \longleftarrow \wedge^3(C^\infty(M)_m^n) \longleftarrow \dots \\ \searrow \\ \dots \longleftarrow 0 \longleftarrow C^\infty(M)_m \longleftarrow C^\infty(M)_m^n \longleftarrow \end{array}$$

where the differential d_0 is now given by

$$\begin{aligned} C^\infty(M)^n &\rightarrow C^\infty(M) \\ \{f_j\}_{1 \leq j \leq n} &\mapsto \sum_{j=1}^n f_j \cdot (g^j \circ dS) = \sum_{j=1}^n f_j \cdot \left(\frac{\partial}{\partial x^j} S\right), \end{aligned}$$

and the higher differentials given similarly such that they combine to a derivation on the given complex.

So we change our complexes from considering formal linear combinations of the g^j (which we originally had chosen as the generating set of $C^\infty(T^*M)$) over $C^\infty(T^*M)$ to considering formal linear combinations over $C^\infty(M)$. We can now identify the g^j with $\frac{\partial}{\partial x^j} \in TM$, as the linear map on T^*M given by inserting $\frac{\partial}{\partial x^j}$ has the same action on T^*M as g^j . Then we obtain that the degree 1 part of the complex is given by the local vector fields on M , parametrized in coordinates by the local frame of the $\frac{\partial}{\partial x^i}$. The similar equivalences hold in higher degrees, such that we obtain

$$\begin{array}{ccccccc} \leftarrow \wedge^2(\Gamma^\infty(M, TM)_m) & \longleftarrow & \wedge^3(\Gamma^\infty(M, TM)_m) & \longleftarrow & \cdots & & \\ & & & & & \nwarrow & \\ \cdots & \longleftarrow & 0 & \longleftarrow & C^\infty(M)_m & \longleftarrow & \Gamma^\infty(M, TM)_m \longleftarrow \end{array} .$$

The differential here is given by the action of dS onto the vector fields through $\vee dS$, or equivalently the insertion of S itself through ι_S .

The last step that remains to be seen is the fact that tensoring over $C^\infty(T^*M)$ is the same as tensoring over $C^\infty(M)$. But this is evident from the fact that we can embed $C^\infty(M)$ into $C^\infty(T^*M)$ as the fiber-wise constant functions. Thus we can bring the complex into the form given in the theorem. $\square$

2 The Complex of Classical Observables

2.1 The Diffeological Set-up for Classical Field Theory

Now that we want to start working in a more physical context, we will always consider the following setting: Let M be a manifold with a fiber bundle $\pi : F \rightarrow M$. We will, just as in Example 1.1.2, consider the space $\mathfrak{F} = \Gamma^\infty(M, F)$ together with an action functional $S : \mathfrak{F} \rightarrow \mathbb{R}$. If M contains only finitely many points, we can just apply the result from that example, which is why we explicitly exclude that case. The same goes for the case $F = M$. But as assuming $\mathfrak{F}$ to be a manifold is much too rigid, we will consider it as a diffeological space with the functional diffeological structure.

To understand what we are working with, first let us review the notion of a diffeological space. The concept, including the main definitions and properties we will use here, will be introduced along the lines of [3] and [23].

Definition 2.1.1. A diffeological space $\mathfrak{D}$ is a set D (as with manifolds, often also denoted $\mathfrak{D}$ itself) together with an association

$$U \mapsto d(U) \subset \text{Hom}_{\text{Set}}(U, D)$$

for every $U \in \text{Eucl}$ (meaning every $U \subset \mathbb{R}^n$ open). The maps contained in $d(U)$ are considered the smooth maps from U to D . Further, the association d must satisfy the following criteria:

- I) $d(U)$ must contain all constant maps.
- II) If $f : U \rightarrow V$ is a smooth map between $U, V \in \text{Eucl}$ and $p \in d(V)$, then $p \circ f \in d(U)$.
- III) If $p : U \rightarrow D$ is a map of sets and $\{V_i\}_{i \in I}$ a cover of U such that $p|_{V_i} \in d(V_i)$ for all V_i , then $p \in d(U)$.

Let $\mathfrak{D}_1$ and $\mathfrak{D}_2$ be two diffeological spaces with underlying sets D_1 and D_2 and diffeologies d_1 and d_2 respectively. A morphism between $\mathfrak{D}_1$ and $\mathfrak{D}_2$ is a map $g : D_1 \rightarrow D_2$ such that for any $U \in \text{Eucl}$ we have that

$$g_* : d_1(U) \subset \text{Hom}_{\text{Set}}(U, D_1) \rightarrow \text{Hom}_{\text{Set}}(U, D_2)$$

has its image contained in $d_2(U)$. We use the notation $g_*(f) = g \circ f$ here.

As we would like to see diffeological spaces as spaces together with a notion of “smooth maps”, we will denote the set of morphisms between $\mathfrak{D}_1$ and $\mathfrak{D}_2$ as $C^\infty(\mathfrak{D}_1, \mathfrak{D}_2)$. Similarly, we denote by $C^\infty(\mathfrak{D})$ all morphisms from $\mathfrak{D}$ to $\mathbb{R}$, when we interpret $\mathbb{R}$ as a diffeological space with diffeology given by $r : U \mapsto C^\infty(U)$ for any $U \in \text{Eucl}$. That this notation is consistent with the pre-existing one is motivated in the following example.

Example 2.1.2. Let M be a manifold. We can define the diffeological space $\mathfrak{M}$ from the underlying set $|M|$ by defining $m : U \mapsto C^\infty(U, M)$ for any $U \in \mathbf{Eucl}$ as the diffeology.

Let N be another manifold, with associated diffeological space $\mathfrak{N}$. Then the morphisms between $\mathfrak{M}$ and $\mathfrak{N}$ are given by $C^\infty(M, N)$.

Example 2.1.3. The space $\mathfrak{G} = C^\infty(M, N)$ is a diffeological space with respect to the functional diffeology: For $U \subset \mathbb{R}^n$, a map $f : U \rightarrow \mathfrak{G}$ is smooth if and only if $\text{ev}_{m \in M} \circ f : M \times U \rightarrow N$ is smooth.

Example 2.1.4. We will from now on work with $\mathfrak{F} = \Gamma^\infty(M, F)$ endowed with the subspace diffeology of the functional diffeology on $C^\infty(M, F)$. This means a function $f : U \rightarrow \mathfrak{F}$ is smooth if and only if it smooth as a function $U \rightarrow C^\infty(M, F)$.

I will omit the proofs here that these are indeed diffeologies.

Remark. Note that if we have a smooth map $p : \mathbb{R}^k \rightarrow C^\infty(M, N)$ of functions with respect the functional diffeology, we can analyse the values these functions take at a fixed point $m \in M$. By definition, they have to assemble into a smooth map $f : \mathbb{R}^k \rightarrow N$ along p . Moreover, the directional derivatives and higher directional derivatives at m (which only can be defined locally with respect to a coordinate chart on N) only vary smoothly as well. This implies that we have a natural smooth map from $\mathfrak{F} \times M$ into any jet bundle $J^k F$ of $F \rightarrow M$, where we use the identification of Example 2.1.2 to view $J^k F$ as a diffeological space.

As in the case of manifolds, it is beneficial to not only be able to talk about the smooth functions globally, but also locally on the stalk of a point φ . Therefore we need to endow $\mathfrak{F}$ with a topology, that should at best arise naturally from the diffeological structure.

Definition 2.1.5. The diffeological topology on a diffeological space $\mathfrak{D}$ is given by the finest (i.e. strongest) topology such that all smooth maps $d(U)$ for $U \in \mathbf{Eucl}$ are continuous.

From now on, we will consider this topology, especially when working on stalks.

Another concept we have been working with on manifolds were the respective tangent spaces. We also want to consider these in a diffeological setting. However, there are two (initially different) possible generalized definitions throughout the literature. The first one we will discuss here arises naturally from a categorical construction. The second definition has a construction that is more explicit.

Definition 2.1.6. (cf. [3], Definition 2.2.1) Let $T : \mathbf{Mfld} \rightarrow \mathbf{Mfld}$ denote the tangent functor on the category of manifolds. Let $I : \mathbf{Mfld} \rightarrow \mathbf{Diffg}$ be

the functor from manifolds to diffeological spaces that arises from Example 2.1.2. Then the left Kan extension of $I \circ T$ along I will be called the tangent functor of diffeological spaces and usually also be denoted by T .

Given a diffeological space $\mathfrak{D}$, we call $T(\mathfrak{D})$ (also denoted $T\mathfrak{D}$) the tangent space of $\mathfrak{D}$.

The natural transformation between T and id given by the projection to the base space on $\mathbf{Mfld}$ further gives rise to a natural map $T\mathfrak{D} \rightarrow \mathfrak{D}$.

Definition 2.1.7. (cf. [23], p.43) Given a diffeological space $\mathfrak{D}$, the tangent space $T\mathfrak{D}$ is given by

$$\{p : \mathbb{R} \rightarrow \mathfrak{D} \text{ smooth}\} / \sim .$$

The equivalence relation is given by $p \sim q$ if $p(0) = q(0)$ and for every $f \in C^\infty(\mathfrak{D})$ we have that

$$\frac{\partial}{\partial t} f(p(t))|_{t=0} = \frac{\partial}{\partial t} f(q(t))|_{t=0}.$$

The diffeological structure on $T\mathfrak{D}$ is given by the push-forward diffeology of $C^\infty(\mathbb{R}, \mathfrak{D})$ (equipped with the functional diffeology) under the natural projection $\pi : C^\infty(\mathbb{R}, \mathfrak{D}) \rightarrow T\mathfrak{D}$. This is the coarsest diffeology in which all maps of the form $\pi \circ g$ are smooth, where $g : \mathbb{R}^k \rightarrow C^\infty(\mathbb{R}, \mathfrak{D})$ denotes a smooth map.

The projection $T\mathfrak{D} \rightarrow \mathfrak{D}$ is induced by the map $\text{ev}_0 : C^\infty(\mathbb{R}, \mathfrak{D}) \rightarrow \mathfrak{D}$. This assignment $[p]_\sim \mapsto p(0)$ is by definition well-defined for each equivalence class.

Remark. In contrast to manifolds, the tangent space at a point φ in a diffeological space is not necessarily equal to the space of derivations of $C^\infty(\mathfrak{D})$ at that point. To be precise, let $\text{Der}_\varphi(C^\infty(\mathfrak{D}))$ denote the space of derivations at φ , i.e. the space of $\mathbb{R}$ -valued derivations of $C^\infty(\mathfrak{F})$ with respect to the evaluation at φ as a morphism of rings $\text{ev}_\varphi : C^\infty(\mathfrak{F}) \rightarrow \mathbb{R}$.

For the first definition, we obtain a natural map $T_\varphi \mathfrak{D} \rightarrow \text{Der}_\varphi(C^\infty(\mathfrak{D}))$, that is neither injective nor surjective in general (cf. [4], Remark 3.21). It arises from the natural transformation of the respective functors on manifolds. For the second definition, it holds that

$$T_\varphi \mathfrak{D} \subset \text{Der}_\varphi(C^\infty(\mathfrak{D})),$$

where $T_\varphi \mathfrak{D}$ is a cone in the vector space of derivations and no longer closed under addition.

However, these definitions are not as different as one might fear. It is shown in [3] (cf. Proposition 2.2.6) that there indeed is a very natural surjective map from $C^\infty(\mathbb{R}, \mathfrak{D})$ to $T\mathfrak{D}$, where $T\mathfrak{D}$ denotes the tangent space with respect to Definition 2.1.6, such that $T\mathfrak{D}$ carries the push-forward topology.

When considering the constructions explicitly, we even find that for the two definitions to be equivalent it is sufficient that $T_\varphi \mathfrak{D} \rightarrow \text{Der}_\varphi(C^\infty(\mathfrak{D}))$ (still with respect to the Definition 2.1.6) is injective for every $\varphi \in \mathfrak{F}$. This is actually a result that holds for $\mathfrak{F}$ (cf. Proposition 2.1.9).

The missing link we still need to prove this is the following theorem. It gives us computational control over the categorical construction of the first definition.

Proposition 2.1.8. *(cf. [3], Theorem 2.3.2) The tangent space of $\mathfrak{F}$ in the sense of Definition 2.1.6 can be naturally identified with*

$$T\mathfrak{F} = \Gamma^\infty(M, VF),$$

where $VF \subset TF$ is defined as $VF = \ker(T\pi_{(F \rightarrow M)} : TF \rightarrow TM)$. The fiber of $T\mathfrak{F}$ over $\varphi \in \mathfrak{F}$ is given by those sections that project to φ under $\pi_{(VF \rightarrow F)}$. The diffeology on $T\mathfrak{F}$ corresponds to the subspace diffeology of $\Gamma^\infty(M, VF)$ in $C^\infty(M, VF)$.

Remark. Note that it is only through this identification that we can understand $T_\varphi \mathfrak{F}$ as a vector space.

This theorem now finally gives us the means to show the following with respect to the first definition of tangent spaces:

Proposition 2.1.9. *The natural map*

$$T_\varphi \mathfrak{F} \rightarrow \text{Der}_\varphi(C^\infty(\mathfrak{F}))$$

is injective. Here, $\text{Der}_\varphi(C^\infty(\mathfrak{F}))$ means the $\mathbb{R}$ -valued derivations of $C^\infty(\mathfrak{F})$ with respect to evaluation at φ .

Proof. The fact that the map is injective follows from applying Theorem 2.1.8: Assume that we had paths γ and p , that are not tangent as elements of $\Gamma^\infty_\varphi(M, VF)$. Then there is some $m \in M$, where they disagree.

Choose local coordinates around $\varphi(m)$ that respect the fiber structure of $F \rightarrow M$, i.e. choose a coordinate chart $U \subset M$ over which the bundle is trivializable and coordinates on $V \subset F_m$, such that $\varphi(m) \in U \times V$. Further choose some $\chi \in C^\infty(F_m)$ with compact support completely inside the coordinate chart V , such that $\chi \equiv 1$ on an open around the point corresponding to $\varphi(m)$.

We can now construct for every coordinate function x^i of $V \cong \mathbb{R}^k$ the functions

$$s_m^i(\psi) = \begin{cases} \chi(\psi(m)) \cdot \psi(m)^i & \text{if } \psi(m) \text{ lies inside the chart} \\ 0 & \text{else,} \end{cases}$$

which we can explicitly check to be smooth.

Now assume we realize γ as a path $\gamma(t) : \mathbb{R} \rightarrow \mathfrak{F}$. We then have that, as $\chi \equiv 1$ around $\varphi(m)$, the function along that path takes the values of $s_m^i(\gamma(t)(m)) = \gamma(t)(m)^i$ for small enough values of t . If we want to consider the action of $\gamma(t)$ as a derivation on s_m^i , we hence need to derive this function, giving us the i -th coordinate component of

$$\gamma(m)(t) \in T_{\varphi(m)}(F_m) \cong \mathbb{R}^k.$$

Note that $(V_m)_{\varphi(m)}F = T_{\varphi(m)}(F_m)$, which means we actually recover $\gamma(m) \in V_mF$. The same construction is of course also applicable to p , also giving us a way to recover $p(m)$ through the coordinates induced by V .

But we assumed $\gamma(m) \neq p(m)$, such that these values have to differ for some $1 \leq i \leq k$. So p and γ do not agree as derivations, which was what we wanted to show. $\square$

Remark. It actually holds that the map is linear with respect to the vector space structures of $\Gamma^\infty(M, VF)$ and $\text{Der}_\varphi(C^\infty(\mathfrak{F}))$. The idea behind this is the following:

We know that for any map $p : \mathbb{R}^2 \rightarrow \mathfrak{F}$ the paths $\gamma_1 : t \mapsto (t, 0) \in \mathbb{R}^2$ and $\gamma_2 : t \mapsto (0, t) \in \mathbb{R}^2$ define paths in $\mathfrak{F}$. These satisfy that their action as derivations on a function $f \in C^\infty(\mathfrak{F})$ is already determined by the values $f \circ p \in C^\infty(\mathbb{R}^2)$. But on manifolds, we know that $\frac{\partial}{\partial \gamma_1} + \frac{\partial}{\partial \gamma_2} = \frac{\partial}{\partial \gamma'}$ for the path induced by $\gamma' : t \mapsto (t, t)$.

On the other hand, we can consider the function on $\mathfrak{F}$ that is given by evaluation at $m \in M$. This induces a smooth map $\text{ev}_m \circ p : \mathbb{R}^2 \rightarrow F_m$, through which the paths γ_1 and γ_2 induce elements $v_1(m)$ and $v_2(m)$ that lie in VF . These assemble into exactly the tangent vectors $v_1, v_2 \in \Gamma^\infty(M, VF)$ that the γ_i represent. But here, we also have that γ' induces exactly the element $v'(m) = v_1(m) + v_2(m)$.

Thus the addition of the tangent vectors represented by the γ_i corresponds exactly to the point-wise addition of the corresponding sections v_i . The only thing we are missing is that we can smoothly extend any tangent vector along one representative path for any second tangent vector, which would prove the existence of a map p as described above for any choice of v_i .

But this follows from a result of [3], which states we can realise any tangent vector at any point φ as a path in a tubular neighborhood of U of $\Gamma(\varphi)$ (cf. Lemma 2.3.7). We can choose such a neighborhood small enough such that each section contained in U locally corresponds, on a fixed cover of M by opens, to a section of a vector bundle. As smoothness of an extension is a property that is local on M as well, and through decomposing and reassembling $\Gamma^\infty(M, VF)$ through a partition of unity on M , we may simply consider the claim on a vector bundle, where the statement is clear.

From this we can also deduce the following:

Lemma 2.1.10. *The natural map*

$$\Gamma^\infty(\mathfrak{F}, T\mathfrak{F}) \rightarrow \text{Der}(C^\infty(\mathfrak{F}))$$

is injective.

Proof. By the point-wise nature of the multiplication on $C^\infty(\mathfrak{F})$, we have that every derivation composed with evaluation at φ gives us an element of $\text{Der}_\varphi(C^\infty(\mathfrak{F}))$. In the case of a derivation by $X \in \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$, this recovers $X(\varphi)$ as a derivation. So if two vector fields X and Y agree as derivations, also $X(\varphi) = Y(\varphi)$ as tangent vectors by 2.1.9. As this holds for all $\varphi \in \mathfrak{F}$, we have $X = Y$. $\square$

However, the tangent space is only one of the ingredients that we need. For the definition of the derived critical locus we also need a definition of the cotangent space of $\mathfrak{F}$.

Definition 2.1.11. The cotangent bundle $T^*\mathfrak{F} \rightarrow \mathfrak{F}$ is defined fiber-wise as

$$T_\varphi^*\mathfrak{F} = \{L : T_\varphi\mathfrak{F} \rightarrow \mathbb{R} \text{ s.t. } L \in C^\infty(T_\varphi\mathfrak{F}) \text{ and } \mathbb{R}\text{-linear}\}.$$

The diffeology is given for $U \in \text{Eucl}$ by those maps $f : U \rightarrow T^*\mathfrak{F}$ that satisfy the following:

- I) The projection $\pi_{(T^*\mathfrak{F} \rightarrow \mathfrak{F})} \circ f : U \rightarrow \mathfrak{F}$ is smooth.
- II) Given a smooth map $g : U \rightarrow T\mathfrak{F}$, that satisfies

$$\pi_{T^*\mathfrak{F} \rightarrow \mathfrak{F}} \circ f = \pi_{T\mathfrak{F} \rightarrow \mathfrak{F}} \circ g = \gamma.$$

Then we have that $\langle g, f \rangle : U \rightarrow \mathbb{R}$ is smooth, which denotes the map

$$z \in U \mapsto f(z)(g(z)).$$

Note that we can apply $f(z) \in T_{\gamma(z)}^*\mathfrak{F}$ to $g(z) \in T_{\gamma(z)}\mathfrak{F}$ by definition.

This definition is not the only possible way to establish the cotangent space. For example, we defined the elements of $T_\varphi^*\mathfrak{F}$ as point-wise linear maps on $T_\varphi\mathfrak{F}$ without further conditions. In other definitions, they can be required to be extendible to a smooth fibre-wise linear map on $T\mathfrak{F}$. However, the following lemma tells us that in this case, both versions of the definition are equivalent.

Lemma 2.1.12. *We have that any $u \in T_\varphi^*\mathfrak{F}$ can be extended to an exact differential 1-form (df) on $\mathfrak{F}$. Here, $(df) \in \Gamma^\infty(\mathfrak{F}, T^*\mathfrak{F})$ denotes the section that at any $\psi \in \mathfrak{F}$ is given by*

$$(df)(\psi)(X) = X(f)$$

for all $X \in T_\psi \mathfrak{F}$. The action of X on f is given by $T_\psi \mathfrak{F} \rightarrow \text{Der}_\psi(C^\infty(\mathfrak{F}))$.

Furthermore, the map $C^\infty(\mathfrak{F}) \times \mathfrak{F} \rightarrow T^*\mathfrak{F}$ is a subduction, i.e. it is surjective and the diffeology on $C^\infty(\mathfrak{F}) \times \mathfrak{F}$ (which is given by the product diffeology of $C^\infty(\mathfrak{F})$ with the functional diffeology and $\mathfrak{F}$) induces the diffeology on $T^*\mathfrak{F}$. By this, we mean that $T^*\mathfrak{F}$ carries the push-forward diffeology with respect to the given projection, i.e. a map $\mathbb{R}^k \rightarrow T^*\mathfrak{F}$ is smooth if and only if it can locally be expressed as the image of a smooth map $U \subset \mathbb{R}^k \rightarrow C^\infty(\mathfrak{F}) \times \mathfrak{F}$.

Remark. We know that for any $f \in C^\infty(\mathfrak{F})$ the corresponding (df) defines a smooth section $(df) \in \Gamma^\infty(\mathfrak{F}, T^*\mathfrak{F})$, as the notions of smoothness are well compatible with each other. So we have actually found an identification between differential 1-forms at φ and cotangent vectors at φ .

Proof. On a vector bundle $W \rightarrow M$, there is a clear intuition of why this should hold: We can identify $\Gamma^\infty(M, W) \cong \Gamma^\infty_\varphi(M, VW)$, as at any $m \in M$ we have a natural isomorphism between W_m and $T(W_m) = V_m W$. So when applying the result of 2.1.8, any function on $T_\varphi \mathfrak{F}$ gives automatically rise to a function on $\mathfrak{F}$, and it turns out that this function has exactly the extension property we want.

Given an element $u \in T_\varphi^* \mathfrak{F}$ for a general fiber bundle $F \rightarrow M$, we will find a similar method to define a function f . The first thing we need to do is to associate to a tangent vector $v \in T_\varphi \mathfrak{F}$ to every $\psi \in \mathfrak{F}$ by applying a variation of the method for vector bundles in local coordinates. So then u as a function on $T_\varphi \mathfrak{F}$ defines a function on $\mathfrak{F}$ just as before. Then lastly, we will prove that indeed this function has the desired property that $(df)(\varphi) = u$. This gives us the first part of the claim.

For the second part of the claim, the surjectivity follows from the first, so it remains to show that $T^*\mathfrak{F}$ indeed carries the induced topology. As we know that the projection $T^*\mathfrak{F} \rightarrow \mathfrak{F}$ is smooth, it is therefore sufficient to show that a smooth map $\mathbb{R}^k \rightarrow T^*\mathfrak{F}$ can locally be expressed as coming from (df_z) , where f_z denotes a smooth homotopy of functions. This is actually already clear for a map fully contained in a single fiber $T_\varphi \mathfrak{F}$, as the constructed map between $\mathfrak{F}$ and $T_\varphi \mathfrak{F}$ is smooth. We will then check that the construction we have used to show the first claim can actually be generalized to yield smooth results for varying basepoints.

So now, let us prove the first claim: Assume we are given $\varphi \in \mathfrak{F}$ and some $u \in T_\varphi^* \mathfrak{F}$. We now aim to define a certain function $f \in C^\infty(\mathfrak{F})$ and then check that it satisfies the property of the lemma.

Define a function $\chi \in C^\infty(F)$ supported on a neighborhood of the graph $\Gamma(\varphi)$, such that on an even smaller neighborhood U of $\Gamma(\varphi)$ we have $\chi|_U \equiv 1$. Further choose χ in a way that there is a cover $\{V_i\}_{i \in I}$ of M , where $\pi^{-1}(V_i) \cap \text{supp}(\chi)$ comes with a fixed open superset admitting global coordinates, which respect the fiber structure. Also fix an underlying locally finite partition of unity $\{\mu_i\}_{i \in I}$

Now assume we have given $\psi \in \mathfrak{F}$ and $m \in M$ with $\psi(m)$ in the range of some coordinate patch around $\text{supp}(\chi)$. Then, for the respective V_i with $m \in V_i$ admitting these coordinates, there is a unique path on $\mathbb{R}$ from $\varphi(m)$ at 0 to $\psi(m)$ at 1 that is linear (in the corresponding coordinates). This then defines an element in $V_{\varphi(m)}F$ which we call $v_i(\psi)(m)$.

Now we can consider $v_\varphi \in \Gamma_\varphi^\infty(M, VF)$ given by

$$v_\varphi = \left(m \mapsto \begin{cases} \chi(\psi(m)) \cdot \sum_{i \in I} \mu_i(m) v_i(\psi)(m) & \text{if } v_i(\psi)(m) \text{ is defined} \\ 0 & \text{else} \end{cases} \right),$$

which we can explicitly check to give us a smooth map $v_\varphi : \mathfrak{F} \rightarrow T_\varphi \mathfrak{F}$. This allows us to define

$$f(\psi) = u(v_\varphi(\psi)),$$

which is indeed a smooth function, as v is constructed smoothly depending on m and ψ and u is smooth.

Now assume we have some $X \in T_\varphi \mathfrak{F}$ represented by the path $p : \mathbb{R} \rightarrow \mathfrak{F}$. Then we have

$$\begin{aligned} X(f) &= \frac{\partial}{\partial t} f(p(t))|_{t=0} \\ &= \frac{\partial}{\partial t} u(v_\varphi(p(t)))|_{t=0} \\ &= u \left(\frac{\partial}{\partial t} v_\varphi(p(t))|_{t=0} \right), \end{aligned}$$

where the last derivation is taken point-wise in VF as a finite dimensional vector space. The derivation can be computed there as u is a linear map and we can write $v(t)$ as

$$v_\varphi(p(t)) = v_\varphi(p(0)) + t \cdot \frac{\partial}{\partial t} v_\varphi(p(t))|_{t=0} + \frac{t^2}{2} \tilde{v}(t)$$

by point-wise analysis on M and smoothness of v . But we can specify this expression further:

$$w_0 = \frac{\partial}{\partial t} v_\varphi(p(t))|_{t=0} = \left(m \mapsto \sum_{i \in I} \mu_i(m) \frac{\partial}{\partial t} v_i(p(t))(m)|_{t=0} \right),$$

as $\chi(p(t)(m)) = 1$ for t in an open around 0.

Here, by our construction of v_φ , we have that $v_i(p(t))(m) \in V_{\varphi(m)}F$ is given by a linear path in the coordinates of V_i of length 1 starting at $\varphi(m)$ and ending at $p(t)(m)$. So as $v_i(p(0))(m)$ is the constant path at $\varphi(m)$ and hence corresponds to 0, we have that $\frac{\partial}{\partial t} v_i(p(t))(m) \in V_{\varphi(m)}F$, when considered as an equivalence class of pathes in F_m , is the limit of the equivalence classes containing the linear paths of length t starting at $\varphi(m)$

and ending at $p(t)(m)$. But this is just a description of constructing a linear tangent to $p(t)(m)$ at 0, such that $p(t)(m)$ itself lies in the same equivalence class. Thus we have $w_0(m) = X(m)$ and $w_0 = X$.

This means that after all,

$$X(f) = u(X),$$

which proves our first claim, as f gives rise to an exact differential 1-form with this action at $T_\varphi \mathfrak{F}$.

We will now show the second claim. Assume we are given a smooth map $u(z) : \mathbb{R}^k \rightarrow T^* \mathfrak{F}$. As the projection from $T^* \mathfrak{F}$ to $\mathfrak{F}$ is smooth, we have a smooth homotopy $\varphi(z)$. What we have seen previously is that the map $v_\varphi : \mathfrak{F} \rightarrow T_\varphi \mathfrak{F}$ is smooth. We now aim to show that the construction is also smooth in φ , giving us a smooth map $v : \mathfrak{F} \times \mathfrak{F} \rightarrow T \mathfrak{F}$. When checking the definition of $f_z : \mathbb{R}^k \rightarrow C^\infty(\mathfrak{F})$, we then see that we have obtained a smooth homotopy.

Consider all open subsets $U \times O \subset F$, where U is a pre-compact open on M with a local trivialization $U \times F_m \rightarrow U$ and we have that $U \times O \cong \mathbb{R}^p \times \mathbb{R}^q$, giving us coordinates respecting the fiber bundle structure. As we can refine the cover first with respect to the U_i covering M and then with respect to F_m , we can choose a locally finite cover of F by sets $U_i \times O_i$ of the given form, together with a partition of unity ξ_i .

Now given any φ , we have that the projections U_i together with the partition of unity

$$\mu_i(m) = \xi_i(\varphi(m))$$

form a cover of M . Even though the cover itself is no longer locally finite, the sum of the χ_i is. Also we can choose some $g \in C^\infty(\mathbb{R})$ with $g(z) = 1$ for $z \in [0, 1]$ and $g(z) = 0$ for $z \geq 2$. Then for m such that $\varphi(m) \in U_i \times O_i$ we could define

$$\chi_i(m, x) = \begin{cases} g(\text{dist}_{O_i}(x, \varphi(m))) & \text{if } x \in O_i \\ 0 & \text{else,} \end{cases}$$

where m denotes the base coordinate and $x \in F_m$ the fiber component of a point in $U_i \times F_m$. By the choice of g , this function is smooth. As we have that $\mu_i(m)$ is supported on a subset of the open where $\varphi(m) \in U_i$ we have that especially $\chi_i(m, x) \cdot \mu_i(m)$ can be smoothly extended by 0 to all of $U_i \times F_m$.

If we now reconsider the definition of $v_\varphi(\psi) \in \Gamma^\infty(M, VF)$ through

$$v_\varphi(\psi) = \left(m \mapsto \begin{cases} \sum_{i \in I} \chi_i(\psi(m)) \cdot \mu_i(m) \cdot v_i(\psi)(m) & \text{if } v_i(\psi)(m) \text{ is defined} \\ 0 & \text{else} \end{cases} \right),$$

we see that the $v_i(m)$ might not be defined everywhere, not even everywhere on U_i . However, they are defined on an open containing the support of

$\chi_i(\psi(m)) \cdot \mu_i(m)$, meaning that the term

$$\chi_i(\psi(m)) \cdot \mu_i(m) \cdot v_i(m)$$

can smoothly be extended by 0 where it is not defined.

This definition indeed does not only depend on ψ smoothly but also on $\varphi \in \Gamma^\infty(M, F)$, as it only depends on $\varphi(m)$ at each $m \in M$. But around each $\varphi(m) \in F$ there is an open W such that if $\varphi(m) \in U_i \times O_i$ also $W \subset U_i \times O_i$ and if $\varphi(m) \notin U_i \times O_i$ we have $\xi_i \equiv 0$ on W .

Hence we indeed have found a smooth way to construct v depending on φ and ψ . If we now consider

$$f_z(\psi) = u(z)(v_{\varphi(z)}(\psi)),$$

we have that as $v : \mathfrak{F} \times \mathfrak{F} \rightarrow T\mathfrak{F}$ is smooth, the corresponding $f_z : \mathbb{R}^k \times \mathfrak{F} \rightarrow \mathbb{R}$ is smooth as well, so that f_z is indeed a smooth homotopy.

The proof that for the f_z holds

$$X(f_z) = u(z)(X)$$

is similar to the proof above and will not be repeated here. Hence we have indeed shown that the diffeology on $T^*\mathfrak{F}$ is induced by the diffeology of $C^\infty(\mathfrak{F}) \times \mathfrak{F}$, proving our claim. $\square$

2.2 Fiber-wise Linear Functions on the Cotangent Space

Another property of the diffeological space $\mathfrak{F}$ is characterised by the theorem we will prove in this section. It essentially tells us that a smooth linear function on $T_\varphi^*\mathfrak{F}$ has to be given by the insertion of a tangent vector. We will use the result later when computing the observables.

Proposition 2.2.1. *Assume that $M = \mathbb{R}^p$ and $F = \mathbb{R}^p \times \mathbb{R}^q$, such that we have $T_\varphi\mathfrak{F} \cong (C^\infty(\mathbb{R}^p))^q$. Choose a strictly monotone function $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $f(0) = 0$ and the derivatives of all orders at 0 vanish. Consider some $\eta \in C_c^\infty(\mathbb{R}^p)$ supported on $B_1(0)$ with $\int \eta = 1$ and $\eta(0) = 1$ and define η_t by*

$$\eta_t(x) = t^{-p} \eta\left(\frac{x}{t}\right)$$

for $t \neq 0$.

Through the identification $T_\varphi\mathfrak{F} \cong (C^\infty(\mathbb{R}^p))^q$, where the q components arise naturally from the global coordinates on $\mathbb{R}^q$, we can define an action of η_t on any $X \in T_\varphi\mathfrak{F}$ through

$$\eta_t * X = \eta_t * \left(\sum_{i=1}^q e_i \cdot X^i \right) = \sum_{i=1}^q e_i (\eta_t * X^i),$$

where $e_i \cdot X^i$ denotes the decomposition of X into its q components as an element of $(C^\infty(\mathbb{R}^p))^q$, with the $X^i \in C^\infty(\mathbb{R}^p)$ and $*$ denoting the usual convolution.

Then for any $s \in T_\varphi^* \mathfrak{F}$, the path $s_t : \mathbb{R} \rightarrow T_\varphi^* \mathfrak{F}$ given by

$$s_t = \begin{cases} s & \text{if } t = 0 \\ s * \eta_{f(|t|)} & \text{else} \end{cases}$$

is a well-defined and smooth path. Here we define the convolutions on the cotangent space given at $X \in T_\varphi \mathfrak{F}$ to be

$$(s * \eta_{f(t)})(X) = s(\eta_{f(t)} * X).$$

Furthermore all of the s_t for $t \neq 0$ can be expressed by integration against a smooth function that is compactly supported on M .

Remark. The result of this Proposition resembles quite closely the well-known result that $C_c^\infty(M)$ is dense in $\mathcal{D}_c(M)$, the space of compactly supported distributions on M . However, while every compactly supported distribution defines a cotangent vector, these can be much more - for example distributions on the directional derivatives.

Proof. As the proposition includes several claims, this proof will consist of a number of steps. We will have to show that s_t for $t \neq 0$ is a cotangent vector at all. Then we have to show that the path of the s_t is indeed smooth and finally, that it can be expressed as an integral.

The first thing that we now aim to prove is that the s_t are indeed cotangent vectors. So we need to show that for any $s \in T_\varphi^* \mathfrak{F}$ and any smooth map $X_x : \mathbb{R}^k \rightarrow T_\varphi \mathfrak{F}$ we have that $s * \eta_{f(t)}(X_x) : \mathbb{R}^k \rightarrow \mathbb{R}$ is smooth. For that it is sufficient to prove that

$$\eta_{f(t)} * X_x : \mathbb{R}^k \rightarrow T_\varphi \mathfrak{F}$$

also is a smooth map.

Let us at first consider the map evaluated at a fixed $y \in M = \mathbb{R}^p$. There we find that

$$(\eta_{f(t)} * X_x)(y) = \int_{\mathbb{R}^p} \eta_{f(t)}(w) X_x(y - w) dw,$$

such that the map

$$v : \mathbb{R}^k \times M \rightarrow VF$$

is continuous by well-known properties of the convolution (cf. [7], p.86 f.). We now have to prove that all directional derivatives are continuous as well, but as

$$\frac{\partial}{\partial p} (\eta_{f(t)} * X_x) = \eta_{f(t)} * \left(\frac{\partial}{\partial p} X_x \right)$$

for any direction p in $T\mathbb{R}^k \times TM$, this is also the case. Similarly, we can argue that the higher derivatives are continuous, such that indeed v is a smooth map and hence $\eta_t * X_x$ is smooth.

Quite similarly, we can show that the s_t assemble into a smooth path outside of $t = 0$, as for any other given t there is an open not containing 0 where the functions $\eta_{f(t)}$ are bounded by a uniform compact, allowing us to interchange integration and differentiation. What we want to show next is that the extension of the path to 0 is smooth as well.

At 0 we have that - as a smooth homotopy of tangent vectors converges in particular uniformly on compacts - if a smooth homotopy of tangent vectors $X_{x,t}$ lies in the fiber of φ , then also

$$\lim_{t \rightarrow 0} \eta_{f(t)} * X_{x,t} = X_{x,0}.$$

So the corresponding homotopy of tangent vectors after the convolution is continuous.

Also we can bound the difference

$$\eta_{f(t)} * X_{x,t}(y) - X_{x,t}(y)$$

by the term

$$f(|t|) \cdot \max_{\substack{z \in B_{f(t)}(y), \\ \frac{\partial}{\partial p} \in T_z M, |\frac{\partial}{\partial p}|=1}} \left| \frac{\partial}{\partial p} X_{x,t}|_z \right|,$$

because this is the maximal possible value that $X_{x,t}(y) - X_{x,t}(z)$ can have at any point $z \in B_{f(t)}(y)$.

The second term itself locally only depends on values on a compact and thus can be bounded by a finite constant for $-1 \leq t \leq 1$. By the nature of f , we have that

$$f(|t|) \cdot c$$

is a smooth function with vanishing derivative at 0. So also

$$\eta_{f(t)}(y) * X_{x,t} - X_{x,t}(y)$$

is differentiable at $t = 0$ with vanishing derivative in direction of t . So we have that $\eta_{f(t)}(y) * X_{x,t}$ is differentiable as well at $t = 0$, with

$$\frac{\partial}{\partial t} (\eta_{f(t)}(y) * X_{x,t}) = \frac{\partial}{\partial t} X_{x,t}$$

It remains to show that

$$\frac{\partial}{\partial t} (\eta_{f(t)}(y) * X_{x,t})$$

is continuous in t at $t = 0$.

To analyse the expression at $t \neq 0$, note that as we have previously remarked, we can interchange integration and differentiation. To simplify the expressions, it is however beneficial to execute a change of coordinates first. For every $t \neq 0$, we can express $\eta_{f(|t|)} * X_{x,t}(y)$ as

$$\int_M \frac{1}{f(|t|)^d} \eta\left(\frac{w}{f(|t|)}\right) X_{x,t}(y - w) \, dw = \int_M \eta(z) X_{x,t}(y - f(|t|) \cdot z) \, dz.$$

This expression is much nicer to derive in direction of t . We can compute

$$\begin{aligned} & \frac{\partial}{\partial t} \int_M \eta(z) X_{x,t}(y - f(|t|) \cdot z) \, dz \\ &= \int_M \frac{\partial}{\partial t} (\eta(z) X_{x,t}(y - f(|t|) \cdot z)) \, dz \\ &= \int_M \eta(z) \left(\frac{\partial}{\partial t} X_{x,t} \right) (y - f(|t|) \cdot z) \, dz \\ & \quad + \int_M \left(\frac{\partial f(|t|)}{\partial t} \right) \cdot \eta(z) \cdot (z \cdot (\nabla X_{x,t})(y - f(|t|) \cdot z)) \, dz \end{aligned}$$

So, as we know that $\lim_{t \rightarrow 0} \frac{\partial}{\partial t} f(|t|) = 0$ and the integral

$$\int_M \eta(z) \cdot (z \cdot (\nabla X_{x,t})(y - f(|t|) \cdot z)) \, dz$$

is compactly supported and hence well-defined for $t = 0$, we can conclude that

$$\frac{\partial}{\partial t} \eta_{f(|t|)} * X_{x,t}$$

converges to $\frac{\partial}{\partial t} X_{x,t}$ for $t \rightarrow 0$, even locally uniformly. Hence the derivative in direction of t is indeed a continuous function.

The arguments for all other directional derivatives (including the higher ones) are analogous to the given arguments, such that we can conclude that

$$\eta_{f(|t|)} * X_{x,t}$$

is indeed a smooth path, such that

$$s(\eta_{f(|t|)} * X_{x,t}) = s_t(X_{x,t})$$

is smooth as well.

Now that we have shown that the path $s_t : \mathbb{R} \rightarrow T_{\varphi}^* \mathfrak{F}$ is smooth, we will turn to the statement that for $t \neq 0$ we can express s_t by integration against a smooth function.

To see that statement, we would like to use that

$$s\left(\int_M \eta_{f(|t|)}(y - w) \cdot X(w) \, dw\right) = \int_M s(\eta_{f(|t|)}(y - w) \cdot X(w)) \, dw.$$

While the first term here is well-defined, the second might not be. But we know that if it is, it can be written as

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \lambda_k s(\eta_{f(|t|)}(y - w_k)X(w_k)) = \lim_{n \rightarrow \infty} s\left(\sum_{k=1}^n \lambda_k \eta_{f(|t|)}(y - w_k)X(w_k)\right),$$

where the w_k and λ_k indicate the partition of $\mathbb{R}$ into intervals to compute the conventional Riemannian integral.

Let us consider the case where X is compactly supported on $C \subset M$. This means that the second integral above is defined and so is the given limit. Then it is sufficient to show that there exists a smooth path Y_r along the functions $y \mapsto \sum_{k=1}^n \lambda_k \eta_{f(|t|)}(y - w_k)X(w_k)$ such that the values of r converge to a point r' where $Y_{r'} = \int_M \eta_{f(|t|)}(y - w)X(w)dw$. This would allow us to compute the limit of $s(\sum_{k=1}^n \lambda_k \eta_{f(|t|)}(y - w_k)X(w_k))$ as the value of $s(Y_{r'})$ and prove our claim.

Without loss of generality we may assume that C , the support of X , is given by a cube of intervals with integer endpoints. Let us consider the partition of C into smaller intervals given by

$$I_k = \left[\frac{1}{a^a} \cdot z_i^k, \frac{1}{a^a} \cdot (z_i^k + 1)\right]^p$$

for some $z_i^k \in \mathbb{Z}$ and $a \in \mathbb{N}$. We choose w_k as the center points and obtain that $\lambda_k = \frac{1}{a^{p \cdot a}}$.

Let us fix a smooth function χ with $\chi(x) = 0$ for $t \leq 0$ and $\chi(x) = 1$ for $t \geq 1$. Consider the functions

$$g_1 : y \mapsto \sum_{k=1}^{n_1} \lambda_k \eta_{f(|t|)}(y - w_k)X(w_k)$$

for the partition with $a = 1$ and

$$g_2 : y \mapsto \sum_{l=1}^{n_2} \lambda_l \eta_{f(|t|)}(y - w_l)X(w_l)$$

for the partition with $a = 2$. Now we can define a homotopy between these functions along $z = x \in [0, \frac{1}{2}]$, which is given by

$$g(y, x) = (1 - \chi(2x))g_1(y) + \chi(2x)g_2(y).$$

In quite a similar fashion, we can construct a homotopy on the interval $z \in [0, \frac{3}{4}]$. We do this by extending the given one along $z = (x + \frac{1}{2}) \in [\frac{1}{2}, \frac{3}{4}]$ by

$$g(y, x + \frac{1}{2}) = (1 - \chi(4x))g_2(y) + \chi(4x)g_3(y),$$

where g_3 is the function that arises from the partition for $a = 3$.

We know that if we continue this for $x \rightarrow 1$ we have that

$$\begin{aligned} & g(y, x + c_a) - \int \eta_{f(|t|)}(y - w)X(w) \, dw \\ &= (1 - \chi(2^a x)) \left(g_a(y) - \int \eta_{f(|t|)}(y - w)X(w) \, dw \right) \\ & \quad + \chi(2^a x) \left(g_{a+1}(y) - \int \eta_{f(|t|)}(y - w)X(w) \, dw \right), \end{aligned}$$

for the appropriate c_a . But on every one of the little cubes of our partition, the values of the function $\eta_{f(|t|)}(y - w)X(w)$ on the cube and the value at the center differ by at most

$$\sqrt{d} \cdot \frac{1}{a^a} \cdot \max_{\substack{z \in C, \frac{\partial}{\partial w} \in T_z M, \\ |\frac{\partial}{\partial w}|=1}} \frac{\partial}{\partial w} (\eta_{f(|t|)}(y - w)X(w)).$$

So each term $g_{a+1}(y) - \int \eta_{f(|t|)}(y - w)X(w)dw$ can be bounded in absolute value by

$$\text{vol}(C) \cdot \sqrt{d} \cdot \frac{1}{a^a} \cdot \max_{\substack{z \in C, \frac{\partial}{\partial w} \in T_z M, \\ |\frac{\partial}{\partial w}|=1}} \frac{\partial}{\partial w} (\eta_{f(|t|)}(y - w)X(w)).$$

As X is compactly supported, we know that the maximum exists for every value of y separately, and by smoothness we can even replace it by a uniform bound for any compact $K \subset M$.

The constructed bound decreases by at least $\frac{1}{a}$ for every new iteration a of the homotopy, we now know that on any compact K the function $g(x, y)$ converges uniformly to $\int \eta_{f(|t|)}(y - w)X(w)dw$ for $x \rightarrow 1$, satisfying that the function is differentiable in direction of x at $x = 1$ with differential given by 0.

If we analyze the derivative in direction of x for $x \neq 1$, we find that it is given by

$$2^a \chi'(2^a x) (g_a(y) - g_{a+1}(y)),$$

where the term in brackets is again bounded by a constant similar to the first. At first, this shows that the derivative in direction of x is continuous at $x = 1$. Further, as again for each iteration a the bound decreases by the factor $\frac{1}{a}$, we have that for $x \rightarrow 1$ the second derivative exists and equals 0. This argument can by iteration be applied also to the higher derivatives in direction of x . Hence the extension of g to $z \in [0, 2]$ by the constant homotopy on $[1, 2]$ is smooth on $(0, 2)$.

Nowhere in our proof have we used the specific form of $\eta_{f(|t|)}(y - w_k)$, which is why all the arguments work as well with

$$\frac{\partial}{\partial y} \eta_{f(|t|)}(y - w_k).$$

Hence we can apply a method quite similar to the above to conclude that all directional derivatives of $g(y, z)$ exist for all $y \in M$ and $z \in (0, 2)$ and that they are continuous, such that indeed g is smooth restricted to this domain.

Thus we have that indeed

$$\lim_{n \rightarrow \infty} s\left(\sum_{k=1}^n \lambda_k \eta_{f(|t|)}(y - w_k) X(w_k)\right) = s\left(\int \eta_{f(|t|)}(y - w) X(w) dw\right).$$

But by our previous considerations, this is sufficient to compute

$$\begin{aligned} & s\left(\int_M \eta_{f(|t|)}(y - w) \cdot X(w) dw\right) \\ &= \int_M s(\eta_{f(|t|)}(y - w) \cdot X(w)) dw \\ &= \int_M \sum_{i=1}^q (s(\eta_{f(|t|)}(y - w) \cdot e_i) \cdot X^i(w)) dw, \end{aligned}$$

which implies that s_t is indeed given by integration against a smooth function for compactly supported X .

Before we continue the proof for non-compact X , let us review why we imposed this condition: So far, we have assumed X to be compactly supported to imply that all of the integrals we encountered converged, especially

$$\int_M s(\eta_{f(|t|)}(y - w) \cdot X(w)) dw = \int_M \sum_{i=1}^q (s(\eta_{f(|t|)}(y - w) \cdot e_i) \cdot X^i(w)) dw.$$

Going forward, we will ensure this instead by proving that the function

$$w \mapsto s(\eta_{f(|t|)}(y - w) \cdot e_i)$$

is compactly supported, which is a result that we aimed to prove anyway.

So assume that the function was not compactly supported for a given i and a given $t \neq 0$. Then we could find a sequence w_n for $n \in \mathbb{N}$, such that

$$s(\eta_{f(|t|)}(y - w_n) \cdot e_i) \neq 0$$

and further each w_n, w_m have distance at least $3 \cdot f(|t|)$. Then we have that the functions $\eta_{f(|t|)}(y - w_n)$ for $n \neq m$ have disjoint support from each other and even that there is a positive distance (of $f(|t|)$) between the respective supports.

Hence the infinite sum

$$\sum_{n \in \mathbb{N}} c_n \cdot \eta_{f(|t|)}(y - w_n) \cdot e_i$$

is locally finite everywhere and thus smooth. This means that

$$|s(\sum_{n \in \mathbb{N}} c_n \cdot \eta_{f(|t|)}(y - w_n) \cdot e_i)| < \infty.$$

However, we can construct a homotopy along the tangent vectors

$$\sum_{n=1}^k c_n \cdot \eta_{f(|t|)}(y - w_n) \cdot e_i$$

towards

$$\sum_{n \in \mathbb{N}} c_n \cdot \eta_{f(|t|)}(y - w_n) \cdot e_i.$$

As we can use the same method that we employed for constructing the previous homotopy (i.e. interpolate between the terms for k and $k+1$), I will not repeat the construction. However, the fact that it can be extended with the constant homotopy is much simpler here, as the sum is locally finite, such that locally the homotopy is already constant for some $[1-\varepsilon, 1]$.

Hence we find that

$$s(\sum_{n \in \mathbb{N}} c_n \cdot \eta_{f(|t|)}(y - w_n) \cdot e_i) = \lim_{k \rightarrow \infty} \sum_{n=1}^k c_n \cdot s(\eta_{f(|t|)}(y - w_n) \cdot e_i).$$

By assumption $s(\eta_{f(|t|)}(y - w_n) \cdot e_i) \neq 0$, so for the appropriate choice of c_n the expression diverges. This leads us to a contradiction, showing that

$$w \mapsto s(\eta_{f(|t|)}(y - w) \cdot e_i)$$

is indeed compactly supported.

This allows us to finish the proof that s_t is given by integration against the compactly supported function

$$s(\eta_{f(|t|)}(y - w) \cdot e_i) \cdot e_i,$$

which we have already succeeded in showing for compactly supported tangent vectors previously.

So assume we have now a general tangent vector X , that does no longer have to be compactly supported. We can find a smooth family of X_t such that for $t \in (0, 1)$ the X_t are compactly supported and for $t \geq 1$ we have that $X_t = X$. We can create this homotopy for example by scaling X on M with $\zeta((1-t) \cdot z)$, where $\zeta(z)$ is a compactly supported function on M that satisfies $\zeta \equiv 1$ on a small open around 0. This is a smooth homotopy of compactly supported functions on M along $t \in (0, 1)$. It can be smoothly extended by the constant 1 function, as locally we have that $\zeta((1-t) \cdot z) \equiv 1$ already for $t \in [1-\varepsilon, 1)$ for some ε .

We now have

$$\lim_{t \rightarrow 1} \left(\int_M \sum_{i=1}^q s(\eta_{f(|t|)}(y-w) \cdot e_i) X_t^i \, dw \right) = \int_M \sum_{i=1}^q s(\eta_{f(|t|)}(y-w) \cdot e_i) X^i \, dw.$$

This holds simply because the functions $s(\eta_{f(|t|)}(y-w) \cdot e_i)$ are compactly supported and hence on their united support $K \subset M$ we have $X \equiv X_t$ for $t \in [1-\varepsilon, 1]$. Also we have that

$$\lim_{t \rightarrow 1} s \left(\int_M \eta_{f(|t|)}(y-w) \cdot X_t(w) \, dw \right) = s \left(\int_M \eta_{f(|t|)}(y-w) \cdot X(w) \, dw \right),$$

as we have seen that the convolution of a smooth path is smooth.

Hence we can apply that

$$\int_M \sum_{i=1}^q s(\eta_{f(|t|)}(y-w) \cdot e_i) X_t^i \, dw = s \left(\int_M \eta_{f(|t|)}(y-w) \cdot X_t(w) \, dw \right),$$

which holds as the X_t are compactly supported, to conclude that

$$\int_M \sum_{i=1}^q s(\eta_{f(|t|)}(y-w) \cdot e_i) X^i \, dw = s \left(\int_M \eta_{f(|t|)}(y-w) \cdot X(w) \, dw \right)$$

as well for general tangent vectors X .

Hence we have shown that we can indeed express s_t for $t \neq 0$ as integration against a smooth and compactly supported function. As this was the last claim that was still open, this concludes the proof. $\square$

Theorem 2.2.2. *Any fiber-wise linear and smooth function on $T^*\mathfrak{F}$ can be identified with a section $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$.*

Proof. This proof can be subdivided in two parts. First, we explicitly construct for any fiber-wise linear function X the section $v' \in \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$, that we claim acts the same on $T^*\mathfrak{F}$ as X . We do so point-wise at any $\varphi \in \mathfrak{F}$ by explicitly constructing functions s_m^i that for any X that was given by a vector field would recover the i -th coordinate of $X_m \in VF$. The second part of the proof is devoted to showing that X and v' actually agree.

A key part of this proof is applying the previous proposition (locally), as any linear and smooth function on $T_\varphi^*\mathfrak{F}$ is then determined by the covectors given through integration against a smooth compactly function. But we can approximate any compact Riemannian integral by evaluating a function on finitely many points, which we can incidentally express through the covectors (ds_m^i) which already played a role earlier. So we find that the fiber-wise linear and smooth function X is uniquely determined by their values on the (ds_m^i) and hence has to agree with v' .

Let us start now with the construction of the vector field v' . We will firstly find for every $\varphi \in \mathfrak{F}$ and $m \in M$ separately an element in VF and then prove smoothness of the constructed map.

So consider any $m \in M$. At a given $\varphi \in \mathfrak{F}$ we can choose a coordinate chart on F around $\varphi(m)$ that respects the fiber structure. Choose a bump function χ on the fiber F_m that is supported inside of the coordinate chart and constantly 1 on an open around $\varphi(m)$. Then define $s_m^i : \mathfrak{F} \rightarrow \mathbb{R}$ by

$$s_m^i(\psi) = \begin{cases} \chi(\psi(m)) \cdot \psi(m)^i & \text{if } \psi(m) \text{ lies inside the chart} \\ 0 & \text{else,} \end{cases}$$

where i denotes the component of the i -th fiber coordinate. By definition of χ , this function is smooth.

Now assume we were given a fiber-wise linear function X . Then we could insert

$$X(ds_m^i) = v_m^i.$$

Now by definition, there is some open $U \subset F_m$ around $\varphi(m)$, such that $\chi|_U \equiv 1$. For all $\psi \in \mathfrak{F}$ with $\psi(m) \in U$, we can reconstruct a vector $v_m = \sum_j v_m^j \frac{\partial}{\partial y^j} \in T(F_m) = V_m F$ that is covering $\psi(m)$.

Let us now argue that this vector is independent of the chosen frame. After any (smooth) change of the frame, we can express the new fiber coordinates as function of the old ones, i.e. we have smooth $f^i \in C^\infty(U' \subset \mathbb{R}^k)$ such where ever both frames are defined it holds that

$$\widetilde{s}_m^i(\psi) = f^i((s_m^j(\psi))_j).$$

When deriving along any tangent vector at ψ we can then apply the chain rule to obtain

$$\left(d\widetilde{s}_m^i \right) (\psi) = F^i(\psi) ((ds_m^j)_j),$$

where $F^i(\psi) \in C^\infty(\mathbb{R}^k)$ is linear.

Firstly, this leads to

$$\widetilde{v}_m^i = F^i(\psi) ((v_m^j)_j)$$

Also, when combining all $F^i(\psi)$ into a matrix $F(\psi)$, we can explicitly compute that $F(\psi)$ corresponds to the linearisation of the coordinate change on F_m at $\psi(m)$. When computing the induced change of coordinates on the frame of the tangent space we furthermore get a second set of linear functions $G^i \in C^\infty(\mathbb{R}^k)$, such that

$$\frac{\partial}{\partial \widetilde{y}^i} = G^i(\psi) \left(\left(\frac{\partial}{\partial y^j} \right)_j \right).$$

Here, the components of the matrix $G(\psi)$ contrarily arise from the linearised change of coordinates that is inverse to $F(\psi)$. But as these matrices act

combined on the sum $v_m = \sum_j v_m^j \frac{\partial}{\partial y^j} \in VF$, the result turns out to be independent of F and G , hence independent of the chosen frame.

Also, if we consider a different χ' , we find that the ds_m^i and hence the v_m^i agree on any point within an open neighborhood that satisfies $\chi \equiv \chi' \equiv 1$.

So via this construction, we have for $m \in M$ define a map $v_m : \mathfrak{F} \rightarrow VF$, where $v_m(\psi)$ covers $\psi(m)$. As we have just proven that the value does not depend on the local frame or the bump function, we may prove smoothness under the assumption that we define the v_m locally around $\varphi(m)$ with respect to a fixed frame.

We also can notice, that we may restrict the consideration to an open of $\mathfrak{F}$, and that for any compact K and open V with $\varphi(K) \subset V$, the set

$$\{\psi \in \mathfrak{F} | \psi(K) \subset V\}$$

is indeed open with respect to the diffeological topology of $\mathfrak{F}$. So if we choose an open U around m on which we have a trivialization of the bundle $F \rightarrow M$ and $\Gamma(\varphi)$ is contained in an open subset O of the fiber where $\chi \equiv 1$ (and especially the fiber chart is still well-defined), we can choose a smaller pre-compact open V with $\bar{V} \subset U$ around m and consider

$$\mathfrak{U} = \{\psi \in \mathfrak{F} | \psi(\bar{V}) \subset O\}.$$

On this open subset $\mathfrak{U}$ of $\mathfrak{F}$, we have that all v_m for $m \in V$ assemble into a map $v : V \times \mathfrak{U} \rightarrow VF$. From our explicit formula we obtain that for any smooth path $p : T \subset \mathbb{R}^k \rightarrow \mathfrak{U}$ with T open the path

$$v \circ (\text{id} \times p) : V \times \mathbb{R}^k \rightarrow VF$$

is smooth.

We can now observe in general how the assembly of all v_m into a map $v : M \times \mathfrak{F} \rightarrow VF$ behaves. For every smooth path $p : \mathbb{R}^k \rightarrow \mathfrak{F}$, we have that

$$v \circ (\text{id} \times p) : M \times \mathbb{R}^k \rightarrow VF$$

is smooth if and only if it is locally smooth. But for any $m \in M$ and $x \in \mathbb{R}^k$, we can find as described above an open $V \subset M$ and open $p^{-1}(\mathfrak{U})$, such that restricted to $V \times p^{-1}(\mathfrak{U})$ the map is smooth.

Hence we know that

$$v : M \times \mathfrak{F} \rightarrow VF$$

and therefore

$$v' : \mathfrak{F} \rightarrow C^\infty(M, VF)$$

is smooth. But by construction, $v'(\varphi)$ is a section that covers φ , such that v' is a vector field.

What remains to show now is that this vector field actually agrees with X on all of $T^*\mathfrak{F}$. Equivalently, we can consider the function $X - v'$. As our

construction is linear, $X - v'$ yields the vanishing vector field. We then have to show that $X - v'$ is vanishing everywhere. So without loss of generality, we may assume $v' = 0$. Hence $X(ds_m^i)$ will be the assumption from now on and to prove $X = 0$ our aim.

Our first step in that direction will be to get to a setting where we can apply Proposition 2.2.1. All we need for that is to use that locally on M and F we actually have a vector bundle structure, which does not change the tangent or cotangent space.

Assume we are given some $\varphi \in \mathfrak{F}$. We can at first choose a sufficiently fine locally finite covering U_i and partition of unity χ_i on M , such that the underlying opens U_i are equipped with atlases, the support of the χ_i is compact for all i and the bundle $F \rightarrow M$ is trivializable on an open extension of every U_i . Also we can choose the U_i such that $\varphi(\overline{U_i})$ is always contained in some coordinate patch on F_m with respect to the trivialization.

If we now consider a cotangent vector ω at φ , we obtain a family of cotangent vectors ω_i given by

$$\omega_i(Y) = \omega(\chi_i \cdot Y),$$

we find that by linearity we have

$$\sum_{i=1}^n \omega_i = \omega((\sum_{i=1}^n \chi_i) \cdot Y).$$

We can construct homotopies for $t \in [0, 1]$ by

$$t \mapsto (\sum_{i=1}^n \chi_i) + \xi(t) \cdot \chi_{n+1}$$

for some $\xi \in C^\infty(\mathbb{R})$ that is monotone increasing and satisfies $\xi(t) = 0$ for $t \leq 0$ and $\xi(t) = 1$ for $t \geq 1$. As the partition of unity is locally finite, we can assemble these into a single smooth homotopy η on $(0, 1)$ that can smoothly be extended on $[1, 2)$ by the constant function 1 on M .

Just as this is a smooth homotopy, we have that the path p_t with

$$t \mapsto \omega(\eta(t) \cdot -)$$

is a smooth path of cotangent vectors. So $X(p_1) = X(\omega)$ is the limit of $X(p_{t_i}) = X(\sum_{i=1}^n \omega_i)$ and to prove $X(\omega) = 0$ for arbitrary ω it is sufficient to prove that $X(\omega_i) = 0$ by fiber-wise linearity.

So we only have to show that $X(\omega)$ where we may assume that the support of ω is contained in $\text{supp}(\chi_i)$ for some i . We will do this through showing that we may even impose much stricter regulations on ω , as $X(\omega)$ will turn out to depend only on $X(ds_m^i)$.

For this, we notice that ω with compact support in U_i extends naturally to a cotangent vector of the sections $\mathfrak{F}'$ of the bundle $\pi^{-1}(U_i) \subset F \rightarrow U_i$ at $\varphi' = \varphi|_{U_i}$. Here we have that the bundle takes the form of

$$\mathbb{R}^p \times F_m \rightarrow \mathbb{R}^p.$$

Also we know that φ' is contained in some coordinate chart of F_m , such that we may assume $F_m = \mathbb{R}^q$, as the tangent and cotangent spaces that arise from these bundles agree.

Now we can apply Proposition 2.2.1 to see that we find a smooth path γ that approximates ω as a cotangent vector of $\mathfrak{F}'$ at φ' through integration against a compactly supported (vector-valued) function. As the restriction $\mathfrak{F} \rightarrow \mathfrak{F}'$ is smooth, we can consider the image of γ in $T^*\mathfrak{F}_\varphi$, which is still smooth and where the images of $\gamma(t)$ for $t \neq 0$ are still given by integration on U against a compactly supported vector-valued function. As $X \circ \gamma$ is a smooth path we especially have that $X(\gamma(t))$ for $t \rightarrow 0$ converges to $X(\omega)$.

So we find that it is sufficient to prove that $X(\omega) = 0$ where ω is given by integration against a smooth function in local coordinates with compact support. What we aim to prove now is that the value of $X(\omega)$ for such an ω is already determined by the (ds_m^i) . Similarly to the proof of Proposition 2.2.1, we will aim to show that if we consider an integral over a compact, we can approximate it through values of the function at finitely many points in terms of a smooth path. But evaluating the values of a function representing a tangent vector can be expressed by linear combination of the (ds_m^i) .

Let us assume ω is given by integration against the vector-valued function f , where we express by f_m^i the values of that function at $m \in M$ with respect to the frame that has been used to define the s_m^i . As $\text{supp}(f)$ is compact, we can extend it to a compact $C \subset U$, that allows us to divide it into cubes (using the identification $U \cong \mathbb{R}^p$) of the form

$$I_k = [\frac{1}{a^a} \cdot z_i^k, \frac{1}{a^a} \cdot (z_i^k + 1)]^p$$

for some $z_i^k \in \mathbb{Z}$ and $a \in \mathbb{N}$. If we take w_k to be the centers of these cubes, we can approximate ω through

$$\nu_a = \sum_{k=1}^n \lambda_k \cdot f_{w_k}^i \cdot ds_{w_k}^i,$$

where $\lambda_k = \frac{1}{a^{a \cdot p}}$.

The proof that the linear homotopies between ν_a and ν_{a+1} assemble into a smooth path γ that can be continued by the constant path at ω is now so similar to the previous of 2.2.1, that I will only sketch it here.

The first thing to do is using the compactness of the support to find that

$$\text{vol}(C) \cdot \sqrt{d} \cdot \frac{1}{a^a} \cdot c_1 \cdot c_2$$

is an estimate for

$$\nu_a(Y_t) - \omega(Y_t)$$

for Y_t a path in of tangent vectors restricted to t in some compact interval J , where c_1 bounds all directional derivatives of f and c_2 bounds the values of Y_t on $C \times J$. Again, we are using the identification $U \cong \mathbb{R}^p$ for these computations.

Further using that for a fixed $a \in \mathbb{N}$ both

$$\frac{\partial}{\partial t} \nu_a(Y_t) = \nu_a\left(\frac{\partial}{\partial t} Y_t\right)$$

because of the definition of the (ds_m^i) and

$$\frac{\partial}{\partial t} \omega(Y_t) = \omega\left(\frac{\partial}{\partial t} Y_t\right)$$

because of compactness of the support, we find that the values of

$$\frac{\partial}{\partial t} (\gamma(t)(Y_t) - \omega(Y_t)) = \left(\frac{\partial}{\partial t} \gamma(t) \right) (Y_t)$$

are again bounded by a term in the order of

$$2^a \cdot \frac{1}{a^a}.$$

As all terms of the form

$$2^{k \cdot a} \cdot \frac{1}{a^a}$$

decrease sufficiently fast, we find in the end, that the path γ we get through the linear homotopies between the ν_a and continuation by ω is indeed smooth.

This now shows that $X(\nu_a) = 0$ already implies $X(\omega) = 0$ and the assumption $X(\nu_a) = 0$ is true by fiber-wise linearity of X . Hence X vanishes everywhere. This means that indeed $X = v'$ for the constructed v' , even in the case that $v' \neq 0$ and we indeed find that every fiber-wise linear function on $T^*\mathfrak{F}$ is given by a tangent vector. $\square$

Remark. The proof even shows more: As all arguments in the proof come from a point-wise consideration, we even have that $(T_\varphi^* \mathfrak{F})^* = T_\varphi \mathfrak{F}$ at every point $\varphi \in \mathfrak{F}$. The construction can also be generalized to fiber-wise linear functions on $\pi^{-1}(\mathfrak{U})$, denoting the pre-image of some open $\mathfrak{U} \subset \mathfrak{F}$ under the projection $\pi : T^*\mathfrak{F} \rightarrow \mathfrak{F}$, which are given by sections $\Gamma^\infty(\mathfrak{U}, T\mathfrak{F})$.

2.3 The Derived Critical Locus of an Action Functional

Our aim in this section will be to understand the derived critical locus given by the following pull-back:

$$\begin{array}{ccc} D & \xrightarrow{\quad\quad\quad} & (\mathfrak{F}, (0 \leftarrow C^\infty(\mathfrak{F}) \leftarrow 0 \leftarrow \dots)) \\ \downarrow & & \downarrow \text{dS} \\ (\mathfrak{F}, (0 \leftarrow C^\infty(\mathfrak{F}) \leftarrow 0 \leftarrow \dots)) & \xrightarrow{0} & (T^*\mathfrak{F}, (0 \leftarrow C^\infty(T^*\mathfrak{F}) \leftarrow 0 \leftarrow \dots)), \end{array}$$

where the notions of smooth functions on these spaces are given by their diffeological structures.

Technically, this is not a derived critical locus in the sense of Definition 1.1.1, as this would require the spaces we consider to be derived smooth manifolds. But as this is no longer the case, we need to consider a different setting.

So we would like to change the categories and work with derived diffeological spaces from now on. Unfortunately, a theory of derived diffeological spaces has not yet been developed. That it might be possible to define such spaces along the lines of derived ringed spaces (c.f. [22], p.31) is indicated by the following lemma:

Lemma 2.3.1. *For any diffeological space $\mathfrak{D}$ and $m \in \mathfrak{D}$ we have that $C^\infty(\mathfrak{D})_m$ is a local ring with unique maximal ideal*

$$\{f \in C^\infty(\mathfrak{D})_m \mid f(m) = 0\}.$$

Remark. These ideals might not be the only maximal ideals of $C^\infty(\mathfrak{D})$, unlike for the ring $C^\infty(M)$ of smooth functions on a manifold M . This means that the category of diffeological spaces, unlike manifolds, does not embed fully faithfully in the category of rings.

Proof. The statement follows from the fact that any $f \in C^\infty(\mathfrak{D})_m$ with $f(m) \neq 0$ is a unit. This can be seen as follows:

Assume $f(m) \neq 0$, where f is a function defined on the open $O' \subset \mathfrak{D}$. Then we can define $O \subset \mathfrak{D}$ by $O = \{x \in O' \text{ s.t. } f(x) \neq 0\}$. We have the pre-image of O under any map in the diffeology $p : U \rightarrow \mathfrak{D}$ is given by the pre-image of O' , restricted to the pre-image of $\mathbb{R} \setminus \{0\}$ under the smooth function $f \circ p|_{O'} : \mathbb{R} \rightarrow \mathbb{R}$. Hence O is open.

Now consider the function $\frac{1}{f|_O} : O \rightarrow \mathbb{R}$. It is well-defined by definition of O and can hence be seen as an element of the stalk $C^\infty(\mathfrak{D})_m$, which on $O \subset O'$ multiplies with f to the constant unit function. $\square$

There is, however, also another possibility. The definition of dg-manifolds, for example from [15], is much easier to adapt to a diffeological setting. What is more, we have an equivalence of dg-manifolds and derived manifolds through the Dold-Kan-correspondence (c.f. [13], Corollary 2.2.10 and

Remark 2.2.11, p.21). Similarly, the notion of a derived diffeological space should be related to the notion of a dg-diffeological space.

Definition 2.3.2. (c.f. [15], Definition 1.11, p.7) A dg-manifold M is a pair $(\overline{M}, \mathcal{O}_M)$, where $\overline{M}$ is a smooth manifold and

$$\mathcal{O}_M = (\mathcal{O}_{M,0} \leftarrow \mathcal{O}_{M,1} \leftarrow \dots)$$

is a sheaf on $\overline{M}$ of differential graded commutative $\mathbb{R}$ -algebras with the following properties:

Firstly, it has to hold that

$$\mathcal{O}_{M,0} = \mathcal{O}_{\overline{M}},$$

i.e. that $\mathcal{O}_{M,0}$ is the sheaf of smooth functions on $\overline{M}$.

Secondly, $\mathcal{O}_M$ has to be a locally semi-free sheaf, i.e. locally

$$\mathcal{O}_M \simeq \mathcal{O}_{\overline{M}} \otimes \text{Symm}_{\mathbb{R}}(V),$$

where $V = V_1 \oplus \dots \oplus V_n$ is a finite dimensional graded vector space.

The subtlety in adapting this definition is that if we do not drop the finiteness condition on V , the category of dg-diffeological spaces would not have a pull-back similar to manifolds, as our later computations will make clear. The problem is, that dg-manifolds are intended to describe spaces that locally look like the zero locus of a function between $\mathbb{R}^n$ and $\mathbb{R}^m$, i.e. like the zero locus of a finite set of functions on $\mathbb{R}^n$. But for diffeological spaces, we do not have a finiteness condition on the dimension, so even locally, the zero locus between two diffeological spaces $\mathfrak{D}$ and $\mathfrak{E}$ might only be described by an infinite number of equations.

Theoretically, to construct the pull-back we are looking for, we could just drop that part of the definition, as we aim to compute that pull-back by considering the rings only, as we have done with manifolds. As the 0-th degree of the result will still be given by $\mathcal{O}_{\mathfrak{F}}$, we can choose $\mathfrak{F}$ to be the underlying diffeological space.

A possible way to replace the local condition on the sheaf would be through recapitulating where it comes from. If we have an embedding of manifolds $\iota : N \hookrightarrow M$, we have that (at least locally) $\mathcal{O}_M \rightarrow \iota^* \mathcal{O}_N$ is surjective. To be precise, locally the embedding takes the form of $\mathbb{R}^n \rightarrow \mathbb{R}^m$ with image the hyperplane where all coordinates after the n -th are 0. If we now, similar to the first chapter, express $C^\infty(\mathbb{R}^n)$ as a quotient of $C^\infty(\mathbb{R}^m)$ through a flat resolution, we obtain a term of the form

$$C^\infty(\mathbb{R})^m \otimes \text{Symm}_{\mathbb{R}}(V).$$

So the condition restricts the category of dg-manifolds to objects, whose rings arise as derived tensor products of manifolds with their natural graded ring of functions $\mathcal{O}_M^{\text{nat}} = \mathcal{O}_M \leftarrow 0 \leftarrow 0 \leftarrow \dots$.

So for the sake of giving a possible definition of a category to work in, we could consider the following:

Definition 2.3.3. A dg-diffeological space $\mathfrak{D}$ is a pair $(\overline{\mathfrak{D}}, \mathcal{O}_{\mathfrak{D}})$, where $\overline{\mathfrak{D}}$ is a diffeological space and

$$\mathcal{O}_{\mathfrak{D}} = (\mathcal{O}_{\mathfrak{D},0} \leftarrow \mathcal{O}_{\mathfrak{D},1} \leftarrow \dots)$$

is a sheaf on $\overline{\mathfrak{D}}$ of differential graded commutative $\mathbb{R}$ -algebras, with the following properties:

Firstly, it has to hold that

$$\mathcal{O}_{\mathfrak{D},0} = \mathcal{O}_{\overline{\mathfrak{D}}},$$

i.e. that $\mathcal{O}_{\mathfrak{D},0}$ is the sheaf of smooth functions on $\overline{M}$.

Secondly, $\mathcal{O}_{\mathfrak{D}}$ has to satisfy that it locally arises from $\mathcal{O}_{\mathfrak{D}_i}^{\text{nat}}$ for a finite number of diffeological spaces $\mathfrak{D}_i$ through the operation of taking derived tensor products.

This now allows us to expand our notion of a derived critical locus.

Definition 2.3.4. The derived critical locus of $S \in C^\infty(\mathfrak{F})$ is the pull-back D in the following diagram of dg-diffeological spaces:

$$\begin{array}{ccc} D & \longrightarrow & (\mathfrak{F}, \mathcal{O}_{\mathfrak{F}}) \\ \downarrow & & \downarrow \text{d}S \\ (\mathfrak{F}, \mathcal{O}_{\mathfrak{F}}) & \xrightarrow{0} & (T^*\mathfrak{F}, \mathcal{O}_{T^*\mathfrak{F}}). \end{array}$$

In the following, however, we will focus our main concern on the sheaf of rings $\mathcal{O}(D)_\varphi$, which we will define here as

$$\mathcal{O}(D)_\varphi = (\mathcal{O}_{\mathfrak{F}})_\varphi \otimes_{(\mathcal{O}_{T^*\mathfrak{F}})_\varphi}^{\mathbb{L}} (\mathcal{O}_{\mathfrak{F}})_\varphi.$$

As this is the method that gives us the complex of observables in finite dimensions, we can expect that the result will help us understand the observables for a diffeological $\mathfrak{F}$ better.

But before we start computing the given term, let us discuss shortly, what generalization of Example 1.1.2 might be expected. The answer $\mathfrak{X}(\mathfrak{F})$ might seem intuitive - but how to define $\mathfrak{X}(\mathfrak{F})$ is not entirely clear. It should in degree 0 be given by $C^\infty(\mathfrak{F})$ and in degree 1 by $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$. But, in contrast to $\Omega(\mathfrak{F})$, the notion is not functorial in manifolds, so there are different ways to continue.

Firstly, we can note that for a manifold M we have that

$$\mathfrak{X}^n(M) = \wedge_{C^\infty(M)}^n (\Gamma^\infty(M, TM)),$$

which is a nice algebraic construction that emphasizes the property that $\mathfrak{X}^*(M)$ is generated by its degree 1 component. So we could just define the following:

Definition 2.3.5. Define $\mathfrak{S}(\mathfrak{F})$ to be the free commutative differential graded algebra over $C^\infty(\mathfrak{F})$ generated in degree 1 by $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$, where the differential is given by $\vee dS$.

However, there are also other properties one might look out for. We could for example define $\wedge^2(T\mathfrak{F})$ fiber-wise over every point $\varphi \in \mathfrak{F}$ and consider $\Gamma^\infty(\mathfrak{F}, \wedge^2(T\mathfrak{F}))$. However, when we compare this method to the generalization of $\Omega(\mathfrak{F})$, we notice that $\Omega^2(\mathfrak{F})$ no longer takes the form of $\Gamma^\infty(\mathfrak{F}, (\wedge^2(T^*\mathfrak{F})))$ for the fiber-wise wedge product, so one could argue that neither is this to be expected for $T\mathfrak{F}$. Rather, we find that $\Omega^n(\mathfrak{F})$ is given by $\Gamma^\infty(\mathfrak{F}, (\wedge^n(T\mathfrak{F}))^*)$, where $(\wedge^n(T\mathfrak{F}))^*$ denotes the bundle, where the fiber at each $\varphi \in \mathfrak{F}$ is given by linear maps on $\wedge^n(T_\varphi\mathfrak{F})$. This brings us to the other option we might consider.

Definition 2.3.6. Define $\mathfrak{T}(\mathfrak{F})$ to be the differential graded algebra over $C^\infty(\mathfrak{F})$ given in degree k by $\Gamma^\infty(\mathfrak{F}, (\wedge^k(T^*\mathfrak{F}))^*)$, where $\wedge^k(T^*\mathfrak{F})$ arises from the fiber-wise vector space constructions (and the corresponding quotient diffeology) on the k -fold product $T^*\mathfrak{F} \times_{\mathfrak{F}} \cdots \times_{\mathfrak{F}} T^*\mathfrak{F}$ of diffeological spaces. The differential on the complex is given by $\iota_{(dS)}$, which denotes the insertion of dS as the first variable.

It is not completely immediate, that this is indeed still a differential graded algebra, but we can explicitly define a multiplication for

$$A \in \Gamma^\infty(\mathfrak{F}, (\wedge^k(T^*\mathfrak{F}))^*) \text{ and } B \in \Gamma^\infty(\mathfrak{F}, (\wedge^l(T^*\mathfrak{F}))^*)$$

as

$$\begin{aligned} & (A \cdot B)(\varphi)(X_1 \wedge \cdots \wedge X_{k+l}) \\ &= \sum_{S, S'} \varepsilon(S, S') A(X_{i_1} \wedge \cdots \wedge X_{i_k}) \cdot B(X_{i_{(k+1)}} \wedge \cdots \wedge X_{i_{(k+l)}}) \end{aligned}$$

for $X_i \in T_\varphi^*\mathfrak{F}$. Here $S = (X_{i_1}, \dots, X_{i_k})$ and $S' = (X_{i_{(k+1)}}, \dots, X_{i_{(k+l)}})$ denote ordered subsets of the ordered set $(X_1, \dots, X_{(k+l)})$ and $\varepsilon(S, S')$ is given by $(-1)^e$, where

$$e = |\{i < j | i \in S', j \in S\}|.$$

Unwinding the definitions and terms, we find that $\mathfrak{S}(\mathfrak{F})$ is actually a subalgebra of $\mathfrak{T}(\mathfrak{F})$ through the graded symmetric action of vector fields on cotangent vectors. Also, through the result 2.2.2, we have that we can identify $\mathfrak{T}^1(\mathfrak{F})$ with $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$.

We will see in the following, that $\mathcal{O}(D)$ agrees with none of these spaces, leaving us none the wiser about the most appropriate choice. The reason why $\mathcal{O}(D)$ differs from the given complexes is deeply entwined with the following fact, that our functions on $T_\varphi\mathfrak{F}$ are no longer linearizable.

Proposition 2.3.7. *Let $\varphi \in \mathfrak{F}$, then the ideal $\{g : g|_{\Gamma(0)} = 0\}_{(\varphi,0)}$ is not generated by the fiber-wise linear functions.*

Proof. Consider any linear function X on the fiber $T_\varphi^* \mathfrak{F}$. We have that $\dim(\text{coim}(X)) = \dim(\text{Im}(X)) \leq 1$ where we regard $T_\varphi^* \mathfrak{F}$ as an $\mathbb{R}$ -vector space. So the kernel of X has co-dimension at most 1 with respect to $T_\varphi^* \mathfrak{F}$.

When we now consider a function g in the ideal generated by linear functions, we have that

$$g = \sum_{i=1}^k f_i \cdot X^i.$$

As $T^* \mathfrak{F}$ is infinite dimensional, we know that the kernels of the X^i have a common intersection that is also infinite dimensional and of co-dimension at most k (as $\mathbb{R}$ -vector spaces). So also g must have a kernel that contains a $\mathbb{R}$ -vector subspace of $T^* \mathfrak{F}$ of finite co-dimension.

We will now aim to construct a function f that does not satisfy this property. Therefore, we will take vector fields $X_i \in \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ as fiber-wise linear functions on $T^* \mathfrak{F}$, such that we can consider arbitrary bounded (and even certain unbounded) sums of all the X_i . We will then consider

$$f(w) = \sum_{i \in \mathbb{N}} \frac{1}{2^i} (X_i(w))^2.$$

The idea why this should be well-defined at all comes from the fact that not only is

$$\sum_{i \in \mathbb{N}} |X_i(w)| < \infty,$$

but the limit is again a continuous function, meaning all terms can be bounded on compacts and thus the squares converge uniformly.

We will then apply a result from real analysis to see that the sum of the infinitely many smooth functions is smooth. Therefore at first we need to analyze how the terms

$$\frac{\partial}{\partial v} X_i(w)$$

behave, depending on a decomposition of v similar to decomposing a vector on a fiber bundle of manifolds into a vertical and a horizontal component. Revisiting the choice of our X_i then shows that these terms also converge in a well-behaved way.

We then finally can come to the concluding argument of this proof, that the given function f does not satisfy the property that we have established at the very beginning for all linearizable functions: The kernel of f at the point φ does not contain a vector subspace of $T^* \mathfrak{F}$ that has only finite co-dimension. Hence f cannot be linearizable.

So we start by choosing the following vector fields X_i that are non-vanishing at the point φ : We choose local coordinates on $U \times V \subset F$ around $\varphi(m)$ for some $m \in M$ that respect the fiber bundle structure. Now we can choose a countable but infinite number of distinct points $m_i \in M$ (which is why we exclude the case that M only consists of finitely many points at the beginning of the chapter), such that $\varphi(m_i)$ lies in the coordinate patch.

Then we can w.l.o.g. assume that we can choose bump functions ζ_i around the m_i that vanish on a fixed O_j open containing m_j for all $j \neq i$, as we can choose such m_i in every $\mathbb{R}^d$ with $d \geq 1$ and M is discrete otherwise. By the same reasoning (and choosing the ζ_i contained in a compact through rescaling and shifting a unique bump function ζ) we may also assume that any bounded infinite sum of the ζ_j is a smooth function, as all values and derivatives converge to 0. For later purposes, we may assume that even after scaling the ζ_j with a factor in the order of c^j for c constant the infinite sums exist.

To be precise: An explicit method for a fiber bundle $F \rightarrow \mathbb{R}^k$ to construct such ζ_i would simply be to take the disjoint balls of radius $\frac{1}{4^{k+1}}$ centered around $e_1 \cdot (1 - \frac{1}{2^k})$ and consider $\zeta_k(z) = \frac{1}{k^k} \cdot \zeta(4^{k+1} \cdot (z - e_1 \cdot (1 - \frac{1}{2^k})))$ for some ζ contained in $B_1(0)$ centered around the origin. For a general bundle $F \rightarrow M$, we can just restrict domain and target to an open of M with a coordinate patch. The compact that corresponds to $B_1(0)$ in this patch and hence contains $\text{supp}(\zeta_j)$ for all j will be denoted by $K_M \subset M$. Whenever I refer to properties of an explicit construction in this proof, they are satisfied by this construction.

Now for any base-point $f \in U \times V \subset F$ we can choose paths on the coordinate patch that are linear with derivative 1 in direction of the first fiber coordinate. This defines a local vector field on F . For every $i \in \mathbb{N}$ we can further rescale the vector field with the product of the bump function on $U \subset M$ around m_i and some fixed bump function ξ on $V \cong \mathbb{R}^d$, where d is the fiber dimension, that has compact support and satisfies $\chi \equiv 1$ on an open containing the fiber coordinates of all $\varphi(m_i)$ (as we assumed to choose the m_i contained in a compact, we may assume their image to be contained in a compact as well, this if for example ensured if $\varphi(U) \subset V$). This allows us to extend the vector field by 0 outside of the coordinate patch.

The global vector field $Y_i \in \Gamma^\infty(F, TF)$ we obtain from a fixed m_i is, by construction, a vertical vector field. For any ψ , we can now consider the composition $Y_i \circ \psi \in \Gamma^\infty_\psi(M, VF) = T\mathfrak{F}_\psi$, which in return yields a vector field $X_i \in \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$.

As we have chosen the ζ_i carefully, we can go through this construction not only for each m_i and ζ_i alone, but also for every finite or infinite bounded sum of the ζ_i . By evaluation on M and F , we can explicitly check that this corresponds to the finite or infinite sums of the X_i , which are also defined because of the local finiteness.

Now consider the function on $T^*\mathfrak{F}$ given by

$$f(w) = \sum_{i \in \mathbb{N}} \frac{1}{2^i} (X_i(w))^2.$$

That this function is well-defined at all is not clear. To show this will now be our first aim.

We have already discussed, that bounded infinite linear combinations of our X_i are still well-defined. Thus we can conclude

$$\left(\sum_{i \in \mathbb{N}} (-1)^{\sigma(i)} X_i \right) (w) < \infty,$$

where we choose the $\sigma(i)$ such that all terms of the form

$$(-1)^{\sigma(i)} X_i(w)$$

are non-negative. It is not immediately clear that

$$\left(\sum_{i \in \mathbb{N}} (-1)^{\sigma(i)} X_i \right) (w) = \sum_{i \in \mathbb{N}} \left((-1)^{\sigma(i)} X_i(w) \right),$$

so let us analyze the situation.

We can define a homotopy from 0 to $\sum_{i \in \mathbb{N}} (-1)^{\sigma(i)} X_i$ along the finite sums $\sum_{i=1}^k (-1)^{\sigma(i)} X_i$, for example by repeatedly applying a fixed homotopy between 0 and ζ along the intervals $[0, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{4}]$ and so on. As we have scaled the X_i with sufficiently fast decreasing constants (we implicitly use convergence of $2^{k \cdot a} \cdot \frac{1}{a^a}$ here, the existence of the bounded sums alone does not suffice), we can extend it by the constant homotopy.

We hence find that indeed

$$\lim_{k \in \mathbb{N}} \left(\sum_{i=1}^k (-1)^{\sigma(i)} X_i(w) \right) = \left(\sum_{i \in \mathbb{N}} (-1)^{\sigma(i)} X_i \right) (w).$$

This shows us, that the limit of $\sum_{i=1}^k (-1)^{\sigma(i)} X_i(w)$ exists and as by assumption all occurring terms are positive, we even have that

$$\sum_{i \in \mathbb{N}} (-1)^{\sigma(i)} X_i(w) = \sum_{i \in \mathbb{N}} |X_i(w)|$$

exists.

If we denote the value we have found here as c_w , we have especially found

$$|X_j(w)| \leq c_w$$

for all $j \in \mathbb{N}$ and

$$\sum_{i \in \mathbb{N}} \frac{1}{2^i} (X_i(w))^2 \leq c_w^2 < \infty.$$

This point-wise bound is already helpful, but we would like to show even more: We aim to show that for any map $p : \mathbb{R}^k \rightarrow T^*\mathfrak{F}$ we have that we can associate to any compact K a constant c_K such that

$$|X_j(p(z))| \leq c_K$$

for $z \in K$.

Therefore it suffices to prove that the map on $\mathbb{R}^k$ given by

$$z \mapsto \sum_{i \in \mathbb{N}} |X_i(p(z))|$$

is lower semi-continuous. As we are on a map on $\mathbb{R}^k$, we can check this on sequences. So assume we are given some $z' \in \mathbb{R}^k$ and a sequence z_n for $n \in \mathbb{N}$ converging to z' such that $\sum_{i \in \mathbb{N}} |X_i(p(z_n))|$ either diverges or converges to a value strictly greater than $\sum_{i \in \mathbb{N}} |X_i(p(z'))|$. This means that the same is also true for each of its subsequences, leading us to a statement we will soon turn into a contradiction.

So consider the $\sigma_n(i) \in \{0, 1\}$ that are associated to the z_n through

$$(-1)^{\sigma_n(i)} X_i(z_n) \geq 0$$

as a sequence parametrised by $n \in \mathbb{N}$. Then we have that either $\sigma_n(i) = 0$ for infinitely many z_n or that $\sigma_n(i) = 1$ for infinitely many z_n . Hence we can define a subsequence of the z_n given by $k_1 : \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing such that $\sigma_{k_1(n)}(1)$ is constant in n . We can then further define a subsequence k_2 of the first subsequence, such that $\sigma_{k_2(n)}(2)$ are constant, while retaining the property that $\sigma_{k_2(n)}(1)$ remains constant as well.

By a diagonal argument (i.e. pick z_1 , then $z_{k_1(2)}$, then $z_{k_2(3)}$) we thus obtain a subsequence z'_n converging against z' , such that for all $i \in \mathbb{N}$ the terms $\sigma'_n(i)$ associated to the z'_n are constant as a sequence after finitely many terms. We especially have that if $X_i(p(z')) > 0$, i.e. $\sigma'(i) = 0$ for z' , that already $X_i(p(z'_n)) > 0$ for almost all $n \in \mathbb{N}$. A similar statement holds for $X_i(p(z')) < 0$. So we simply may choose the $\sigma'(i)$ as the limit of the $\sigma'_n(i)$.

Now we can without loss of generality assume, that the z'_n are chosen in a way such that there exists a smooth path $\gamma : \mathbb{R} \rightarrow \mathbb{R}^k$ which satisfies that

$$\gamma(1 - \frac{1}{2^n}) = z'_n$$

and $\gamma(t) = z'$ for $t \geq 1$. Otherwise, we may restrict to yet another subsequence (for example one, in which the distance to z' decreases by n with

every new element z'_n , where we can take the straight paths between the z'_n and smooth them out, as the derivative is bounded).

Similarly, we can define a homotopy between the vector fields given by

$$X'_n = \sum_{i \in \mathbb{N}} (-1)^{\sigma'_n(i)} X_i,$$

for example through fixing a homotopy on $[0, 1]$ between ζ and $-\zeta$ that can be continued by the smooth homotopy, or simply by interpolating between 1 and -1 smoothly. They also assemble into a path $X' : \mathbb{R} \rightarrow T\mathfrak{F}$ projecting to $\pi \circ p \circ \gamma$, where π denotes the projection $\pi : T^*\mathfrak{F} \rightarrow \mathfrak{F}$, such that

$$X'(1 - \frac{1}{2^n}) = X'_n(\pi(p(z'_n)))$$

and

$$X'(t) = \sum_{i \in \mathbb{N}} (-1)^{\sigma'(i)} X_i(\pi(p(z')))$$

for $t \geq 1$. The smoothness of this construction is ensured on the support of each X_i as the $\sigma'_n(i)$ are constant after a finite number of iterations and smoothness on the points where the sum is not locally finite (by assumption all X_i vanish on these points) is assured as the scaling factors of ζ_i with respect to the X_i decrease sufficiently fast.

Now we can evaluate γ along X' and by smoothness of both of them, we have that

$$t \mapsto X'(t)(p(\gamma(t)))$$

is smooth and especially continuous. So we have that

$$\lim_{n \rightarrow \infty} X'(1 - \frac{1}{2^n})(p(\gamma(1 - \frac{1}{2^n}))) = X'(1)(p(\gamma(1))),$$

meaning

$$\lim_{n \rightarrow \infty} X'_n(p(z'_n)) = X'(1)(p(z')).$$

But by construction, this means

$$\lim_{n \rightarrow \infty} \sum_{i \in \mathbb{N}} |X_i(p(z_n))| = \sum_{i \in \mathbb{N}} |X_i(p(z'))|,$$

contradicting our original assumption and proving that the map

$$z \mapsto \sum_{i \in \mathbb{N}} |X_i(p(z))|$$

is lower semi-continuous (actually, the proof even shows it is continuous).

Hence for any map $p : \mathbb{R}^k \rightarrow T^*\mathfrak{F}$, we have the existence of some c_K with

$$|X_j(p(z))| \leq c_K$$

for every $j \in \mathbb{N}$.

Now that we have established this, we can return to the question if f is smooth. Note that, as we are on a diffeological space, it suffices to check smoothness on maps of the form $p : \mathbb{R}^k \rightarrow T^*\mathfrak{F}$. If we would only have a finite sum in the definition of f , we could now simply compute

$$\frac{\partial}{\partial v} \frac{1}{2^i} (X_i(w))^2 = 2 \cdot \frac{1}{2^i} \left(\frac{\partial}{\partial v} X_i(z)|_w \right) \cdot X_i(w).$$

To show that f is indeed smooth and f' agrees with the sum of these terms, we will aim to show that the convergence of this sum is uniform on every compact subset $K \subset \mathbb{R}^k$ along smooth map $\mathbb{R}^k \rightarrow T^*\mathfrak{F}$. This does not yet prove that f is smooth, as it just shows that f is differentiable along the given map, by a result from standard real analysis (cf. [20], Theorem 1). However, the same result implies that, if we can show that the convergence of all second derivatives is uniform as well, we find that f' is differentiable and inductively we find that f is smooth.

First, let us take a closer look at the term

$$\frac{\partial}{\partial v} X_i(z)|_w$$

for any path $v : \mathbb{R} \rightarrow T^*\mathfrak{F}$ with $v(0) = x$. Assume that v was fully contained in the fiber $T_\psi^*\mathfrak{F}$, where $\psi \in \mathfrak{F}$ is the projection of x . Then we would have that

$$X_i(v(t)) = v(t)(X_i) = v(0)(X_i) + t \cdot v'(t)(X_i),$$

where $v'(t)$ is given on any $Y \in T_\psi\mathfrak{F}$ by the corresponding formula. This is smooth, because $v(t)$ is and hence

$$\frac{\partial}{\partial t} X_i(v(t))|_{t=0} = X_i(v'(0))$$

due to the linearity of X_i . So for any vertical path, we find that derivation in a certain direction is given by evaluation at a cotangent vector and the construction satisfies that any homotopy of vertical paths gives us a smooth map of cotangent vectors.

So it remains to be seen what happens in the case where the path v projects to a non-constant path $p_v : \mathbb{R} \rightarrow \mathfrak{F}$. We would like to reduce this to the previous case. Hence we consider the open coordinate patch $U \subset M$ that contains the m_i and their disjoint opens O_j in the compact $K_M \subset U$ and the corresponding coordinate patch $U \times V \subset F$. Let us further denote by φ the section given by $\varphi = p_v(0)$.

For any $\psi \in \mathfrak{F}$, we now have that $X_i(\psi)$ is defined with respect to the coordinates of $U \times V$, meaning that especially $X_i(\psi)(m) = 0$ if $\psi(m)$ is not contained in the coordinate frame. This allows us to define a new path p given, where this is well-defined, by the vectors $\tilde{X}_i \in T_\varphi\mathfrak{F}$ that agree

with $X_i(p_v(t))$ in the fiber coordinates of $VF \cong TV \cong \mathbb{R}^d \times \mathbb{R}^d$. We do not have that this construction is well-defined for every ψ , as we need that $X_i(\psi)(m) = 0$ if $X_i(\varphi)(m) \notin U \times V$.

But for $t \in (-\varepsilon, \varepsilon) \subset \mathbb{R}$, we have that the construction works for $p_v(t)$, as it only depends on the values of ψ on K_M , which either have to be outside of $T_1 = \text{supp}(\zeta)$, which is closed, or in the open subset T_2 of points of F that project to m with $\varphi(m) \in U \times V$. But we have that

$$\{\psi \in \mathfrak{F} | \psi(K_M) \subset (T_1)^c \cup T_2\}$$

is open, such that their pre-image under $p_v(t)$ must contain a small open around 0.

Similarly, for $v : \mathbb{R} \rightarrow T^*\mathfrak{F}$ we can locally define a path $\tilde{v} : \mathbb{R} \rightarrow T_\varphi^*\mathfrak{F}$, where $\tilde{v}(t)$ acts on $T_\varphi\mathfrak{F}$ similar to $v(t)$ acting on $T_{p_v(t)}\mathfrak{F}$. Therefore, we need to fix a map $T_\varphi\mathfrak{F} \rightarrow T_{p_v(t)}\mathfrak{F}$. At first let us fix a function $g \in C^\infty(V)$ that is compactly supported and satisfies $g \equiv 1$ on $\text{supp}(\xi)$. Also fix a function $\mu \in C^\infty(U)$ that is compactly supported and satisfies $\mu \equiv 1$ on K_M .

For every tangent vector $Y \in T_\varphi\mathfrak{F}$, we would like to consider $\tilde{Y} \in T_{p_v(t)}\mathfrak{F}$ such that $\tilde{Y}(m)$ agrees with $Y(m)$ in the fiber coordinates of TV or we have $\tilde{Y}(m) = 0$ and $Y(m) = 0$. Of course, such a $\tilde{Y}$ would not be well-defined in general, which is why we have to rescale Y previously at each point $m \in M$ with $g(p_v(t)(m)) \cdot \mu(m)$. Still, this definition is only well-defined if the case $p_v(t)(m) \in \text{supp}(\mu) \times \text{supp}(g)$ only occurs if $\varphi(m) \in U \times V$ for $m \in K_M$ (otherwise we could not express $Y(m) \neq 0$ in our coordinates). But again, the set $T_1 = \text{supp}(\mu) \times \text{supp}(g)$ is closed, and the set T_2 of points in F that project to m with $\varphi(m) \in U \times V$ is open, such that

$$\{\psi \in \mathfrak{F} | \psi(K_M) \subset (T_1)^c \cup T_2\}$$

is open and hence contains $p_v(t)$ for $t \in (-\varepsilon, \varepsilon)$ for an appropriate ε .

The point of this construction is that we can check

$$\widetilde{X_i(t)}(\tilde{v}(t)) = \tilde{v}(t)(\widetilde{X_i(t)}) = v(t)(X_i).$$

This holds, as for $X_i(p_v(t))$, which is only non-zero for $m \in M$ where $p_v(t)(m) \in K_M \times \text{supp}(\xi)$, the constructions on the tangent spaces are inverses to each other. Hence we can compute that

$$\frac{\partial}{\partial t} X_i(v(t))|_{t=0} = \frac{\partial}{\partial t} \widetilde{X_i(t)}(\tilde{v}(t)).$$

Now we are almost in the previous case again, where we have a map contained in a single fiber. However, we now have that the path $\widetilde{X_i(t)}$ is not constant any more. Hence we find that

$$\frac{\partial}{\partial t} X_i(v(t))|_{t=0} = X'_i(v(0)) + X_i(v'),$$

by a similar argument to the above one. X'_i is given in coordinates on $U \times V$ by the derivatives of the fiber coordinates along $p_v(t)$, such that by definition, we can compute them to be

$$X'_i = X_i \cdot \frac{\partial}{\partial t} \xi(p_v(t)(m))|_{t=0}.$$

So finally, let us return to the question of the smoothness of f . Assume we have given a map $p : \mathbb{R}^k \rightarrow T^*\mathfrak{F}$ with projection $\gamma : \mathbb{R}^k \rightarrow \mathfrak{F}$. Our aim is to show that

$$\sum_{i \in \mathbb{N}} \frac{\partial}{\partial v} \frac{1}{2^i} (X_i(p(z)))^2 = \sum_{i \in \mathbb{N}} 2 \cdot \frac{1}{2^i} \left(\frac{\partial}{\partial v} X_i(p(x))|_z \right) \cdot X_i(p(z)).$$

converges locally uniformly.

As the sums contain the scaling factors of $\frac{1}{2^i}$, we are finished if we can show that for any compact K we have

$$\left| \left(\frac{\partial}{\partial v} X_i(p(x))|_z \right) \cdot X_i(p(z)) \right| < C_K.$$

The second factor is already bounded, as we have shown above. Further, when inserting

$$\frac{\partial}{\partial v} X_i(w)|_{p(z)} = X'_i(p(z)) + X_i(v'(z)),$$

we find that

$$z \mapsto v'(z)$$

is another smooth map, such that

$$|X_i(v'(z))| < c'_K$$

is also bounded by a fixed constant.

As our proof of the boundedness here did not depend on the specifics of the X_i , only on the existence of certain infinite sums, we would like to say, that the same argument applies to the X'_i as well. This is true, as long as we can bound the term

$$\frac{\partial}{\partial t} \xi(p_{v(z)}(t)(m))$$

as well as all of its derivations for $z \in K$ and $m \in K_M \subset U$. But this is clear, as we can continue ξ outside of K_M by $\xi \cdot \mu$ and then outside of the coordinate patch by 0, such that we are just evaluating a smooth function on $\mathbb{R}^k \times M$ on the compact subset $K \times K_M$.

The argument for bounding all further derivatives is quite similar and I will omit it here. But we can finally draw the conclusion that f is indeed a smooth function.

Now we can finally analyze the kernel of the given f in the fiber at φ , where we find that $f((\varphi, dg)) = 0$ if and only if we also have $X_i((\varphi, dg)) = 0$ for all $i \in \mathbb{N}$. This is a sub-vector space of $T^*\mathfrak{F}_\varphi$ and hence the maximal sub-vector space that is a subspace of the kernel of the given function. However, this sub-vector space has an infinite co-dimension in relation to $T^*\mathfrak{F}$, as all the tangent vectors X_i are non-zero and linearly independent at φ .

We can conclude that the given function f is not contained in the ideal generated by the fiber-wise linear functions, even though it is a function that vanishes on $\Gamma(0) \subset T^*\mathfrak{F}$. Hence we have shown the claim of the Proposition. $\square$

This implies that we cannot use the resolution that we established for the case of manifolds in Proposition 1.1.4 and compute the functions on the derived critical locus in the same way for diffeological spaces. A priori that is not sufficient to say the result would change, but unfortunately the following holds.

Theorem 2.3.8. *The chain complex $\mathcal{O}(D)_m$, which we obtain by identifying linear functions on $T^*\mathfrak{F}$ with elements of $T\mathfrak{F}$, is not in general naturally quasi-isomorphic to $\mathfrak{S}(\mathfrak{F})_m$ or $\mathfrak{T}(\mathfrak{F})_m$ or any complex given in degree 1 by $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})_m$.*

Remark. This theorem does not necessarily mean that there is a case where $\mathcal{O}(D)_m$ and $\mathfrak{S}(\mathfrak{F})_m$ do not agree as abstract vector spaces, $C^\infty(\mathfrak{F})$ -modules of even diffeological spaces, as we do not argue about dimensions here. It specifically applies to the method of identifying linear functions with tangent vectors, as we have done in Theorem 1.1.5 regarding manifolds.

Proof. Assume that $\mathcal{O}(D)_m$ and $\mathfrak{S}(\mathfrak{F})_m$ or $\mathfrak{T}(\mathfrak{F})_m$ are quasi-isomorphic for the function $S = 0$, such that $dS = 0$. Then their 1st homology group should in both cases be given by $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ as the differential on the chain complexes vanish. But we will show that this is not a form that $H_1(\mathcal{O}(D))$ takes naturally. To prove this, we will at first establish a free and hence flat resolution of $C^\infty(\mathfrak{F})$ as a $C^\infty(T^*\mathfrak{F})$ -module.

We will then define a map from $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ into the first homology group of the obtained complex, that might not seem to extend to a morphism of chain complexes at first. We will however even sketch as a nice additional result that there is a very natural map $\mathfrak{S}(\mathfrak{F})_m \rightarrow \mathcal{O}(D)_m$ as derived chain complexes, i.e. after a quasi-isomorphic replacement of $\mathcal{O}(D)_m$.

Coming back to the claim of the theorem, we will then consider an element 1_h and show that its homology class is not contained in the image of $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$. We do this by explicitly identifying the only possible pre-image of this element as 0 and then showing that $1_h \not\sim 0$ by using the fact that its second derivative is not linearizable.

Let us now commence by computing the $H_1(\mathcal{O}(D))$, which we do by choosing the beginning of a free resolution of $C^\infty(\mathfrak{F})$ as a $C^\infty(T^*\mathfrak{F})$ -module. (The type of resolution that I am going to use for this is also known as a Koszul or Koszul-Tate resolution, it is much more general than the one I used for manifolds.)

We can consider the complex

$$\begin{array}{c} \bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(T^*\mathfrak{F})_m \longleftarrow \bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g_f \cdot f=0}} C^\infty(T^*\mathfrak{F})_m \longleftarrow \dots \\ \curvearrowright \\ \dots \longleftarrow 0 \longleftarrow \left(\frac{C^\infty(T^*\mathfrak{F})}{\{f: f|_{\Gamma(0)}=0\}} \right)_m \longleftarrow C^\infty(T^*\mathfrak{F})_m \longleftarrow \dots \end{array},$$

with differentials given by mapping the unit in one component of its direct sum to the element of its corresponding index. Note that the notation “ f s.t. $f|_{\Gamma(0)} = 0$ ” in the index refers to $f \in C^\infty(T^*\mathfrak{F})_m$, such that we do not intend to list functions that agree on the stalk separately. Similarly, the indexing $\{g_f\}_f$ is in this case a shorthand notation for

$$\{g_f\}_f \in \bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(T^*\mathfrak{F})_m$$

meaning that there are especially only finitely many $g_f \neq 0$. This makes the complex exact and free apart from the first term by construction.

Now as before, we can tensor the resolution with $C^\infty(\mathfrak{F})_m$ given by the module structure with respect to $dS = 0$. Thus we obtain the complex

$$\begin{array}{c} \bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g_f \cdot f=0}} C^\infty(\mathfrak{F})_m \longleftarrow \bigoplus_{\dots} C^\infty(\mathfrak{F})_m \longleftarrow \dots \\ \curvearrowright \\ \dots \longleftarrow 0 \longleftarrow C^\infty(\mathfrak{F})_m \longleftarrow \bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(\mathfrak{F})_m \longleftarrow \dots \end{array},$$

which has first homology group

$$\begin{aligned}
H_1(\mathcal{O}(D)_m) &= \frac{\left(\ker(d_1) \left(\bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(\mathfrak{F})_m \right) \right)}{\left(\text{Im}(d_2) \left(\bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g_f \cdot f=0}} C^\infty(\mathfrak{F})_m \right) \right)} \\
&= \frac{\left(\bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(\mathfrak{F})_m \right)}{\left(\text{Im}(d_2) \left(\bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g_f \cdot f=0}} C^\infty(\mathfrak{F})_m \right) \right)},
\end{aligned}$$

as the differential vanishes on the degree 1 elements.

We can now map $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ into this $C^\infty(\mathfrak{F})$ -module by sending any tangent vector $X \in T\mathfrak{F}$ to the sum with only one non-zero entry, which is 1 indexed by f_X , where f_X describes the (fiber-wise linear) function on $T^*\mathfrak{F}$ that is given by $f_X(v) = v(X)$ for all $v \in T^*\mathfrak{F}$. In the following, we will also denote this function as X .

We find that this map is, in a sense, canonical, when we consider that the given complex is quasi-isomorphic to the complex

$$\begin{array}{c}
\left(\bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(\mathfrak{F})_m \right) / \sim \longleftarrow \left(\bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g_f \cdot f=0}} C^\infty(\mathfrak{F})_m \right) / \sim \longleftarrow \dots \\
\swarrow \hspace{15em} \searrow \\
\dots \longleftarrow 0 \longleftarrow C^\infty(\mathfrak{F})_m \longleftarrow \dots
\end{array},$$

where the $\sim$ is the equivalence relation identifying two elements of

$$\bigoplus_{n \in \mathbb{N}} C^\infty(T^*\mathfrak{F}) \times \dots \times C^\infty(T^*\mathfrak{F}) \times C^\infty(\mathfrak{F}),$$

if they agree under the canonical projection to

$$C^\infty(T^*\mathfrak{F}) \otimes_{C^\infty(\mathfrak{F})} \dots \otimes_{C^\infty(\mathfrak{F})} C^\infty(T^*\mathfrak{F}) \otimes_{C^\infty(\mathfrak{F})} C^\infty(\mathfrak{F}).$$

This uses the fact, that two elements, that agree under this identification will have (as the differential is $C^\infty(\mathfrak{F})$ -linear) the same image under the

differential, such that the complex is still well-defined. Also, when regarding the homology groups, we notice that the relations we introduce are either relations of the form $(h \cdot f)_I - f_{(h \cdot I)} \sim 0$ and $1_{g_{f_I}} + 1_{h_{f_I}} - 1_{g_f + h_{f_I}} \sim 0$ or identify an element with 0 that does not lie in the kernel of the differential.

We can also embed the higher degrees of $\mathfrak{S}(\mathfrak{F})$ into the given complex now, by considering $X \wedge Y$ as $1_{X_Y - Y_X}$, i.e. the constant function on $\mathfrak{F}$ indexed by $X_Y - Y_X$, where X and Y are understood as functions on $T^*\mathfrak{F}$ that evidently satisfy $X \cdot Y - Y \cdot X = 0$.

This method generally models the method we have been using with regard to manifolds, to identify linear functions with tangent vectors. But while in the case of manifolds we could use a local frame to algebraically obtain that these complexes are identical, we unfortunately cannot conclude the same here.

When regarding the induced map on homology, we can observe that it is not surjective, at least in degree 1. This has to do with the result of Proposition 2.3.7. For the proof we consider a function h , that vanishes on $\Gamma(dS) = \Gamma(0)$, has vanishing first derivative at 0 in all fiber directions of $T_\varphi^*\mathfrak{F}$. Its most important property is that it has a second derivative (which means the function fiber-wise defined as $v \rightarrow \frac{\partial^2}{\partial t^2} h(t \cdot v)|_0$) that is non-linearizable at φ . We can for example take the function h described in Proposition 2.3.7.

Then

$$1_h \in \frac{\ker(d_1) \left(\bigoplus_{\substack{f \text{ s.t.} \\ f|_{\Gamma(0)}=0}} C^\infty(\mathfrak{F})_m \right)}{\text{Im}(d_2) \left(\bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g \cdot f=0}} C^\infty(\mathfrak{F})_m \right)},$$

and we will show that 1_h cannot be equivalent to any element X in the image of $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$.

Otherwise, there would have to be an element

$$x \in \text{Im}(d_2) \left(\bigoplus_{\substack{\{g_f\}_f \text{ s.t.} \\ \sum_f g \cdot f=0}} C^\infty(\mathfrak{F})_m \right),$$

such that

$$1_h - 1_X = d_2(x).$$

Now we can use that we assumed $dS = 0$. We can without loss of generality assume that the element x is given by 1_I for some index I , as any degree two element has the same image as an element of this form.

This holds as we can firstly multiply the top level of the indices with the occurring elements of $C^\infty(\mathfrak{F})$, which does not change the image, and then add all elements of the form 1_{I_j} by summing the respective I_j (also meant as a construction on the top level of the indices). So there would be some finite number of non-zero $\{g_f\}_f$ s.t. $f|_{\Gamma(0)}=0$ such that

$$\sum g_f \cdot f = 0$$

and

$$g_f(0) = \begin{cases} 1 & \text{if } f = h \text{ or } f = X \\ 0 & \text{else.} \end{cases}$$

Not let us analyze this on the $T_\varphi^* \mathfrak{F}$. If we consider the first derivative of the linear combination in any fiber direction v , we know that is has to be given by

$$0 = \frac{\partial}{\partial v} \left(\sum g_f \cdot f \right) |_0 = \sum g_f(0) \cdot \left(\frac{\partial}{\partial v} f |_0 \right),$$

as all f vanish at 0.

But this only leaves us with a sum of two terms, as $g_f(0) = 0$ for most f . So

$$0 = 1 \cdot X(v) + 1 \cdot \left(\frac{\partial}{\partial v} h |_0 \right),$$

but as we assumed that h has vanishing first derivative at 0 in all fiber directions, such that we can deduce $X = 0$ at φ .

So at φ we can omit X from our considerations and get to a new equation

$$\sum g_f \cdot f = 0,$$

where $g_f(0) = 0$ for all $f \neq h$.

If we now consider the second derivative at 0 in any direction v , we find that

$$\frac{\partial^2}{\partial v^2} \left(\sum g_f \cdot f \right) |_0 = \left(\sum \left(\frac{\partial}{\partial v} g_f |_0 \right) \cdot \left(\frac{\partial}{\partial v} f |_0 \right) \right) + 1 \cdot \left(\frac{\partial^2}{\partial v^2} h |_0 \right),$$

as all terms that contain the factor $g_f(0)$ or $f(0)$ will vanish with exception of g_h .

But this is a contradiction, as each function that is contained in the sum as one of the factors is a linear function in v . Hence we have found a linearization of the second derivative of h at 0, which we assumed would not exist.

So we may conclude that indeed, 1_h is an element of $H_1(\mathcal{O}(D)_m)$, that is not contained in the image of $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ under the map described above. Hence the map between these two vector spaces can especially not be an isomorphism. But as both $\mathfrak{S}(\mathfrak{F})_m$ and $\mathfrak{T}(\mathfrak{F})_m$ admit this group as their first homology group, it follows that they cannot be quasi-isomorphic to $\mathcal{O}(D)_m$, which concludes the proof. $\square$

So the complex that arises by $\mathcal{O}(D)_m = (\mathcal{O}_{\mathfrak{F}})_m \otimes_{(\mathcal{O}_{T^*\mathfrak{F}})_m}^{\mathbb{L}} (\mathcal{O}_{\mathfrak{F}})_m$ has the property of admitting more symmetries than can be expressed by $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$. This goes against the physical intuition, that a symmetry should correspond to a physical symmetry of $\mathfrak{F}$, i.e. a flow, which is considered a symmetry of the field theory if and only if it restricts to a flow on the subspace that is “on-shell”.

This goes to show that it depends on the context which of the three different dg-algebras $\mathfrak{S}(\mathfrak{F})_m$, $\mathfrak{T}(\mathfrak{F})_m$ and $\mathcal{O}(D)_m$ is the most appropriate object to study. For example, the naturality of its construction makes $\mathcal{O}(D)_m$ a definition we can hope to generalise to other contexts. When considering the constructions of [5] on the other hand, it is more beneficial to employ the viewpoint of considering “linear observables” and consider options just as $\mathfrak{S}$ or $\mathfrak{T}$ to express these objects.

2.4 The Derived Critical Locus with respect to a Lagrangian

There is a problem that sometimes arises when mathematically modelling the physics of classical field theory: The action function, given as the integral over the Lagrangian, is not always actually well-defined on all fields. It may happen that even for a not well-defined function S , we can find a section from $\mathfrak{F}$ to $T^*\mathfrak{F}$ that act as some kind of “formal derivative” to models dS . But it would be beneficial to find a model that is dependant of the Lagrangian itself instead.

To work with the Lagrangian field theory from a mathematical perspective, it is important to be familiar with the notion of the jet bundles (of finite and infinite order) of $F \rightarrow M$, which we denote by $J^k F$ or $J^\infty F$. Its construction and properties can be found for example in [17].

Definition 2.4.1. A Lagrangian field theory is a fiber bundle $F \rightarrow M$ onto an manifold M , that usually comes equipped with an orientation form, together with an order k and a smooth bundle map $\bar{L} : J^k M \rightarrow \wedge^n(T^*M)$.

Remark. This automatically defines a smooth function $L : \mathfrak{F} \rightarrow \Omega^n(M)$ given by $L(\varphi)(m) = \bar{L}(j_m^k \varphi)$. This is referred to as the Lagrangian form and the property that it factors through the jet bundle is referred to as locality.

While in physics, most action functionals come from a Lagrangian, not every Lagrangian is integrable to an action functional. Considering Lagrangians is strictly upgrading the field theories we can express and consider, as for example the following result shows.

Proposition 2.4.2. *Consider a Lagrangian field theory on a manifold M with an orientation form, where the space of fields $\mathfrak{F}$ is path-connected and path-connected relative to any compact K , i.e. for any φ we have that*

$$\{\psi \in \mathfrak{F} | \psi \equiv \varphi \text{ on } K\}$$

is path-connected.

Consider a Lagrangian with the property that $\int_M L(\varphi)$ is defined for every φ . Assume that the point-wise derivative (for $m \in M$) of the Lagrangian L along any path $p : \mathbb{R} \rightarrow \mathfrak{F}$ is integrable. Then there exists a compact $K \subset M$ such that the $\int_M L(\varphi)$ only depends on $\varphi|_K$.

Remark. If we assume that the point-wise derivative of the Lagrangian is always integrable, without the condition, that the integral over L itself is well-defined, we can consider the Lagrangian $\tilde{L} = L - L(\psi_*)$, for some arbitrary $\psi_* \in \mathfrak{F}$. L and $\tilde{L}$ have the same derivative along any path. If we assume that $\mathfrak{F}$ is path-connected, we have that for any $\psi \in \mathfrak{F}$ we can recover $\tilde{L}(\psi)$ through integrating the derivative of $\tilde{L}$ along a path from ψ_* to ψ , which is finite. So we find that there is a somewhat equivalent $\tilde{L}$ that is everywhere well-defined.

Proof. Assume this was not the case. (This especially means that M is not compact.) We will now firstly construct an infinite number of tangent vectors X_{K_i} in $T_{\varphi_{K_i}} \mathfrak{F}$ for some φ_{K_i} such that the action functional has non-vanishing derivative along these tangent vectors. Further, we will choose the tangent vectors with disjoint and compact support on M .

By the locality of the Lagrangian, the derivative of the action functional is also non-zero in direction of the induced tangent vectors at any section that agrees with φ_{K_i} on the respective (compact) support of X_{K_i} on M . So we will construct a section ψ such that there are an infinite number of these disjoint induced tangent vectors in $T_\psi \mathfrak{F}$. By rescaling of the X_{K_i} , we obtain that the Lagrangian then has an infinite, i.e. non-integrable derivative along some infinite (but locally finite on M , hence well-defined) sum of the tangent vectors.

So let us begin our construction: Choose a fixed locally finite cover of M by pre-compact opens U_i , such that the bundle $F \rightarrow M$ is trivializable over the U_i and such that the $U_i \subset M$ admit local coordinates. Further choose a locally finite refinement cover of pre-compact V_i such that $\overline{V_i} \subset U_i$. For each V_i we can also choose a function χ_i with $\text{supp}(\chi_i) \subset U_i$ and $\chi_i \equiv 1$ on V_i .

By our assumption, for every compact K there are two sections $\varphi_1, \varphi_2 \in \mathfrak{F}$ with $\varphi_1|_K = \varphi_2|_K$, but $\int_M L(\varphi_1) \neq \int_M L(\varphi_2)$. We may just assume that $\int_M L(\varphi_1) < \int_M L(\varphi_2)$.

We have assumed $\mathfrak{F}$ to be path-connected relative to K . Hence we can choose a path between φ_1 and φ_2 . As $\int_M L(-)$ is smooth along this path and its value at φ_1 is lower than its value at φ_2 , there thus must be a φ_K , where $\frac{\partial}{\partial X} \int_M L(\varphi_K) > 0$, where X is the tangent vector at φ_K that is corresponding to p .

We can now choose a partition of unity according to the V_j and decompose X into a countable number of X_j , where every X_j has support in

$V_j \subset M$. As the notion is linear and the partition locally finite, we have some X_K with support in the respective $V_K \subset U_K$ such that

$$\frac{\partial}{\partial X_K} \int_M L(\varphi_K) > 0.$$

Let us define a sequence of K_i with the corresponding X_{K_i} as following: Start by fixing any complete Riemannian metric on M . That this is always possible is shown for example in [12]. Choose any compact K_1 with the associated φ_{K_1} and X_{K_1} . Then consider any compact C_1 that is containing U_{K_1} and all U_j with $\text{dist}(U_j, U_{K_1}) \leq \varepsilon$ as a subset and set $K_2 = K_1 \cup C_1$. That there exists such a C_1 is ensured as there are only finitely many U_j with that property (as the cover is locally finite and $\overline{U_{K_1}} + \varepsilon$ is closed and bounded, hence compact) and the U_j are pre-compact. With K_2 iterate the procedure. That we choose the C_i with so broadly around U_{K_i} has the consequence that no sequence of the form $x_i \in U_i$ has a converging subsequence.

Now we can find homotopies on $[0, 1]$ between φ_{K_1} and φ_{K_i} on U_i for all $i \in \mathbb{N}$, as we have that $\mathfrak{F}$ is path-connected. Evaluating the homotopies along the values of the χ_{K_i} we had earlier results in a section ψ_i each, which agrees with φ_{K_1} outside of U_{K_i} and with φ_{K_i} inside of V_{K_i} . By the choice of the K_i all the occurring U_{K_i} have at least distance ε between each other, so we can combine the ψ_i into a single smooth section ψ , that agrees with ψ_i on U_{K_i} and with φ_{K_1} outside of the U_{K_i} .

As L is local and the support of X_{K_i} is contained in V_{K_i} , we have that

$$\frac{\partial}{\partial X_{K_i}} \int_M L(\varphi_K) = \frac{\partial}{\partial X_{K_i}} \int_{V_{K_i}} L(\varphi_K) = \frac{\partial}{\partial X_{K_i}} \int_{V_{K_i}} L(\psi).$$

Especially, we have

$$\frac{\partial}{\partial X_i} \int_M L(\psi) = \frac{\partial}{\partial X_i} \int_{V_{K_i}} L(\psi) > 0.$$

Now consider λ_i given by the positive constant such that

$$\frac{\partial}{\partial X_i} \int_M L(\psi) = 1.$$

Because of the way the U_i were chosen, we have that

$$X = \sum_{i \in \mathbb{N}} \lambda_i X_i$$

is a well defined smooth tangent vector (the positive distance between the U_{K_i} ensures smoothness). So as the sum is locally finite, we have that

$$\frac{\partial}{\partial X} \int_M L(\psi) = \sum_{i \in \mathbb{N}} \frac{\partial}{\partial \lambda_i X_i} \int_M L(\psi) = \sum_{i \in \mathbb{N}} 1,$$

such that the derivative is not well defined. This contradicts the assumption. $\square$

Outlook. The requirement that $\mathfrak{F}$ is path-connected relative to any given $K \subset M$ seems relatively hard. But firstly note that it is only really necessary to be path-connected relative to the K_i of the previous proof (and path-connected in general) for the proof to work. Secondly note that if the fiber F_m admits a Riemannian metric that allows smooth unique shortest paths, we can apply the method of the following sketch:

Assume we are given a compact K and $\varphi_1, \varphi_2 \in \mathfrak{F}$ that agree on K . We can for example choose a cover as given in the previous proof, and for every U_i define a ψ_i together with a homotopy between $\psi_{(i-1)}$ and ψ_i , where we set $\psi_0 = \varphi_1$. Therefore choose a homotopy on $U_i \times [0, 1]$ between $\psi_{(i-1)}$ and φ_2 , that is at each $m \in U_i$ given by the shortest path in F_m , normalised by its length.

At every point $m \in U_i$ we can re-parametrise this path with some compactly supported function (with values between 0 and 1) that equals 1 on all of V_i . This implies we can continue the homotopy outside of U_i by the constant homotopy. The endpoint of the resulting homotopy will be called ψ_i and inductively we have $\psi_i \equiv \varphi_2$ on $\bigcup_{j=1}^i V_j$. As the cover of the U_i is locally finite, we have that the homotopies along the ψ_i combine into a homotopy between φ_1 and φ_2 .

Similar constructions might also be possible under different requirements. For example, any Lagrangian on $F \rightarrow M$ induces a Lagrangian on the universal coverings $\widehat{F} \rightarrow \widehat{M}$. So if the fibers of F are not simply connected (which would imply there are no unique shortest paths), but M is, we could hope that $\widehat{F} \rightarrow \widehat{M}$ has fibers admitting shortest paths.

This means that we now have to define the observables as a derived critical locus with respect to different spaces and functions than before.

We could aim to find a model of the observables, that returns the original physical intuition described in Example 1.1.2 that a symmetry of the observables should be given by a vector field. This has the nice interpretation, that we could integrate this vector field (point-wise on M) to a flow on $\mathfrak{F}$ which maps the solutions of $dS = 0$ onto themselves.

But there are also valid reasons to look for different solutions. For example, in [5], the aim is to construct a factorisation algebra out of 1.1.2. $U \hookrightarrow V$ induces natural maps

$$\mathfrak{F}|_V \rightarrow \mathfrak{F}|_U$$

and

$$T(\mathfrak{F}|_V) \rightarrow T(\mathfrak{F}|_U),$$

which both combine into a contravariant functor. But they do not assemble into a single covariant functor $\Gamma^\infty(\mathfrak{F}|_U, T(\mathfrak{F}|_U)) \rightarrow \Gamma^\infty(\mathfrak{F}|_V, T(\mathfrak{F}|_V))$, which

would be necessary if we wanted to assemble the observables into a factorisation algebra.

When searching for an alternative here, we should first look to the methods that have already been developed to mathematically express Lagrangian field theory without using the action function. These methods depend on the cohomological action principle. An explanation of this principle is to find in [3], but we will highlight the relevant aspects of the theory when we need them.

Our aim in the following is to experiment with this approach and to find a possible way to generalize the objects we have identified as models of the observables in the case of a well-defined action functional.

Definition 2.4.3. For every $\varphi \in \mathfrak{F}$, there is a vector bundle $\pi : W_\varphi \rightarrow M$ of pro-manifolds, defined as a set by $\pi^{-1}(m) = \wedge^n(T_m^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F$. It can be constructed as

$$W_\varphi = \wedge^n(T^*M) \otimes (M \times_{J^\infty F} V^*(J^\infty F \rightarrow M)),$$

where $V^*(J^\infty F \rightarrow M)$ denotes the vertical cotangent bundle on $J^\infty F$. The map from $V^*(J^\infty F \rightarrow M)$ to $J^\infty F$ that we use in the second factor is given by the bundle projection and the map from M to $J^\infty F$ is given by the jet prolongation $j^\infty \varphi : M \rightarrow J^\infty F$.

The vertical cotangent bundle $V^*(J^\infty F \rightarrow M)$ is given as the dual bundle to the vertical tangent bundle $V(J^\infty F) \subset T(J^\infty F)$ that is characterised as

$$V(J^\infty F) = \ker(T\pi : T(J^\infty F) \rightarrow TM).$$

The sections $\Gamma^\infty(M, W_\varphi)$ will suggestively be denoted by

$$\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F) = \Gamma(M, W_\varphi)$$

from here on. The reason for this notation is that we would like to remind ourselves that W_φ is dependent of $\varphi \in \mathfrak{F}$.

Let $\mathfrak{L} \rightarrow \mathfrak{F}$ be the fiber bundle of diffeological spaces given by the following construction: At every point $\varphi \in \mathfrak{F}$, we have that $\mathfrak{L}_\varphi$ is given by $\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F)$.

Further a map $p : \mathbb{R}^k \rightarrow \mathfrak{L}$ is smooth if and only the corresponding map

$$\text{ev}_{m \in M} \circ p : \mathbb{R}^k \times M \rightarrow \wedge^n(T^*M) \otimes V^*(J^\infty F \rightarrow M)$$

is smooth.

For the sections $\Gamma^\infty(\mathfrak{F}, \mathfrak{L})$ we will also use the more suggestive notation $\Gamma^\infty(\mathfrak{F}, \Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))$.

Remark. At each $j_m^\infty \varphi$, we then have that $T_{j_m^\infty \varphi}^* J_m^\infty F$ can be projected to T_m^*M by the push-forward along $j^\infty \varphi : M \rightarrow J^\infty F$. This can be used to interpret $T_{j_m^\infty \varphi}^* J_m^\infty F$ as the direct sum of $V_{j_m^\infty \varphi}^* J_m^\infty F$ and T_m^*M . Under this identification, we have that any section $A \in \Gamma^\infty(\mathfrak{F}, \mathfrak{L})$ is smooth if it corresponds to a smooth map $\omega_A : \mathfrak{F} \times M \rightarrow \wedge^{(n+1)}(T^*J^\infty F)$.

The equation that determines if a field is “on-shell” according to the cohomological action principle is given by $P\delta L(\varphi) = 0$. $P\delta$ (also denoted E for Euler operator) is an operator that when evaluated on a Lagrangian, returns a local map $\mathfrak{F} \times M \rightarrow \wedge^{(n+1)}(T^*J^\infty F)$ that can be factored as

$$J^\infty F \rightarrow (J^\infty F \times_M \wedge^n(T^*M)) \otimes V^*(J^\infty F \rightarrow M).$$

Due to this property, we can check that especially $P\delta L$ can be interpreted as an element of $\Gamma^\infty(\mathfrak{F}, \mathfrak{L})$. The cohomological action principle might then lead us to the following definition.

Definition 2.4.4. The generalized observables of a Lagrangian field theory are given by $\mathcal{O}_D$ of the derived pull-back of

$$\begin{array}{ccc} D & \longrightarrow & (\mathfrak{F}, \mathcal{O}_{\mathfrak{F}}^{\text{nat}}) \\ \downarrow & & \downarrow P\delta L \\ (\mathfrak{F}, \mathcal{O}_{\mathfrak{F}}^{\text{nat}}) & \xrightarrow{0} & (\mathfrak{L}, \mathcal{O}_{\mathfrak{L}}^{\text{nat}}). \end{array}$$

in the category of dg-diffeological spaces. Here, $\mathcal{O}_{\mathfrak{F}}^{\text{nat}}$ and $\mathcal{O}_{\mathfrak{L}}^{\text{nat}}$ denotes the sheaf of differential graded $\mathbb{R}$ -algebras concentrated in degree 0, where it is given by $\mathcal{O}_{\mathfrak{F}}$ and $\mathcal{O}_{\mathfrak{L}}$ respectively.

This definition might be useful because of the following properties: Firstly, if we have Lagrangians with the same action functional, we obtain the same observables. Secondly, the definition is functorial in the category $\mathbf{Open}(M)$, which denotes the category of the open subsets of M with the respective inclusions. As described above, this is a property that we do not find in $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$.

Lemma 2.4.5. *Let M be compact and orientable with a fixed orientation form. The derived critical locus is independent of the Lagrangian that we are working with, i.e. if we have two Lagrangian forms L and L' , such that for all $\varphi \in \mathfrak{F}$ we have*

$$\int_M L(\varphi) = \int_M L'(\varphi),$$

then the generalized observables on their respective field theories agree.

Remark. The fact that M is compact is used here for nothing more than the fact that the integral is well-defined for every φ . We could also simply assume that the integral was well-defined and smooth instead.

Proof. We can define the field theory corresponding to the Lagrangian form $\tilde{L} = L - L'$. By assumption, we have that

$$\int_M \tilde{L}(\varphi) = \int_M L(\varphi) - L'(\varphi) = 0$$

on all of $\mathfrak{F}$, such that every $\varphi \in \mathfrak{F}$ is a critical point of the action. As shown in [3] (cf. Theorem 7.6.1), we then have $P\delta(L - L') = 0$.

But as described in [2] (cf. Theorem 2.12), P is a linear operator, so that

$$P\delta(L - L') = 0$$

implies that $P\delta L = P\delta L'$. Hence the Definition 2.4.4 is independent of the choice of the Lagrangian. $\square$

Lemma 2.4.6. *Assume we are given a Lagrangian field theory on $F \rightarrow M$. Then the generalized observables of the theory assemble into a functor $\text{Open}(M) \rightarrow \text{DerRing}$ to the category of derived rings.*

Here, $\text{Open}(M)$ denotes the category of open subsets $U \subset M$ of the base manifold together with inclusions. The bundles with respect to which we can define the generalized observables is given by $(F \cap \pi^{-1}(U)) \rightarrow U$ and the Lagrangian is obtained by restriction of the jet bundle.

Proof. We have that

$$U \mapsto (\mathfrak{L}|_U)_\varphi = \Gamma^\infty(U, \wedge^n(T^*U) \otimes T_{j_m^\infty \varphi}^* J_m^\infty(F \cap (\pi^{-1}(U))))$$

is a contravariant functorial assignment, as for every $m \in U$ we have that

$$T_m U = T_m M,$$

hence

$$T_m^* U = T_m^* M$$

and

$$J_m^\infty(F \cap (\pi^{-1}(U))) = J_m^\infty F.$$

Hence for $V \subset U$ the restrictions on base space and fiber of $\mathfrak{L} \rightarrow \mathfrak{F}$ induce a projection $\mathfrak{L}|_U \rightarrow \mathfrak{L}|_V$.

Also, $P\delta$ is a local operator on the jet bundles, such that we have that $P\delta(L|_U) = (P\delta L)|_U$. Hence for every $V \subset U$ we have a commutative diagram of the form

$$\begin{array}{ccccc}
D_U & \xrightarrow{\quad} & & & (\mathfrak{F}|_U, \mathcal{O}_{\mathfrak{F}}) \\
\downarrow & \searrow \text{dashed} & D_V & \xrightarrow{\quad} & (\mathfrak{F}|_V, \mathcal{O}_{\mathfrak{F}}) \\
& & \downarrow & & \downarrow (P\delta L)|_V \\
& & (\mathfrak{F}|_V, \mathcal{O}_{\mathfrak{F}}) & \xrightarrow{0} & (\mathfrak{L}|_V, \mathcal{O}_{\mathfrak{L}}) \\
& \nearrow & & \nwarrow & \\
(\mathfrak{F}|_U, \mathcal{O}_{\mathfrak{F}}) & \xrightarrow{0} & & & (\mathfrak{L}|_U, \mathcal{O}_{\mathfrak{L}})
\end{array}$$

where the dashed arrow exists due to the pull-back property of the derived critical locus. This implies that there is a morphism $\mathcal{O}_{D_V} \rightarrow \mathcal{O}_{D_U}$ in the category of derived rings. $\square$

However, we have already seen when comparing the results of Theorem 2.3.8 with Example 1.1.2, that the structure sheaf of the derived critical locus of diffeological spaces might differ from the physical intuition for a field theory that we intend to model. The question that remains is, which sub-algebra of the generalized observables would be an appropriate physical model to work with. This is a question that cannot be answered from the mathematical perspective alone, but we can define some options and analyze their suitability. Following [5], one of the most important properties we would like is a homotopy Poisson algebra structure or some similar property.

Let us make a few first attempts here.

Definition 2.4.7. Let $\mathfrak{L}$ be given as in 2.4.3. We can define the diffeological space $\mathfrak{L}^*$ as the fiber-wise dual to $\mathfrak{L} \rightarrow \mathfrak{F}$, i.e. $\mathfrak{L}^* \rightarrow \mathfrak{F}$ has fibers given by $(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))^*$ with the diffeology that is dual to $\mathfrak{L}$.

This means that a map $d : \mathbb{R}^l \rightarrow \mathfrak{L}^*$ is smooth if and only if it combines with all compatible smooth maps $d' : \mathbb{R}^l \rightarrow \mathfrak{L}$ to a map $\langle d, d' \rangle \in C^\infty(\mathbb{R}^l)$. Here, compatible means that the projections of d and d' to $\mathfrak{F}$ agree and $\langle d, d' \rangle$ denotes the map

$$z \mapsto d(z)(d'(z))$$

for any $z \in \mathbb{R}^l$, as the fibers of $\mathfrak{L}^*$ are dual to the fibers of $\mathfrak{L}$.

As in 2.4.3, we will also use the suggestive notation

$$\Gamma^\infty(\mathfrak{F}, (\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))^*)$$

instead of $\Gamma^\infty(\mathfrak{F}, \mathfrak{L}^*)$.

Definition 2.4.8. Consider the notational conventions of Definition 2.4.3 and Definition 2.4.7. We can define $\mathfrak{S}_*^1(\mathfrak{F})$ as the free commutative differential graded algebra over $C^\infty(\mathfrak{F})$ generated in degree 1 by

$$\Gamma^\infty(\mathfrak{F}, (\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))^*).$$

The diffeology on $\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))^*$ in this context is given through evaluation at smooth maps $C^\infty(U, \mathfrak{L})$ and the differential is generated by

$$d : \Gamma^\infty(\mathfrak{F}, (\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))^*) \rightarrow C^\infty(\mathfrak{F}),$$

where d is given through $\vee P\delta L$, i.e. through evaluation at $P\delta L \in \Gamma^\infty(\mathfrak{F}, \mathfrak{L})$.

We can define $\mathfrak{T}_*^1(\mathfrak{F})$ as the differential graded algebra over $C^\infty(\mathfrak{F})$ given in degree k by

$$\Gamma^\infty(\mathfrak{F}, (\wedge^k(\mathfrak{L}))^*) = \Gamma^\infty\left(\mathfrak{F}, \left(\wedge^k\left(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F)\right)\right)^*\right).$$

The differential is given through $\iota_{(P\delta L)}$.

These complexes have the property that our “symmetries” in degree 1 can act on $P\delta L$. However, when bearing in mind we will be looking for Poisson algebra structures on these complexes later, there is no evident way how these symmetries might act on ordinary functions. This is because the space $\Gamma^\infty(\mathfrak{F}, (\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi J_m^\infty F))^*)$ is defined with the locality of L in mind. This leaves us two options, if we aim for a Poisson structure: Either restrict the space of functions $C^\infty(\mathfrak{F})$, or consider different symmetries.

We can analyze, that restricting the function space to functions similar to L will imply that we also need to adjust our space of symmetries. For example, we might want to restrict it to only those functions that, like L , arise from integrating local functions over all of M . But to do this we would need that the differential on the rest of the complex is still well-defined, i.e. that every closed element of the function space is expressible by an integral over all of M . Hence we would need to exclude all degree 1 elements $A \in \Gamma^\infty(\mathfrak{F}, (\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi J_m^\infty F))^*)$, that at some $\varphi \in \mathfrak{F}$ act through evaluating $v \in \Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi J_m^\infty F)$ at a single fixed point $m \in M$.

If we intend on not restricting the function space, we have one immediate option. The sections $\Gamma_c^\infty(M, VJ^\infty F)$ are easily classifiable elements, that can act on $\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi J_m^\infty F)$ by point-wise pairing and integration. But we only know that sections of the form $\Gamma^\infty(M, VJ^\infty F)$ can act on any $f \in C^\infty(\mathfrak{F})$ when it comes in form of a tangent vector $\Gamma^\infty(M, VF)$. For any other section we do not have a uniform way to extend this action. Hence we can define the following

Definition 2.4.9. We can define $\mathfrak{S}_*^2(\mathfrak{F})$ as the free commutative dg-algebra over $C^\infty(\mathfrak{F})$ generated in degree 1 by $\Gamma^\infty(\mathfrak{F}, \Gamma_c^\infty(M, VF))$. The diffeology on $\Gamma_c^\infty(M, VF)$ is given such that the action on $\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi J_m^\infty F)$ via integration over M is smooth. This implies a map to $\Gamma_c^\infty(M, VF)$ is smooth if and only if it is smooth as a map to $T\mathfrak{F}$ that is locally supported on a uniform compact on M . The differential on the complex is given by $\vee P\delta L$.

We can define $\mathfrak{T}_*^2(\mathfrak{F})$ as the differential graded algebra over $C^\infty(\mathfrak{F})$ given in degree k by $\Gamma^\infty(\mathfrak{F}, (\wedge^k(\Gamma_c^\infty(M, VF)^*))^*)$. The differential is given through $\iota_{(P\delta L)}$.

The other option we have is choosing that the symmetries that we are interested in should be given at a point by $\Gamma_c^\infty(M, TJ_m^\infty F) = \Gamma_c^\infty(M, VJ^\infty F)$. As previously mentioned, these symmetries cannot act on all $f \in C^\infty(\mathfrak{F})$, although they can act on all functions that, like L , arise from a local function on $\mathfrak{F} \times M$. This leads to the following definitions.

Definition 2.4.10. Let $\mathfrak{D}$ be a diffeological space. We call a map $f : \mathfrak{F} \rightarrow \mathfrak{D}$ local, if it factors as

$$\mathfrak{F} \rightarrow \mathfrak{J} \rightarrow \mathfrak{D},$$

where $\mathfrak{J}$ is the diffeological space arising from the set $J = \Gamma^\infty(M, J^\infty F)$, that can be equipped with the functional diffeology.

Remark. The notion of smooth maps from M to $J^\infty F$ here means smooth maps as Pro-manifolds, which is not a problem with regards to the definition of the functional diffeology.

Definition 2.4.11. We can define $\mathfrak{S}_*^3(\mathfrak{J})$ as the free commutative dg-algebra over $C^\infty(\mathfrak{J})$ generated in degree 1 by $\Gamma^\infty(\mathfrak{J}, \Gamma_c^\infty(M, VJ^\infty F))$, where we define sections with respect to the natural projection of $\Gamma_c^\infty(M, VJ^\infty F) \rightarrow \mathfrak{J}$.

The diffeology on $\Gamma_c^\infty(M, VJ^\infty F)$ is given such that the action on $\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* J_m^\infty F)$ via integration over M is smooth. This implies that a map to $\Gamma_c^\infty(M, VJ^\infty F)$ is smooth if it is smooth as a map to $T\mathfrak{J}$ that is locally supported on a uniform compact on M . The differential on the complex is given by $\vee P\delta L$.

We can define $\mathfrak{T}_*^2(\mathfrak{F})$ as the differential graded algebra over $C^\infty(\mathfrak{J})$ given in degree k by $\Gamma^\infty(\mathfrak{J}, (\wedge^k(\Gamma_c^\infty(M, VJ^\infty F)^*))^*)$. The differential is given through $\iota_{(P\delta L)}$.

Let us shortly analyze some properties of the previously defined objects that can motivate further studies of these.

Definition 2.4.9 agrees well with our original Example 1.1.2 and the natural map $\mathfrak{S}(\mathfrak{F}) \rightarrow C^\infty(\mathfrak{F}) \otimes_{T^*\mathfrak{F}}^{\mathbb{L}} C^\infty(\mathfrak{F})$ constructed in Theorem 2.3.8. It only changes the complex minimally for non-compact M , but this small change makes it functorial in opens $U \subset M$, as we can extend compactly supported tangent vectors (here meaning compactly supported in U) by 0 to tangent vectors on all of M .

The Definition 2.4.11 changes the space of functions on $\mathfrak{F}$ to local ones. This can, depending on the application, both be an advantage as it reflects more accurately the actual measurements that are of interest in many cases, or a disadvantage, as not all objects of $\mathfrak{J}$ can be encountered in reality.

2.5 The Homotopy Poisson Algebra Structure

In [5], the Example 1.1.2 is taken as the motivation for the conjecture, that the complex of classical observables always comes with a homotopy Poisson algebra structure.

Working on diffeological spaces makes the definition of such a bracket on vector fields more complicated. On manifold the Poisson bracket is defined via derivations, which, when studying diffeological spaces, do no longer agree with elements of the tangent space. However, Lemma 2.1.9 indicates that we might still be able to adapt the construction.

Firstly, let us note that given derivations X and Y of $C^\infty(\mathfrak{F})$, a simple computation shows that the term

$$[X, Y](f) = X(Y(f)) - Y(X(f)),$$

is still a derivation. As on manifolds, this defines a bracket on derivations that is anti-symmetric in X and Y , $\mathbb{R}$ -multi-linear and satisfies the Jacobi identity. That this acts nicely on the tangent space is given by the following result:

Theorem 2.5.1. *The image of the inclusion $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F}) \rightarrow \text{Der}(C^\infty(\mathfrak{F}))$ is closed under the commutator bracket.*

Especially, the complex given by $\mathfrak{S}(\mathfrak{F}) = \bigwedge_{C^\infty(\mathfrak{F})}^ \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ with differential given by $\vee dS = \iota_S$ is a Poisson $\mathbb{R}$ -algebra with respect to the brackets this induces on the complex.*

Proof. Let us start by proving the second claim under assumption of the first:

We can define the Lie brackets on $\mathfrak{S}(\mathfrak{F})$ as follows: Let $g, h \in C^\infty(\mathfrak{F})$ and $X, Y \in \mathfrak{X}^1(\mathfrak{F})$. Then

$$\begin{aligned} [g, h] &= 0, \\ [X, f] &= X(f), \\ [X, Y] &= XY - YX, \end{aligned}$$

where the last line considers $XY - YX$ as a derivation on $C^\infty(\mathfrak{F})$.

It is only through our assumption that this term can actually be expressed by a tangent vector (uniqueness then follows by 2.1.9). But before turning to this issue we can conclude that, by definition, if the bracket is well-defined it can be uniquely extended to all higher degrees of $\mathfrak{S}(\mathfrak{F})$, such that it is compatible with multiplication. The extension also satisfies the Jacobi-identity. That bracket and differential are compatible can be seen by writing the differential as $[S, -]$, similarly to the case in [5] (cf. p. 46). Hence we have indeed defined a Poisson algebra structure.

It remains to prove the first claim. To do this, we will at every point $\varphi \in \mathfrak{F}$ construct a tangent vector that acts on any function $f \in C^\infty(\mathfrak{F})$ the same as $[X, Y]$ composed with the evaluation at φ . We will at first consider X and Y as smooth homotopies of paths. This will allow us to identify the action of $[X, Y]$ on f with a term which only depends on the second derivative of f along two paths $\mathbb{R} \rightarrow \mathfrak{F}$, that happen to be tangent at 0.

That the two paths are tangent means that, when evaluating them pointwise for every $m \in M$ separately and in local coordinates, their difference is given by a path p with vanishing derivative at 0. We can thus find a unique tangent vector on F_m that is given by the second derivative of p . As a tangent vector, this object acts on functions the same (up to a constant factor) as when the second derivative at 0 along p is evaluated. We could hope that these elements in $V_m F$ combine into a global section $s \in \Gamma^\infty(M, VF)$, that similarly recovers information about the second derivatives by acting as a tangent vector.

Before we can check this, we would like to restrict the consideration to the case where we have a trivial vector bundle by decomposing X and Y into a sum of vector fields on some open cover of M . Therefore, we need to go through some technicalities to find an open cover $\{O_j\}$ of M such that M is trivializable over every pair of indices. The final result then follows from an explicit computation and the fact that the second derivative of a function along a path through φ only depends on the first and second derivative of that path.

To start, let us remark, that Y is smooth as a vector field. Hence also the composition $Y \circ X(\varphi) : \mathbb{R} \rightarrow T\mathfrak{F}$ is smooth when we regard $X(\varphi)$ as a path in $\mathfrak{F}$. This then gives us a smooth map from $\mathbb{R}^2$ to $\mathfrak{F}$ by evaluating $Y(X(\varphi)(s))$ as path in $\mathfrak{F}$ itself.

We can compute

$$\begin{aligned}
& [X, Y](f)(\varphi) \\
&= \frac{\partial}{\partial s} \frac{\partial}{\partial t} (f(Y(X(\varphi)(s))(t)) - f(X(Y(\varphi)(s))(t)))|_{t=0, s=0} \\
&= \frac{\partial}{\partial s} \frac{\partial}{\partial t} f(Y(X(\varphi)(s))(t))|_{t=0, s=0} - \frac{\partial}{\partial s} \frac{\partial}{\partial t} f(X(Y(\varphi)(t))(s))|_{t=0, s=0} \\
&= \frac{1}{2} \frac{\partial^2}{\partial(s+t)^2} f(Y(X(\varphi)(s))(t))|_{t=0, s=0} - \frac{1}{2} \frac{\partial^2}{\partial(s+t)^2} f(X(Y(\varphi)(t))(s))|_{t=0, s=0} \\
&= \frac{1}{2} \frac{\partial^2}{\partial r^2} (f(Y(X(\varphi)(r))(r)) - f(X(Y(\varphi)(r))(r)))|_{t=0, s=0},
\end{aligned}$$

as the terms like $f(Y(X(\varphi)(s))(0)) = f(X(\varphi)(s))$ or

$$\frac{\partial^2}{\partial s^2} f(Y(X(\varphi)(s))(0)) = \frac{\partial^2}{\partial s^2} f(X(Y(\varphi)(0))(s))$$

cancel nicely and the arrangement of derivatives of real functions on $\mathbb{R}^2$ are interchangeable. The last line of this equation will be better for the computation we have to make. The remaining question is now how this can be parametrized as a derivation along a smooth path.

For this, we can consider the evaluation at any point $m \in M$. Then both paths $Y(X(\varphi)(r))(r)$ and $X(Y(\varphi)(r))(r)$ locally define a path in F_m , which can be interpreted as tangent vectors in $T(F_m) = VF_m$. By shifting the parametrisation by a constant to

$$p_1(c) : t \mapsto \text{ev}_m \circ Y(X(\varphi)(t+c))(t+c),$$

and similarly for p_2 , we furthermore get two paths in VF_m . At $c = 0$ both path are a representation of $X + Y$ at $\varphi(m)$, such that p_1 and p_2 intersect at $(\varphi(m), X + Y) \in T(F_m)$. Thus we obtain tangent vectors $v_1(0)$ and $v_2(0)$ as elements of $T(T(F_m))$ at that given point. Also, through the reparametrization, we obtain tangent vectors for other base points at other constants.

As locally on any coordinate chart it holds that $T(F_m) \cong F_m \times \mathbb{R}^k$ we have $T(T(F_m)) \cong T(F_m) \times T\mathbb{R}^k$. By construction, we already know that the projection of the $v_i(0)$ to the first factor will be given by $X + Y$. The projection to the second factor, however, is a tangent vector of a vector space, and can be identified naturally with an element of the vector space itself. So we have obtained two new elements $w_1(0), w_2(0) \in T(F_m)$ that lie in the fiber of $\varphi(m)$.

We can explicitly check that $w_1(0) - w_2(0)$ (although not necessarily $w_1(0)$ and $w_2(0)$) is independent of the choice of coordinates: We can check that $w_1(0)$ is given by the second derivative at 0 of the path

$$\text{ev}_m \circ Y(X(\varphi)(r))(r)$$

and $w_2(0)$ similarly with respect to the other path. When applying now a change of coordinates $\rho : \mathbb{R}^k \rightarrow \mathbb{R}^k$ to any path $\gamma : \mathbb{R} \rightarrow \mathbb{R}^k$, we can compute that $\frac{\partial^2}{\partial t^2} \rho_i(\gamma(t))|_0$ consists of terms of the form

$$\frac{\partial}{\partial a_{j_1}} \frac{\partial}{\partial a_{j_2}} \rho_i(a_1, \dots, a_k)|_{\gamma(0)} \cdot \left(\frac{\partial}{\partial t} \gamma_{j_1}(t)|_0 \right) \cdot \left(\frac{\partial}{\partial t} \gamma_{j_2}(t)|_0 \right)$$

and terms of the form

$$\frac{\partial}{\partial a_j} \rho_i(a_1, \dots, a_k)|_{\gamma(0)} \cdot \left(\frac{\partial^2}{\partial t^2} \gamma_j(t)|_0 \right).$$

Terms of the first form agree for tangent paths, such as $\text{ev}_m \circ Y(X(\varphi)(r))(r)$ and $\text{ev}_m \circ X(Y(\varphi)(r))(r)$ (So they cancel when taking the difference). Terms of the second form precisely correspond to the change of coordinates when applied to a tangent vector at $\gamma(0)$.

We can interpret $w_1(0), w_2(0)$ and their difference $w_1(0) - w_2(0)$ as elements of $V_m F$ that project to $\varphi(m)$. We can even construct them locally in a smooth way. Therefore consider an open $U \subset M$ which admits a trivialization of the bundle and where $\varphi(U)$ is contained in a fixed coordinate chart of F_m .

When applying the previous construction to all points of U , the elements $w_1(0), w_2(0)$ and $w_1(0) - w_2(0)$ each define a section from U to VF covering φ . These have the property that, when altering m smoothly, we obtain homotopies of paths on our coordinate chart and thus the sections arising from their second derivatives are smooth. Henceforth, we will denote with s_1, s_2 and $s_U = s_1 - s_2$ these sections, which we can then interpret as elements of $T_\varphi(\mathfrak{F}|_U)$.

But as we have seen that s is indeed independent of the choice of the coordinates we do not only get a section $s_U \in \Gamma^\infty(U, VF)$ for every $U \subset M$ with the required properties, but a global section $s \in \Gamma^\infty(M, VF)$, that we can check locally to be smooth. Thus finally, we have found an element $T_\varphi \mathfrak{F}$, that we can hope acts the same as $[X, Y](\varphi)$. It turns out, that this is

indeed true. If we succeed to show this, we can already conclude here, that we are finished, as the paths $\text{ev}_m \circ Y(X(\varphi)(r))(r)$ and $\text{ev}_m \circ X(Y(\varphi)(r))(r)$ (hence also all their derivatives in local coordinates) depend smoothly on φ . Thus the point-wise constructed values of s assemble into a smooth section $S \in \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$.

What we have not seen yet is the fact that s is also independent of the paths chosen to represent X and Y . So let us choose local coordinates on F_m , in which we then find that similarly to our very first computations, the expression

$$\frac{\partial^2}{\partial r^2} \text{ev}_m \circ Y(X(\varphi)(r))(r)$$

in these coordinates comes from a sum of terms either of the form

$$\frac{\partial}{\partial s} \frac{\partial}{\partial t} \text{ev}_m \circ Y(X(\varphi)(s))(t) = \frac{\partial}{\partial t} \frac{\partial}{\partial s} \text{ev}_m \circ Y(X(\varphi)(s))(t)$$

or terms of the form

$$\frac{\partial}{\partial r^2} \text{ev}_m \circ Y(X(\varphi)(r))(0) = \frac{\partial}{\partial r^2} \text{ev}_m \circ X(\varphi)(r)$$

and

$$\frac{\partial}{\partial r^2} \text{ev}_m \circ Y(X(\varphi)(0))(r) = \frac{\partial}{\partial r^2} \text{ev}_m \circ Y(\varphi)(r)$$

respectively, which are eliminated by taking the difference between s_1 and s_2 .

But when we compute terms of the first form, we have that under the identification $V_m F = T(F_m) \cong \mathbb{R}^k$ given through our coordinates that

$$\left. \frac{\partial}{\partial t} (\text{ev}_m \circ Y(\psi)(t)) \right|_0 = Y(\psi)(m) \in \mathbb{R}^k.$$

This does no longer depend on the choice of the path of Y , and as $Y(\psi)(m)$ is defined locally on $\mathfrak{F}$ as a smooth (vector-valued) function, neither does the derivative

$$\frac{\partial}{\partial X} Y(\psi)(m)|_\varphi.$$

This argument has even shown more than that s is independent of the choice of paths. It has also shown that s is indeed linear in X and Y , which will benefit us later. This can be seen, as the term $\frac{\partial}{\partial X} Y(\psi)(m)|_\varphi$ is linear in both X and Y and addition on $\Gamma^\infty(M, VF)$ is defined point-wise.

Before we turn to the proof that s and $[X, Y](\varphi)$ agree, we have to make one more little digression. What follows now is the technical construction I already foreshadowed in the beginning, that aims to find a cover of M such that the fiber bundle $F \rightarrow M$ is trivializable over each pair of opens.

Consider pre-compact coordinate patches U_i on M that admit an extension to a coordinate patch $(\overline{U_i} \subset \tilde{U_i}) \times V_i$ on F , where this coordinate patch

respects the fiber bundle structure of F and fully contains $\varphi(\overline{U_i})$. Choose a locally finite sub-cover and a partition of unity underlying the U_i .

Consider now for every point p an open O_p satisfying

$$p \in U_i \Rightarrow O_p \subset U_i$$

as well as

$$p \in \partial U_i \Rightarrow \chi_i|_{O_p} \equiv 0$$

and

$$p \notin \overline{U_i} \Rightarrow O_p \cap U_i = \emptyset.$$

We now have that $O_p \cap O_q \neq \emptyset$ implies that both O_p and O_q are contained in the same U_i for some $i \in I$, as at any point $\chi_i \neq 0$ for some $i \in I$. Let us again choose a finite sub-cover and a partition of unity μ_j of these O_j for $j \in \mathbb{N}$.

We can now use this to simplify the proof that the s we have constructed acts the same on functions as $[X, Y]$. We can explicitly see that both constructions (i.e. both the bracket as derivations and the construction of s) are smooth in X and Y . This means that by approximation of X and Y through compact tangent vectors, we may assume that X and Y have compact support for the proof that follows. We also can write

$$X = \sum_{j \in \mathbb{N}} \mu_j \cdot X,$$

and similarly for Y , where the sum is actually finite, such that we can use linearity of both terms to restrict the proof further to assume that X is compactly contained in some O_{j_X} and Y is compactly contained in some O_{j_Y} .

As these opens are either disjoint, where $F \rightarrow M$ is trivializable over each component, or both contained in some U_i , where $F \rightarrow M$ is trivializable, we now have a trivializable bundle and by the choice of the U_i we also have local fiber coordinates around $\Gamma(\varphi|_{O_{j_X} \cup O_{j_Y}})$. We can of course shift these coordinates fiber-wise by a constant, such that $\varphi(m) = 0$ in these coordinates for appropriate m .

This assumption that X and Y are supported in an open with such coordinates benefits us for our computations, as it gives us a vector bundle structure on all fields over $O_{j_X} \cup O_{j_Y}$ that are completely contained in the coordinate chart. On these field, we have in particular a notion of addition. Note that if we consider the addition of a section with compact support, we can continue that addition outside of $O_{j_X} \cup O_{j_Y}$ by the identity.

Also we can identify tangent vectors, such as $(X + Y)$ with either some element of the vector bundle or some linear path. By our assumption that the support of X and Y in M is compactly contained in one of the O_j respectively, we have that all these paths can be extended outside of $O_{j_X} \cup$

O_{j_Y} by the constant paths. As we have shown that both the action of $[X, Y]$ and the definition of s do not depend on the choice of the paths that we use to realise X and Y , we simply may assume that they are given by exactly these linear paths. This would ensure that the restriction of $Y(X(\varphi)(r))(r)$ to $O_{j_X} \cup O_{j_Y}$ is completely contained in our local frame.

Consider now any function $f \in C^\infty(\mathfrak{F})$. When we review our previous construction of $w_1(0)$, we had noted that we can identify $s_1(m)$ with the second derivative of the path $\text{ev}_m \circ Y(X(\varphi)(r))(r)$. Intuitively, we might already assume that the first and second derivation of the path is sufficient to classify $\frac{\partial^2}{\partial r^2} (f(Y(X(\varphi)(r))(r)))|_{r=0}$. This is indeed the case.

When we consider the given path in our local frame, we can compute the difference between $Y(X(\varphi)(r))(r)$ and $\varphi + r \cdot (X + Y) + r^2 \cdot s_1$. At any point $m \in M$, we can write

$$\text{ev}_m \circ Y(X(\varphi)(r))(r) = \varphi(m) + r \cdot (X + Y)(m) + \frac{1}{2}r^2 \cdot s_1(m) + r^3 \cdot h_m(r).$$

The construction of $h_m(r)$ here is smooth in $m \in O_{j_X} \cup O_{j_Y}$. This allows us to extend it to a homotopy of sections $h(r) \in \Gamma^\infty(M, VF)$, as outside of the frame we can smoothly continue it by 0.

Now we find that we can transfer the given problem into a problem on $\mathbb{R}^2$ by defining $\tilde{f} : \mathbb{R}^2 \rightarrow \mathbb{R}$ as

$$\tilde{f}(a, b) = f(\varphi + a \cdot (X + Y) + \frac{1}{2}a^2 \cdot s_1 + b \cdot h(a)).$$

We can recover $Y(X(\varphi)(r))(r)$ by embedding $\mathbb{R}$ into $\mathbb{R}^2$ as $\gamma : r \mapsto (r, r^3)$.

This allows us to make the following computation:

$$\begin{aligned} & \frac{\partial^2}{\partial r^2} (f(Y(X(\varphi)(r))(r)))|_{r=0} \\ &= \frac{\partial^2}{\partial r^2} \left(f(\varphi + r \cdot (X + Y) + \frac{1}{2}r^2 \cdot s_1 + r^3 \cdot h(r)) \right)|_{r=0} \\ &= \frac{\partial^2}{\partial r^2} \tilde{f}(r, r^3)|_0 \\ &= \frac{\partial^2}{\partial a^2} \tilde{f}(a, b)|_{(0,0)} \\ &= \frac{\partial^2}{\partial r^2} \left(f(\varphi + r \cdot (X + Y) + \frac{1}{2}r^2 \cdot s_1) \right)|_{r=0}. \end{aligned}$$

The equality of deriving in direction of γ and a comes from applying the chain rule for derivations (at $r = 0$, most occurring terms vanish). Also, note that the analogous statement for $X(Y(\varphi)(r))(r)$ and s_2 holds for exactly the same reasoning.

Finally, we can now aim to prove that the derivation along $s = (s_1 - s_2)$ at φ acts on f just the same as $[X, Y]$ at φ . But as f was arbitrary, we can

conclude then that $[X, Y]$ can be indeed written as a vector field, showing that the Poisson structure is indeed well-defined. Let us first present the calculation here and then argue the equality of all arising terms.

$$\begin{aligned}
s(f) &= \frac{\partial}{\partial r} f(\varphi + r \cdot s) \\
&= 2 \cdot \frac{\partial}{\partial r} f\left(\varphi + \frac{1}{2} r \cdot s\right) \\
&= 2 \cdot \lim_{r^2 \rightarrow 0} \frac{f(\varphi + \frac{1}{2} r^2 \cdot s) - f(\varphi)}{r^2} \\
&= \frac{\partial^2}{\partial r^2} f\left(\varphi + \frac{1}{2} r^2 \cdot s\right) \\
&= \frac{\partial^2}{\partial r^2} \left(f(\varphi + r \cdot (X + Y) + \frac{1}{2} r^2 \cdot (s + s_2)) \right. \\
&\quad \left. - f(\varphi + r \cdot (X + Y) + \frac{1}{2} r^2 \cdot s_2) \right) \Big|_{r=0} \\
&= \frac{\partial^2}{\partial r^2} f\left(\varphi + r \cdot (X + Y) + \frac{1}{2} r^2 \cdot s_1\right) \Big|_{r=0} \\
&\quad - \frac{\partial^2}{\partial r^2} f\left(\varphi + r \cdot (X + Y) + \frac{1}{2} r^2 \cdot s_2\right) \Big|_{r=0} \\
&= \frac{\partial^2}{\partial r^2} f(Y(X(\varphi)(r))(r)) \Big|_{r=0} - \frac{\partial^2}{\partial r^2} f(X(Y(\varphi)(r))(r)) \Big|_{r=0} \\
&= [X, Y](f)(\varphi).
\end{aligned}$$

That all these terms are indeed equal follows for one thing from the fact that

$$\frac{\partial}{\partial r} \left(f\left(\varphi + \frac{1}{2} r^2 \cdot s\right) - f(\varphi) \right) = 0,$$

as similar to a previous argument, the first derivative of f along any path does just depend on the tangent this path represents, i.e. 0. Only now can we apply that for any function we have

$$g(r) = g(0) + r \cdot g'(0) + \frac{1}{2} r^2 \cdot g''(0) + r^3 \cdot g_{\text{rem}}(r),$$

and as for $f(\varphi + r^2 \cdot s) - f(\varphi)$, we have that the constant and linear terms are 0, the second derivative can be reconstructed by taking the limit divided by r^2 .

The other equality in the equation that does not follow from previous

statements is given by

$$\begin{aligned} & \frac{\partial^2}{\partial r^2} f(\varphi + \frac{1}{2}r^2 \cdot s) \\ &= \frac{\partial^2}{\partial r^2} \left(f(\varphi + r \cdot (X + Y) + \frac{1}{2}r^2 \cdot (s + s_2)) \right. \\ & \quad \left. - f(\varphi + r \cdot (X + Y) + \frac{1}{2}r^2 \cdot s_2) \right) \Big|_{r=0}. \end{aligned}$$

For this, we can define on $\mathbb{R}^2$ the smooth function

$$\begin{aligned} & \bar{f}(a, b) \\ &= \frac{\partial^2}{\partial a^2} \left(f(\varphi + ab \cdot (X + Y) + \frac{1}{2}a^2 \cdot s + \frac{1}{2}(ab)^2 \cdot s_2) \right. \\ & \quad \left. - f(\varphi + ab \cdot (X + Y) + \frac{1}{2}(ab)^2 \cdot s_2) \right), \end{aligned}$$

which has the property that

$$\begin{aligned} & \frac{\partial}{\partial b} \bar{f}|_{(0,t)} \\ &= \frac{\partial^2}{\partial a^2} \left(a \cdot \frac{\partial}{\partial b} (f(\varphi + b \cdot (X + Y) + at \cdot (X + Y) + \dots) \right. \\ & \quad \left. - f(\varphi + b \cdot (X + Y) + at \cdot (X + Y) + \dots)) \Big|_{b=0} \right) \Big|_{a=0}, \end{aligned}$$

as every occurring b comes with a respective factor a .

From rewriting the function like this we can see that the second derivation of the given term at $a = 0$ that we seek depends only on the first derivation in direction of a of the expression

$$\begin{aligned} & \frac{\partial}{\partial b} \left(f(\varphi + b \cdot (X + Y) + at \cdot (X + Y) + \frac{1}{2}a^2 \cdot s + \frac{1}{2}(at)^2 \cdot s_2 + bt \cdot s_2) \right. \\ & \quad \left. - f(\varphi + b \cdot (X + Y) + at \cdot (X + Y) + \frac{1}{2}(at)^2 \cdot s_2 + bt \cdot s_2) \right) \Big|_{b=0}. \end{aligned}$$

As we can exchange the order of the derivatives again, we can eliminate all dependencies on a of second degree. Hence $\frac{\partial}{\partial b} \bar{f}|_{(0,t)}$ only depends on the term

$$\begin{aligned} & \frac{\partial}{\partial b} \frac{\partial}{\partial a} (f(\varphi + b \cdot (X + Y) + at \cdot (X + Y)) \\ & \quad - f(\varphi + b \cdot (X + Y) + at \cdot (X + Y))) \Big|_{a=0} \Big|_{b=0}, \end{aligned}$$

which obviously vanishes.

So we have found that $\bar{f}(0, t)$ is actually constant and thus that $\bar{f}(0, 0) = \bar{f}(0, 1)$. But these correspond to the two consecutive terms in the equation for that we still had to show equality. This concludes the proof. $\square$

Remark. To be precise, we are not only interested in the homotopy Poisson structure on the complex itself, but on the homotopy Poisson structure on any open. This descends to the localised complex however as the action of all brackets is local, i.e. the value at a given base-point is determined by the values of the inputs on any open around that point.

Outlook. This means that $\mathfrak{S}$ has a homotopy Poisson structure, where the bracket is modelled after the bracket on vector fields of a manifold, which was one of the key criteria we had for the suitability of our “observables”. But the theorem can even be applied more flexibly, as the construction of the bracket gives us control over for example the support of the bracket.

If we for example consider the Definition 2.4.9, we see that the bracket defined on $\Gamma^\infty(\mathfrak{F}, T\mathfrak{F})$ restricts to a bracket on $\Gamma^\infty(\mathfrak{F}, \Gamma_c^\infty(M, VF))$ when endowed with the diffeology of the given definition. This is because the construction for $[X, Y]$ given in 2.5.1 at φ only depends on the values of X and Y at $m \in M$ along certain smooth maps $\mathbb{R}^2 \rightarrow \mathfrak{F}$ in a neighborhood of 0. But by assumption on our diffeology, the support in M along a smooth map is locally bounded by a uniform compact, such that $[X, Y] = 0$ outside of this compact. So we find that $\mathfrak{S}^2$ from Definition 2.4.9 with the constructed bracket defines a Poisson algebra.

Quite similarly, the same construction yields a bracket $[X, Y]$ for any $X, Y \in \Gamma^\infty(\mathfrak{J}, \Gamma_c^\infty(M, VJ^\infty F))$ with the diffeology given in Definition 2.4.11. The thing we need to be careful about here is that $\mathfrak{J}$ does not arise as the sections of a vector bundle of manifolds, but is defined with respect to the Pro-manifold $J^\infty F$. But the construction of the tangent vector s in the proof can be adapted without problems, even if the relation to derivations might not be the same as for a usual vector bundle. So $\mathfrak{S}^3(\mathfrak{F})$ from Definition 2.4.11 also defines a Poisson algebra.

The next question that might be of interest to us, is if this homotopy Poisson structure extends to $\mathfrak{T}(\mathfrak{F})$. We could, for example, make an attempt to rewrite the definition of the bracket as

$$[X, Y](\varphi)(df(\varphi)) = X(dY(df))(\varphi) - Y(dX(df))(\varphi),$$

where we interpret X and Y as elements of $\Gamma^\infty(\mathfrak{F}, (T^*\mathfrak{F})^*)$.

The fact that the bracket is a tangent vector now tells us that this expression is independent of the choice of f to represent $(df)(\varphi) \in T_\varphi^*\mathfrak{F}$. If we now apply Proposition 2.1.12, we have found a way to rephrase the bracket in a way that we can extend to higher degrees: For $X \in \mathfrak{T}_k(\mathfrak{F})$ and

$Y \in \mathfrak{T}_l(\mathfrak{F})$ we have

$$\begin{aligned}
& [X, Y](\varphi)(df_1(\varphi) \wedge \cdots \wedge df_{(k+l-1)}(\varphi)) \\
&= \sum_{T, T'} \varepsilon(T, T') X(d(Y(df_{i_1} \wedge \cdots \wedge df_{i_l})) \wedge df_{i_{(l+1)}} \wedge \cdots \wedge df_{i_{(l+k-1)}})(\varphi) \\
&\quad - \sum_{S, S'} \varepsilon(S, S') Y(d(X(df_{i_1} \wedge \cdots \wedge df_{i_l})) \wedge df_{i_{(l+1)}} \wedge \cdots \wedge df_{i_{(l+k-1)}})(\varphi).
\end{aligned}$$

Here, as in previous definitions, we have that T, T' and S, S' respectively denote ordered subsets of the ordered set $(f_1, \dots, f_{(l+k-1)})$ that constitute a decomposition of that set. We additionally require $|T| = l$, $|T'| = k - 1$ and similar conditions on S and S' .

Due to my limitations on research time I cannot present a complete proof here, that this construction equips $\mathfrak{T}(\mathfrak{F})$ with a homotopy Poisson structure. I conjecture that it is possible to do so under utilisation of 2.1.12 and 2.2.2.

What is more, I even suspect this construction works for $\mathfrak{T}^2(\mathfrak{F})$ as well, as I believe the result corresponding to 2.2.2 can be proved for the space $\Gamma_c^\infty(M, VF)$ as well. The result corresponding to 2.1.12 does not hold, but instead every element of the dual can be decomposed into some exact 1-form and a remainder with support outside of a fixed compact K , which is good enough for our considerations.

So many of the possible options that we considered in the previous sections bear a Poisson algebra structure (or at least admit a bracket in degree 1) and hence are still suited as models for the physical observables under this criterion.

3 Observables in Relation with L_∞ -algebras

3.1 Formal Moduli Problems and Related Concepts

In [5], another approach to the complex of observables was explored by studying the relations of commutative differentially graded algebras with other concepts, such as formal moduli problems and L_∞ -algebras. Under consideration of these connections, Costello and Gwilliam describe a notion of classical field theories that solves the problem of finding the “right” complex of observables. Even in the case that $\mathfrak{F}$ is not a manifold, the information on its form is (implicitly) already contained in the way the field theory is expressed. This flexibility comes at the cost of a less intuitive definition of a classical field theory, which to explain requires some set-up.

Hence the goal of this section is to review the concepts we need to motivate and express the necessary constructions. Firstly, we will consider formal moduli problems. To do that, we need to start off by fixing some terminology.

Definition 3.1.1. Let k be a field. Let $\mathbf{CAlg}_k$ denote the category of commutative differential graded algebras over k . We call an element $A \in \mathbf{CAlg}_k$ with a specified morphism $A \rightarrow k$ an augmented commutative differential graded algebra. The category of these will be denoted $\mathbf{CAlg}_k^{\text{aug}} = \mathbf{CAlg}_k/k$.

Example 3.1.2. On any stalk $C^\infty(\mathfrak{F})_\varphi$, we have the evaluation at φ ,

$$\text{ev}_\varphi : C^\infty(\mathfrak{F}) \rightarrow \mathbb{R}.$$

Hence we can interpret for example $\mathfrak{S}(\mathfrak{F}), \mathfrak{T}(\mathfrak{F})$ and the similar differential graded algebras we encountered as augmented differential graded algebras.

To define formal moduli problems, however, we would prefer to have a subset of $\mathbf{CAlg}_k^{\text{aug}}$ that is “well-behaved”. This leads to the notion of small morphisms as in [11] (cf. Example 1.1.4 and Definition 1.1.5), which arise from a finite number of pull-backs along morphisms of the form $k \rightarrow k \oplus k[n]$.

Definition 3.1.3. The ∞ -category $\mathbf{CAlg}_k^{\text{sm}}$ is defined as the full sub- ∞ -category of $\mathbf{CAlg}_k^{\text{aug}}$ that is spanned by the objects with small augmentation morphisms.

Definition 3.1.4. Let k be a field. A formal moduli problem over k is a functor

$$F : \mathbf{CAlg}_k^{\text{sm}} \rightarrow \mathbf{An},$$

where $\mathbf{An}$ denotes the category of anima (or ∞ -groupoids), also known as the category of spaces. It has to satisfy the following properties:

- I) The Kan complex $F(k)$ (where k is augmented via the identity morphism) is contractible.

II) If $\phi : B' \rightarrow B$ is a small morphism and

$$\begin{array}{ccc} A' & \longrightarrow & B' \\ \downarrow & & \downarrow \phi \\ A & \longrightarrow & B \end{array}$$

is a pull-back diagram in $\mathbf{CAlg}_k^{\text{sm}}$, then the image

$$\begin{array}{ccc} F(A') & \longrightarrow & F(B') \\ \downarrow & & \downarrow F(\phi) \\ F(A) & \longrightarrow & F(B) \end{array}$$

is a pull-back diagram in $\mathbf{An}$.

The category of formal moduli problems will be denoted $\mathbf{FMP}$.

Remark. There is also the possibility of defining this functor on the level of categories instead of ∞ -categories. Then one has to take into account model structures, this definition is used for example in [5].

To understand why formal moduli problems are related with $\mathbf{CAlg}_k^{\text{aug}}$, not only with $\mathbf{CAlg}_k^{\text{sm}}$, the key construction is given through the following definition.

Definition 3.1.5. The functor

$$\text{Spec} : (\mathbf{CAlg}_k^{\text{aug}})^{\text{op}} \rightarrow \mathbf{FMP}$$

is defined to map any augmented commutative differential graded algebra A to the formal moduli problem given by

$$B \mapsto \text{Hom}_{\mathbf{CAlg}_k^{\text{aug}}}(A, B),$$

where we consider the simplicially enriched Hom-sets.

Remark. The name Spec comes of course from the notion of interpreting an (affine) scheme as a functor of points in algebraic geometry. However, it is usually regarded as a functor from $\mathbf{Schemes}$ or $\mathbf{AffSchemes}$ (in contrast to pointed schemes, which correspond to augmented algebras). In many cases, the functor of points in that context sends a ring or algebra to

$$B \mapsto \text{Hom}_{\mathbf{CAlg}_k}(A, B),$$

which subtly but importantly differs from the definition used here.

There is also another concept closely related to both commutative differential graded algebras and formal moduli problems: The concept of L_∞ -algebras. It is attributed to Stasheff, who introduced it under the name of strong homotopy Lie algebra.

Definition 3.1.6. (cf. [10]) An L_∞ -algebra $\mathcal{L}$ over a field k is given by an underlying graded k -vector space L_* together with graded anti-symmetric k -linear maps

$$l_i : L^{\otimes i} \rightarrow L$$

of degree $i - 2$ for $i \geq 1$. These maps have to satisfy that for homogeneous elements $x_i \in L$ for $1 \leq i \leq n$ it holds that

$$\sum_{S, S'} \varepsilon(S, S') (-1)^{d(S) \cdot d(S')} l_{d(S') + 1}(l_{d(S)}(x_{i_1}, \dots, x_{i_m}), x_{i_{m+1}}, \dots, x_{i_n}) = 0.$$

Here $S = (x_{i_1}, \dots, x_{i_m})$ and $S' = (x_{i_{m+1}}, \dots, x_{i_n})$ are ordered subsets of the ordered set $(x_1, \dots, x_n)$, such that each element is contained in exactly one of them (i.e. $S \cup S' = \{x_1, \dots, x_n\}$ as unordered sets).

The term $\varepsilon(S, S')$ in the above formula is given by

$$\varepsilon(S, S') = \prod_{\substack{i < j \text{ s.t.} \\ x_i \in S', x_j \in S}} (-1)^{\deg(x_i) \cdot \deg(x_j)}.$$

The function $d(S)$ that also occurs is given by the sum of the degrees of the $x_i \in S$ or $x_i \in S'$ respectively. The term l_0 will be defined as 0.

Spelling the equation of the previous definition out for $n = 1$, we obtain that l_1 is a map of degree -1 that satisfies $l_1^2 = 0$. So every L_∞ -algebra has an underlying graded vector space L_* that is a differential graded vector space together with l_1 .

There is a direct link between L_∞ -algebras over a field k (we will always consider $k = \mathbb{R}$ in this context and omit it from notation) and differential graded k -algebras. The construction arises from universal enveloping algebras (cf. [11], p. 37), but we will consider another explicit description here.

Definition 3.1.7. (cf. [11], p.37, p. 41) Let $\mathcal{L}$ be a L_∞ -algebra. Then the homological Chevalley-Eilenberg complex $C_*(\mathcal{L})$ is defined on the graded vector-space given by

$$C_n = \text{Sym}^n(\mathcal{L}[1]),$$

with differential given by

$$\begin{aligned}
& d(x_1 \wedge x_2 \wedge \cdots \wedge x_n) \\
&= \sum_{i=1}^n (-1)^{k(i)} (l_1(x_i)) \wedge x_1 \wedge \cdots \wedge x_{i-1} \wedge \widehat{x_i} \wedge \cdots \wedge x_n \\
&\quad + \sum_{1 \leq i < j \leq n} (-1)^{k(i,j)} (l_2(x_i, x_j)) \wedge x_1 \wedge \cdots \wedge x_{i-1} \wedge \widehat{x_i} \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge x_n \\
&\quad + \sum_{1 \leq i < j < l \leq n} (-1)^{k(i,j,l)} (l_3(x_i, x_j, x_l)) \wedge x_1 \wedge \cdots \wedge \widehat{x_i} \wedge \cdots \wedge \widehat{x_j} \wedge \cdots \wedge \widehat{x_l} \wedge \cdots \wedge x_n \\
&\quad + \dots,
\end{aligned}$$

where $\widehat{x_i}$ indicates the removal of x_i from the wedge product. The values of $k(i)$ and $k(i, j)$ are given by

$$k(i) = p_i \cdot (p_1 + \cdots + p_{(i-1)})$$

and

$$k(i, j) = p_i \cdot (p_1 + \cdots + p_{(i-1)}) + p_j \cdot (p_1 + \cdots + p_{(i-1)} + p_{(i+1)} + \cdots + p_n),$$

where this definition can be extended in an apparent way for $k(i, j, l)$ and for all higher numbers of indices.

Its dual $C^*(\mathfrak{g})$, the cohomological Chevalley-Eilenberg complex, is a differential graded algebra (where the differential has degree +1) with multiplication given for $\lambda \in C^p(\mathfrak{g})$ and $\mu \in C^q(\mathfrak{g})$ by

$$(\lambda \cdot \mu)(x_1 \wedge \cdots \wedge x_n) = \sum_{S, S'} \lambda(x_{i_1} \wedge \cdots \wedge x_{i_m}) \cdot \mu(x_{i_{m+1}} \wedge \cdots \wedge x_{i_n}),$$

where $S = (x_{i_1}, \dots, x_{i_m})$ and $S' = (x_{i_{m+1}}, \dots, x_{i_n})$ are ordered subsets of the ordered set $(x_1, \dots, x_n)$, such that each element is contained in exactly one of them (i.e. $S \cup S' = \{x_1, \dots, x_n\}$ as unordered sets). Further the degrees of the elements of S have to sum up to p and the degrees of the elements of S' have to sum up to q .

The term $\varepsilon(S, S')$ in the above formula is given by

$$\varepsilon(S, S') = \prod_{\substack{i < j \text{ s.t.} \\ x_i \in S', x_j \in S}} (-1)^{\deg(x_i) \cdot \deg(x_j)}.$$

There is also a notion of the Chevalley-Eilenberg-complexes when considering differential graded Lie algebras. But these notions are not conflicting. We have that every differential graded Lie algebra is especially a L_∞ -algebra, where the brackets l_i for $i \geq 3$ vanish (l_1 is given by the differential and l_2 is given by the bracket). The Chevalley-Eilenberg-complex of a differential

graded Lie algebra is constructed in the same way as that of an L_∞ -algebra, the vanishing brackets just allow us to simplify the construction.

The following theorem, which is formulated in the terminology of operads, states that also conversely, any L_∞ -algebra is quasi-isomorphic to some differential graded Lie algebra. As we will mainly be interested in the Chevalley-Eilenberg complex of our L_∞ -algebras up to homotopy, this means that we can already draw some conclusions based on the theory of differential graded Lie algebras.

Theorem 3.1.8. (*[9], p. 37, Corollary 1.6*) *Let k be a field of characteristic zero and let $\mathcal{J}$ be a strong homotopy Lie operad of differential graded k -modules. Then there is a functor W that assigns a quasi-isomorphic differential graded Lie algebra WL to a $\mathcal{J}$ -algebra L .*

When we keep this in mind, we can also see why the following theorem of Lurie about the equivalence of formal moduli problems and dg-Lie algebras might prove useful for our considerations, even if L_∞ -algebras do not directly appear. Costello and Gwilliam for example use it as a motivation to consider L_∞ -algebras instead of formal moduli problems without the risk of losing information about the field theory the problem represents (cf. [5], p. 26).

Theorem 3.1.9. (*cf. [11], Theorem 2.0.2*) *There is a functor*

$$\Psi : \text{Lie}_k \rightarrow \text{FMP},$$

that is an equivalence of ∞ -categories.

I will not attempt to prove this theorem or any of the propositions that we encounter until the end of this section. The proofs, originally by Lurie, can be found in [11] (they can also be found in [14], which reviews the more general theory of Lurie only with respect to the context we work in). However, the construction of the functor Ψ in Theorem 3.1.9 and some of its properties are of importance here, because they will show why the approaches to classical field theory presented in this thesis and [5] are closely related.

Proposition 3.1.10. (*cf. [11], p.42*) *The functor of ∞ -categories induced by the cohomological Chevalley-Eilenberg-complex*

$$C^* : \text{Lie}_k \rightarrow (\text{CAlg}_k^{\text{aug}})^{\text{op}}$$

admits a right adjoint $\mathfrak{D}$. It is called the Koszul duality functor.

Proposition 3.1.11. (*cf. [11], p. 29, 44*) *Let An denote the ∞ -category of anima (or ∞ -groupoids), also known as the ∞ -category of spaces. Let j denote the Yoneda embedding*

$$j : \text{Lie}_k \rightarrow \text{Hom}_{\text{Lie}_k}((\text{Lie}_k)^{\text{op}}, \text{An}).$$

Then a functor Ψ with the property of Theorem 3.1.9 can be constructed as follows:

$$\mathfrak{g} \mapsto j(\mathfrak{g}) \circ \mathfrak{D}.$$

More explicitly, it can be described by

$$\Psi(\mathfrak{g})(A) = \text{Hom}_{\text{Lie}_k}(\mathfrak{D}(A), \mathfrak{g}).$$

The inverse of the equivalence is given by its left adjoint Φ . It has the property that $\Phi(\text{Spec}(A)) = \mathfrak{D}(A)$.

Remark. The last statement is a little hidden in both [11] and [14]. It is contained in the proofs of the corresponding theorems. It also follows from the final theorem, by a few properties of both occurring adjunctions.

Remark. The functor Ψ arises from a physical context. It is modelled to express the formal moduli problem that arises from Maurer-Cartan elements of a differential graded Lie algebra (cf. [11], p.31).

Proposition 3.1.12. (cf. [11], Proposition 1.3.5) Assume $A \in \text{CAlg}_k^{sm}$ is a small augmented commutative differential graded algebra. Then the unit map $A \rightarrow C^*(\mathfrak{D}(A))$ (of the adjunction of C^* and $\mathfrak{D}$) is an equivalence in CAlg_k^{aug} .

3.2 The Perturbative Observables

Now we can finally study the definition given in [5] for a classical field theory together with its complex of observables. The notions are based on L_∞ -algebras and the Chevalley-Eilenberg complex construction. In this subsection, we will also find a relation between these definitions and our previous approach. They will be linked through an L_∞ -algebra with an underlying graded vector space that is in each degree equipped with a diffeological structure.

Definition 3.2.1. In the BV formalism, a perturbative classical field theory on a manifold M is a local elliptic L_∞ -algebra $\mathcal{L}$ together with an invariant pairing of degree 3. That $\mathcal{L}$ is local means in this context, that the underlying complex arises as the sections of a graded fiber bundle $L \rightarrow M$. That $\mathcal{L}$ is elliptic refers to the corresponding property on the underlying complex together with the differential (cf. [5], Definition 4.1.3.1).

That there is an invariant pairing means that there is, fiber-wise on L , a non-degenerate pairing. Further, when considered on $\mathcal{L}$ it pairs compactly supported sections to an integral over a density on M , with the property that the expression

$$\langle l_i(\alpha_1, \dots, \alpha_i), \alpha_{i+1} \rangle$$

is graded anti-symmetric (cf. [5], Definition 5.2.0.2).

Remark. When comparing this thesis to [5], please note that the grading conventions there are different than here.

Definition 3.2.2. The observables (of a perturbative classical field theory) with support in the open subset U is the commutative dg-algebra

$$\text{Obs}^{cl}(U) = C^*(\mathcal{L}(U)).$$

We will refer to them as the perturbative observables in this section.

The motivation behind this transformation from commutative algebra into L_∞ -algebra can be described by the following diagram:

$$\begin{array}{ccc} \text{CAlg}^{\text{op}} & & \\ \downarrow \text{Spec} & \searrow \mathfrak{D} & \\ \Phi : \text{Moduli} & \longleftrightarrow & \text{Lie} : \Psi \\ & & \downarrow C^* \\ & & \text{CAlg}^{\text{op}} \end{array}$$

We can assume the original observables to be a commutative differential graded algebra, as we have motivated by Example 1.1.2, that we can now turn into a differential graded Lie algebra via the functor $\mathfrak{D}$. To recover the original algebra, we might try to recover it by applying C^* again.

Unfortunately, outside of the circumstances described in Proposition 3.1.12, we may not assume that this returns a meaningful result. The problem here is, that the algebra $C^\infty(\mathfrak{F})_\varphi$ is not finite dimensional as a $\mathbb{R}$ -algebra.

This matter of infinite dimensionality poses a real problem when trying to find a L_∞ -algebra, whose cohomological Chevalley-Eilenberg complex is given by some model of the observables. Under these circumstances, it is not evident that even only $C^\infty(\mathfrak{F})_\varphi$ can be expressed as the dual to some other vector space.

The consideration of the formal moduli problems can give us some insight on how to resolve this conflict here. When considering the physical intuition of a perturbative classical field theory, we choose a base-point φ that is a solution of the given field equation. Then we expect to associate $\mathbb{R}[\varepsilon]/\varepsilon^k$ with an infinitesimal variation in our solution of our field equation, in the sense that we obtain a direction X such that any path in that direction parametrized by ε satisfies the field equations up to a term in the order of ε^k .

However, when we consider the formal moduli problem that arises from $\mathfrak{S}(\mathfrak{F})_\varphi$ or another of our constructions, we can actually see that the moduli problem we obtain through the Spec functor differs from this physical intuition.

Lemma 3.2.3. *The ring $\mathbb{R}[\varepsilon]/\varepsilon^2$ is mapped under $\text{Spec}(\mathfrak{S}(\mathfrak{F})_\varphi)$ to the simplicial set with 0-simplices*

$$\{\gamma \in \text{Der}(C^\infty(\mathfrak{F})_\varphi) \mid \gamma(\frac{\partial}{\partial X} S) = 0 \text{ for all } X \in \Gamma^\infty(\mathfrak{F}, T\mathfrak{F})\}.$$

Proof. By definition, the Spec functor takes $\mathbb{R}[\varepsilon]/\varepsilon^2$ to the set of morphisms

$$\begin{array}{ccccccc} \dots & \longleftarrow & 0 & \longleftarrow & C^\infty(\mathfrak{F}) & \longleftarrow & \Gamma^\infty(\mathfrak{F}, T\mathfrak{F}) & \longleftarrow & \dots \\ & & \downarrow & & \downarrow & & \downarrow & & \\ \dots & \longleftarrow & 0 & \longleftarrow & \mathbb{R} \oplus \mathbb{R} \cdot \varepsilon & \longleftarrow & 0 & \longleftarrow & \dots \end{array}$$

that composed with $\text{ev}_0 : \mathbb{R}[\varepsilon]/\varepsilon^2 \rightarrow \mathbb{R}$ recover the evaluation at φ .

We hence need a linear map $\gamma : C^\infty(\mathfrak{F})_\varphi \rightarrow \mathbb{R}$ that is mapping to the coefficient of ε . As it is supposed to be a morphism of commutative algebras, we obtain the explicit conditions

$$\text{ev}_\varphi(f \cdot g) = \text{ev}_\varphi(f) \cdot \text{ev}_\varphi(g),$$

which is trivial, and

$$\gamma(f \cdot g) = \text{ev}_\varphi(f) \cdot \gamma(g) + \gamma(f) \cdot \text{ev}_\varphi(g).$$

This, however, is equivalent to saying that γ is a derivation.

The only condition that is imposed on γ now is that the diagram from above commutes. So $\gamma \circ \text{vd}S$ needs to vanish, which is equivalent to the condition presented in the Lemma. $\square$

Remark. The proof just depended on degree 0 and 1 of $\mathfrak{S}$ and hence works exactly the same for $\mathfrak{T}$ or other complexes we might want to consider.

As in general, the algebraic derivations of $C^\infty(\mathfrak{F})$ do no longer agree with paths on $\mathfrak{F}$, this can be taken as a motivation that we should adjust our view of the arising concepts to the perturbative method that we want to use.

Proposition 3.2.4. *Assume we are considering a field theory given by a vector bundle $F \rightarrow M$. Recall the notational conventions of Definitions 2.4.3 and 2.4.7. Then both of the following complexes are local L_∞ -algebras that come with a pairing of degree 3:*

$\mathcal{L}_1$ given by

$$\dots \rightarrow 0 \rightarrow \Gamma_\varphi^\infty(M, VF) \longrightarrow \Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F) \rightarrow 0 \rightarrow \dots$$

as well as $\mathcal{L}_2$ given by

$$\dots \rightarrow 0 \rightarrow \Gamma_\varphi^\infty(M, TJ_m^\infty F) \longrightarrow \Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F) \rightarrow 0 \rightarrow \dots,$$

where the highest graded non-zero vector spaces lie in degree -1 and the l_i from the Definition 3.1.6 are given by

$$l_i(\psi_1, \dots, \psi_i) = \frac{\partial}{\partial \psi_1} \dots \frac{\partial}{\partial \psi_i} P \delta L.$$

The derivation here is meant with respect to the canonical identification of $T_j^* J^\infty F$ with $T_{j_m^*}^* J^\infty F$ for any given jet $j \in J^\infty F$.

Proof. We know that all maps l_i are of degree $i - 2$ and hence only can have a non-zero value when all of the inputs $v_1, \dots, v_i$ are of degree -1 , such that the degree of $l_i(v_1, \dots, v_i)$ is -2 . This means inserting the result in any l_j will cause the term to vanish. Hence the strong homotopy Jacobi identity is satisfied.

The pairing of an element X of degree -1 with an element A of degree -2 to a differential n -form on M can be given on both complexes at $m \in M$ simply by inserting the $X_m \in T_{j_m^*}^* J^\infty F$ induced by X into $A(m)$. $\square$

Remark. Note that only on $\mathcal{L}_2$ the pairing is induced by a pairing on the fibers, as the tangent vector X_m at $T_{j_m^*}^* J^\infty F$ induced by some tangent vector $X \in T_\varphi \mathfrak{F}$ is not only dependent on the value $X(m) \in VF$, but also on the stalk of the section. We can still check on both complexes that the only non-vanishing condition of invariance is given by

$$\langle l_i(\psi_1, \dots, \psi_i), \psi_{i+1} \rangle = \int_M \frac{\partial}{\partial \psi_1} \cdots \frac{\partial}{\partial \psi_i} \frac{\partial}{\partial \psi_{i+1}} L \Big|_\varphi$$

which is clearly graded anti-symmetric (i.e. symmetric in the ψ_j).

The fact that these L_∞ -algebras are not always elliptic means it is a priori not clear if they give rise to suitable physical models. By this, we refer to their ability to express the corresponding perturbative Lagrangian field theory from a physical perspective. To analyze this, we will for the rest of the section assume the bundle $F \rightarrow M$ to be a vector bundle. We can already find first hints that important notions of [5] still hold.

Lemma 3.2.5. *A free field theory, in the sense that the Euler-Lagrange equations are linear, corresponds under the construction of $\mathcal{L}_1$ to an Lie-algebra with vanishing Lie brackets, i.e. the conventional definition of a free field theory agrees with the notion given in [5] (cf. p.66).*

Proof. This is an immediate result, as the Euler-Lagrange equations are a way to reformulate $P\delta L$. To be precise, following [3] (cf. Definition 7.1.3 f.), the Euler-Lagrange equations are given by

$$E_\alpha = 0,$$

where E_α are the functions locally on $J^k F$ such that

$$P\delta L = E_\alpha \delta u^\alpha \wedge d_M x_1 \wedge \cdots \wedge d_M x_n,$$

where the x_i are local coordinates on M and u^α are the corresponding jet coordinates.

So the assumption is equivalent to $P\delta L$ being linear, such that all l_i for $i \geq 2$ vanish. $\square$

So we have now found two possible candidates $\mathcal{L}_1$ and $\mathcal{L}_2$ to give us back something related to our observables when constructing the Chevalley-Eilenberg complex. However, as the structures that we have are so closely tied to the diffeological structure, we would prefer to use the following construction.

Definition 3.2.6. Let k be a field with a diffeological structure. Let $\mathcal{L}$ be an L_∞ -algebra, where each on the vector spaces is equipped with a diffeological structure and all the l_i are smooth. Then we define the diffeological Chevalley-Eilenberg complex $C_{\text{diff}}^*(\mathcal{L})$ as the graded sub-vector-space of $C^*(\mathcal{L})$ given by

$$\{A : C_*(\mathcal{L}) \rightarrow \mathbb{R} \mid A \text{ is smooth}\},$$

where the diffeological structure on C_* is induced by the diffeologies on each vector space. This is in the sense that A is smooth if and if it is smooth on every component of the direct sums occurring in C_* .

Lemma 3.2.7. *The differential graded algebra structure of $C^*(\mathcal{L})$ induces a differential graded algebra structure on $C_{\text{diff}}^*(\mathcal{L})$.*

Proof. What is to show here is that both the differential and the multiplication map smooth maps to smooth maps. But the differential on C^* comes from the differential on C_* , which is smooth as all the l_i are smooth. That the product of smooth maps is smooth follows directly from its definition as the sum and product of smooth maps. $\square$

The complex C_{diff}^* still has a minor detail that does not match our “complex of observables”: It is a cohomological complex with a differential of degree $+1$. But there is an easy solution for that.

Definition 3.2.8. We define $\overline{C}_*^{\text{diff}}$ as the differential graded algebra (with differential of degree -1) that arises from C_{diff}^* by reversing the grading.

We will in the following only analyze the perturbative observables of $\mathcal{L}_1$. When carefully considering the proofs that follow, we can see that all of the constructions that we go through for $F \rightarrow M$ together with $\mathfrak{F}$ can also be considered with respect to $J^\infty F \rightarrow M$ and $\mathfrak{J} = \Gamma^\infty(M, J^\infty F)$.

Theorem 3.2.9. *Let $F \rightarrow M$ be a vector bundle. There is a natural projection from the complexes $\mathfrak{S}^1(\mathfrak{F})$ and $\mathfrak{T}^1(\mathfrak{F})$ as defined in definition 2.4.8 to the diffeological Chevalley-Eilenberg complex that arises from the first L_∞ algebra $\mathcal{L}_1$, when reversing the grading to $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$.*

Proof. The proof, although lengthy, is actually very straightforward. We will first define the projection, which arises by sending a function to their derivatives at φ and then check, that it is a morphism of differential graded

algebras, i.e. that differential and algebra structure are compatible with the given map.

To define the projection, let us begin with analyzing the $C_{\text{diff}}^0(\mathcal{L}_1)$. It is given by smooth maps from

$$\mathbb{R} \oplus \Gamma_{\varphi}^{\infty}(M, VF) \oplus \Gamma_{\varphi}^{\infty}(M, VF) \wedge \Gamma_{\varphi}^{\infty}(M, VF) \oplus \dots$$

to $\mathbb{R}$. However, every smooth function $f \in C^{\infty}(\mathfrak{F})_{\varphi}$ gives us such a map that acts on the component of $\mathbb{R}$ by

$$\begin{aligned} \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto f(0) \cdot x, \end{aligned}$$

on the component of $\Gamma_{\varphi}^{\infty}(M, VF)$ by

$$\begin{aligned} \Gamma_{\varphi}^{\infty}(M, VF) &\rightarrow \mathbb{R} \\ X &\mapsto X(f) = \frac{\partial}{\partial X} f, \end{aligned}$$

as well as by

$$\begin{aligned} \Gamma_{\varphi}^{\infty}(M, VF) \wedge \Gamma_{\varphi}^{\infty}(M, VF) &\rightarrow \mathbb{R} \\ X_1 \wedge X_2 &\mapsto \frac{\partial}{\partial X_1} \frac{\partial}{\partial X_2} f, \end{aligned}$$

where the second derivative is defined due to the vector bundle structure of F , which gives us a flat connection on $VF_m = T(F_m) \rightarrow F_m$. Also the map is well defined on $\Gamma_{\varphi}^{\infty}(M, VF) \wedge \Gamma_{\varphi}^{\infty}(M, VF)$ as the wedge product in degree 0 induces the relation $X_1 \wedge X_2 = X_2 \wedge X_1$ and the term $\frac{\partial}{\partial X_1} \frac{\partial}{\partial X_2} f$ is symmetric due to the flatness of the connection.

Quite similarly, we get a map on all higher wedge products of $\Gamma_{\varphi}^{\infty}(M, VF)$.

When analyzing $C_{\text{diff}}^{-1}(\mathcal{L}_1) = \overline{C}_1^{\text{diff}}(\mathcal{L}_1)$ we can observe a similar phenomenon. As

$$C_{-1}(\mathcal{L}_1) = \Gamma^{\infty}(M, \wedge^n(T^*M) \otimes T^*J_m^{\infty}F) \otimes C_0(\mathcal{L}_1),$$

we have that any $A \in \Gamma^{\infty}(\mathfrak{F}, (\Gamma^{\infty}(M, \wedge^n(T^*M) \otimes T^*J_m^{\infty}F))^*)_{\varphi}$ together with a value $x \in \Gamma^{\infty}(M, \wedge^n(T^*M) \otimes T^*J_m^{\infty}F)$ gives us a function on $\mathfrak{F}$ by inserting the constant section

$$\text{const}_x \in \Gamma^{\infty}(\mathfrak{F}, \Gamma^{\infty}(M, \wedge^n(T^*M) \otimes T^*J_m^{\infty}F))$$

into A , where constant is referred to here with respect to the vector bundle structure of F giving us flat connections on the bundles $TJ_m^{\infty}F \rightarrow J_m^{\infty}F$ and $T^*J_m^{\infty}F \rightarrow J_m^{\infty}F$. Now we can analyze $A(\text{const}_x)$ as before.

For every other $C_{\text{diff}}^{-n}(\mathcal{L}_1) = \overline{C}_n^{\text{diff}}(\mathcal{L}_1)$ with $n > 1$ we can take quite a similar course of action, as for every $x_1 \wedge \cdots \wedge x_n$ we can now create $\text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_n}$. For any element $A \in \mathfrak{T}_n^1(\mathfrak{F})$, we then can evaluate $A(\text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_n}) \in C^\infty(\mathfrak{F})$. But $\mathfrak{S}(\mathfrak{F})$ is a sub-algebra of $\mathfrak{T}(\mathfrak{F})$, where for $A_1 \wedge A_2 \in \mathfrak{S}_2^1(\mathfrak{F})$ the function can be explicitly determined to be given by

$$A_1(\text{const}_{x_1})A_2(\text{const}_{x_2}) - A_1(\text{const}_{x_2})A_2(\text{const}_{x_1}),$$

and in the same fashion for more values x_i .

So we have found natural maps of the vector spaces of the chain complexes between $\mathfrak{S}_*^1$ and $\overline{C}_*^{\text{diff}}$ as well as $\mathfrak{T}_*^1$ and $\overline{C}_*^{\text{diff}}$. It remains to be seen that these maps are compatible with the differential and the product structure. We will show the explicit computations here for $\mathfrak{T}^1(\mathfrak{F})$, because, as we have already stated, $\mathfrak{S}^1(\mathfrak{F})$ is only a sub-algebra of $\mathfrak{T}^1(\mathfrak{F})$.

Let us now analyze the differentials. Assume that we are given an element $\lambda_A \in \overline{C}_a^{\text{diff}}$ which is the image of some $A \in \mathfrak{T}_a^1(\mathfrak{F})$. Firstly, consider the differential on $\mathfrak{T}^1(\mathfrak{F})$. We know it is given by $\iota_{(P\delta L)}$.

If we now are given any collection of $x_j \in \Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m}^* \varphi J_m^\infty F)$ for $1 \leq j \leq (a-1)$, we can recover how the image of $\iota_{(P\delta L)}(A)$ acts on the x_j by analysis of the function

$$f_A = A(P\delta L \wedge \text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_{(a-1)}}).$$

We know that for this function holds

$$f_A(\varphi) = 0,$$

as φ is an element of the derived critical locus and thus has to satisfy $P\delta L(\varphi) = 0$. Also because of the standard properties of differentiation and the fiber-wise linearity of A , we find that

$$\begin{aligned} & \frac{\partial}{\partial \psi_1} \cdots \frac{\partial}{\partial \psi_k} f_A|_\varphi \\ &= \sum_{I \cup J = \{1, \dots, k\}} \left(\frac{\partial^{|I|}}{\partial \psi_I} A \right) \Big|_\varphi \left(\frac{\partial^{|J|}}{\partial \psi_J} (P\delta L \wedge \text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_{(a-1)}}) \right) \Big|_\varphi. \end{aligned}$$

We can actually omit $J = \emptyset$ from the sum as we know $P\delta L(\varphi) = 0$, similar to the previous argument.

But as all the terms apart from $P\delta L$ in the second expression are constant, we can express

$$\left(\frac{\partial^{|J|}}{\partial \psi_J} (P\delta L \wedge \text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_{(a-1)}}) \right) \Big|_\varphi$$

simply as

$$y_J \wedge x_1 \wedge \cdots \wedge x_{(a-1)}$$

for

$$y_J = \left(\frac{\partial^{|J|}}{\partial \psi_J} P\delta L|_\varphi \right).$$

We can now write

$$\begin{aligned} & \frac{\partial}{\partial \psi_1} \cdots \frac{\partial}{\partial \psi_k} f_A|_\varphi \\ &= \sum_{I \cup J = \{1, \dots, k\}} \left(\frac{\partial^{|I|}}{\partial \psi_I} A \right) \Big|_\varphi (y_J \wedge x_1 \wedge \cdots \wedge x_{(a-1)})|_\varphi \\ &= \sum_{I \cup J = \{1, \dots, k\}} \frac{\partial^{|I|}}{\partial \psi_I} \left(A(\text{const}_{y_J} \wedge \text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_{(a-1)}}) \right) \Big|_\varphi \\ &= \sum_{I \cup J = \{1, \dots, k\}} \frac{\partial^{|I|}}{\partial \psi_I} (A(y_J \wedge x_1 \wedge \cdots \wedge x_{(a-1)})) \Big|_\varphi, \end{aligned}$$

where the last line arises similar to the first computation, as all derivatives of the constant section

$$(\text{const}_{y_J} \wedge \text{const}_{x_1} \wedge \cdots \wedge \text{const}_{x_{n-1}})$$

vanish.

So we have found an expression of the derivatives of $d_{\mathfrak{T}}A = \iota_{(P\delta L)}A$, when we evaluate it at constant sections given by the x_j in tangent directions given by the ψ_i . But this is exactly the information that constitutes the image of $d_{\mathfrak{T}}A$ in $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$. So we can now hope to compare this expression to how the differential on $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$ acts.

But if we want to analyze the differential on $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$, we need to analyze the differential on $C_*(\mathcal{L}_1)$ first. There we find that

$$d(x_1 \wedge \cdots \wedge x_{(a-1)} \otimes \psi_1 \wedge \cdots \wedge \psi_k) = (-1)^{(a-1)} \cdot x_1 \wedge \cdots \wedge x_{(a-1)} \otimes d(\psi_1 \wedge \cdots \wedge \psi_k).$$

This is due to the fact that the x_j come from degree -2 of the L_∞ -algebra, but every l_i vanishes if one of the (-2) -degree elements is inserted (they have to for reasons of degree).

We have that

$$\begin{aligned}
& d(\psi_1 \wedge \cdots \wedge \psi_k) \\
&= \sum_{1 \leq i \leq k} d(\psi_i) \wedge \psi_1 \wedge \cdots \wedge \widehat{\psi_i} \wedge \cdots \wedge \psi_k \\
&\quad + \sum_{1 \leq i < j \leq k} l_2(\psi_i, \psi_j) \wedge \psi_1 \wedge \cdots \wedge \widehat{\psi_i} \wedge \cdots \wedge \widehat{\psi_j} \wedge \cdots \wedge \psi_k \\
&\quad + \sum_{1 \leq i < j < l \leq k} l_3(\psi_i, \psi_j, \psi_k) \wedge \psi_1 \wedge \cdots \wedge \widehat{\psi_i} \wedge \cdots \wedge \widehat{\psi_j} \wedge \cdots \wedge \widehat{\psi_l} \wedge \cdots \wedge \psi_k \\
&\quad + \dots,
\end{aligned}$$

as the factors arising from the grading vanish.

If we consider the symmetries of all the terms, we can rewrite this sum in a way that suits us better:

$$d(\psi_1 \wedge \cdots \wedge \psi_k) = \sum_{\emptyset \neq J \subset \{1, \dots, k\}} \left(\frac{\partial^{|J|}}{\partial \psi_J} P \delta L \right) \wedge \psi_1 \wedge \cdots \wedge \widehat{\psi_J} \wedge \cdots \wedge \psi_k,$$

where $\widehat{\psi_J}$ is supposed to indicate that we omit all ψ_i with $i \in J$.

So when we want to compute the differential of λ_A , which was defined as the image of A , we obtain that

$$\begin{aligned}
& d\lambda_A(x_1 \wedge \cdots \wedge x_{(a-1)} \otimes \psi_1 \wedge \cdots \wedge \psi_k) \\
&= \sum_{\emptyset \neq J \subset \{1, \dots, k\}} (-1)^{(a-1)} \cdot \lambda_A(x_1 \wedge \cdots \wedge x_{(a-1)} \wedge \left(\frac{\partial^{|J|}}{\partial \psi_J} P \delta L \right) \otimes \psi_1 \wedge \cdots \\
&\quad \cdots \wedge \widehat{\psi_J} \wedge \cdots \wedge \psi_k) \\
&= \sum_{\emptyset \neq J \subset \{1, \dots, k\}} \lambda_A \left(\left(\frac{\partial^{|J|}}{\partial \psi_J} P \delta L \right) \wedge x_1 \wedge \cdots \wedge x_{(a-1)} \otimes \psi_1 \wedge \cdots \wedge \widehat{\psi_J} \wedge \cdots \wedge \psi_k \right) \\
&= \sum_{\emptyset \neq J \subset \{1, \dots, k\}} \lambda_A(y_J \wedge x_1 \wedge \cdots \wedge x_{(a-1)} \otimes \psi_1 \wedge \cdots \wedge \widehat{\psi_J} \wedge \cdots \wedge \psi_k).
\end{aligned}$$

This, however, is exactly the same as

$$\sum_{I \cup J = \{1, \dots, k\}, J \neq \emptyset} \frac{\partial^{|I|}}{\partial \psi_I} (A(y_J, x_1, \dots, x_{(a-1)}))|_{\varphi}$$

and as we have already noted that the condition $J \neq \emptyset$ is redundant in this term, we have now found that $d\lambda_A$ acts the same as the image of $\iota_{(P\delta L(A))}$. Hence the map we have created at the beginning is a morphism of chain complexes.

To check that it is also compatible with the multiplication given on both chain complexes is only a matter of combinatorial rearrangements and application of the product rule as well. Assume we are given $A \in \mathfrak{T}^1(\mathfrak{F})$ and $B \in \mathfrak{T}^1(\mathfrak{F})$, with images λ_A and λ_B . We know by definition that

$$\begin{aligned}
& (\lambda_A \cdot \lambda_B)(x_1 \wedge \cdots \wedge x_a \otimes \psi_1 \wedge \cdots \wedge \psi_k) \\
&= \sum_{S, S'} \varepsilon(S, S') \lambda_A(x_{i_1} \wedge \cdots \wedge x_{i_p} \otimes \psi_{j_1} \wedge \cdots \wedge \psi_{j_q}) \\
&\quad \cdot \lambda_B(x_{i_{(p+1)}} \wedge \cdots \wedge x_{i_a} \otimes \psi_{j_{(q+1)}} \wedge \cdots \wedge \psi_{j_k}) \\
&= \sum_{S, S'} \varepsilon(S, S') \frac{\partial}{\partial \psi_{j_1}} \cdots \frac{\partial}{\partial \psi_{j_q}} (A(x_{i_1} \wedge \cdots \wedge x_{i_p})) \Big|_{\varphi} \\
&\quad \cdot \frac{\partial}{\partial \psi_{j_{q+1}}} \cdots \frac{\partial}{\partial \psi_{j_k}} (B(x_{i_{(p+1)}} \wedge \cdots \wedge x_{i_a})) \Big|_{\varphi},
\end{aligned}$$

where S and S' are chosen as in Definition 3.1.6. We know that $\varepsilon(S, S')$ does not depend on the distribution of the ψ_j , as they are of degree 0. If we now apply the product rule, we can write the same term also as

$$\frac{\partial}{\partial \psi_1} \cdots \frac{\partial}{\partial \psi_k} \left(\sum_{T, T'} \varepsilon(T, T') A(x_{i_1} \wedge \cdots \wedge x_{i_p}) \cdot B(x_{i_{(p+1)}} \wedge \cdots \wedge x_{i_a}) \right) \Big|_{\varphi}$$

Now T and T' only refer to ordered subsets of the ordered set $(x_1, \dots, x_a)$, such that each element is contained in exactly one of them.

But this term, by definition, agrees exactly with the term

$$\frac{\partial}{\partial \psi_1} \cdots \frac{\partial}{\partial \psi_k} ((A \cdot B)(x_1 \wedge \cdots \wedge x_a)) \Big|_{\varphi},$$

i.e. with the action of the image of $(A \cdot B)$ on $x_1 \wedge \cdots \wedge x_a \otimes \psi_1 \wedge \cdots \wedge \psi_k$.

Hence we have found that indeed, the maps that we created are morphisms of commutative differential graded algebras, which was what we intended to prove. $\square$

The map of complexes that we previously defined is apparently not injective: For example, in degree 0 the image of $f \in C^\infty(\mathfrak{F})$ only depends on the value of f and derivatives of f at φ . So a (compactly supported) functional that vanishes at φ and takes values in the order of $e^{-\frac{1}{x^2}}$ around every $\varphi(m)$ integrates to a smooth function with vanishing derivative, that is non-zero in any open neighborhood of φ . It also is an interesting question if the map is surjective for $\mathfrak{T}^1(\mathfrak{F})$, but at least we have a slightly weaker result.

Lemma 3.2.10. *Assume we are given a smooth and linear map*

$$\lambda : \wedge^l \left(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi^* J_m^\infty F) \right) \otimes V \rightarrow \mathbb{R}.$$

Here,

$$V = (\mathbb{R} \oplus \Gamma_\varphi^\infty(M, VF) \oplus \Gamma_\varphi^\infty(M, VF) \wedge \Gamma_\varphi^\infty(M, VF) \oplus \dots),$$

where the finite sum ends after $\wedge^m(\Gamma_\varphi^\infty(M, VF))$ for some $m \in \mathbb{N}$. Then λ agrees with the action of some element $A \in \mathfrak{T}_l^1(\mathfrak{F})$, which is denoting the vector space in degree l of that complex.

Proof. We can simply define

$$A \in \Gamma^\infty(\mathfrak{F}, (\wedge^l(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F)))^*)$$

by its action at ψ on $x \in \wedge^l(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \psi}^* J_m^\infty F))$. To that aim we can identify ψ with the tangent vector

$$p : t \mapsto \varphi + t \cdot (\psi - \varphi)$$

and x with an element $\bar{x} \in \wedge^l(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))$ at φ by the natural identification of $T_{j_m^\infty \psi}^* J_m^\infty F$ and $T_{j_m^\infty \varphi}^* J_m^\infty F$. Then we can set

$$A(\psi)(x) = \lambda(\bar{x} + \bar{x} \otimes p + \dots + \frac{1}{k!} \bar{x} \otimes p \wedge \dots \wedge p),$$

where the last wedge product contains k occurrences of p .

We trivially have that $A(\varphi)(x) = \lambda(x)$, which ensures that the action of A and λ on

$$\wedge^l(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F))$$

(i.e. the first summand of our domain) agrees. Also we know that A is smooth, as λ is smooth, the identification $x \mapsto \bar{x}$ is smooth and the identification $\psi \mapsto p$ is smooth.

When deriving this term in any direction ρ , we have that

$$\begin{aligned} A(\psi + \varepsilon \rho)(x) = & \lambda \left(\bar{x} + \bar{x} \otimes p + \varepsilon \cdot \bar{x} \otimes p' + \dots + \frac{1}{k!} \bar{x} \otimes p \wedge \dots \wedge p \right. \\ & \left. + \frac{\varepsilon \cdot 1}{(k-1)!} \bar{x} \otimes p \wedge \dots \wedge p \wedge p' + \varepsilon^2 \cdot R \right). \end{aligned}$$

Here p' is referring to the tangent vector that corresponds to ρ . Hence we have that

$$\left(\frac{\partial}{\partial \rho} A \right) (\psi)(x) = \lambda(\bar{x} \otimes p' + \dots + \frac{1}{(k-1)!} \bar{x} \otimes p \wedge \dots \wedge p \wedge p'),$$

where the last term now is only $(k-1)$ times containing p .

Evaluating this term for $p = 0$ and hence $\psi = \varphi$ only leaves us with $\lambda(x \wedge p')$, such that indeed

$$\frac{\partial}{\partial \rho} (A(\text{const}_x))|_0 = \lambda(x \wedge p'),$$

such that the action of A and λ agree also on

$$\wedge^l \left(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F) \right) \otimes \Gamma_\varphi^\infty(M, VF).$$

With the same argument we can explicitly compute the derivatives up to the order of k (all higher derivatives vanish) to agree with the action of λ as well, proving the lemma. $\square$

This leads us now to a new way to formulate the perturbative observables purely in terms of $\mathfrak{T}^1(\mathfrak{F})$. We will discuss the benefits of this shortly afterwards.

Proposition 3.2.11. *The complex $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$ is isomorphic to the limit*

$$(\mathfrak{T}^1(\mathfrak{F}) / \sim_1) \leftarrow (\mathfrak{T}^1(\mathfrak{F}) / \sim_2) \leftarrow (\mathfrak{T}^1(\mathfrak{F}) / \sim_3) \leftarrow \dots,$$

in the category cdgAlg of commutative differential graded $\mathbb{R}$ -algebras. The equivalence relation $\sim_k$ on $\mathfrak{T}^1(\mathfrak{F})$ for any $k \in \mathbb{N}$ is imposed by $A \sim_k B \in \mathfrak{T}_l^1(\mathfrak{F})$ if and only if for all $d : \mathbb{R}^l \rightarrow \mathfrak{F}$ smooth with $f(0) = \varphi$ we have that $A \circ d$ and $B \circ d$ agree at 0 up to the k -th derivative, when evaluated at some smooth $\omega \in \Gamma^\infty(\mathfrak{F}, \wedge^l(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F)))$.

Proof. For this proof, we will evaluate the projection of $\mathfrak{T}^1(\mathfrak{F})$ to $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$ and find that every element of $\mathfrak{T}^1 / \sim_k$ has a well-defined image in $\overline{C}_*^{\text{diff}} / \sim_{\pi_k}$, where this relation identifies elements in $\overline{C}_*^{\text{diff}}$ with their action on the first k factors of the direct sum occurring in $C_*(\mathcal{L}_1)$. Under application of the previous lemma, we will then show that this is a bijective correspondence, which then yields the claim.

To begin, let us note that $\overline{C}_*^{\text{diff}}$ is isomorphic to the limit of

$$\overline{C}_*^{\text{diff}} / \sim_{\pi_1} \leftarrow \overline{C}_*^{\text{diff}} / \sim_{\pi_2} \leftarrow \overline{C}_*^{\text{diff}} / \sim_{\pi_3} \leftarrow \dots$$

The maps π_i denote the natural projections that identify the elements of $\overline{C}_*^{\text{diff}}(\mathcal{L}_1)$ in each degree m with their action on

$$\wedge^m \left(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F) \right) \otimes V^i,$$

for

$$V^i = (\mathbb{R} \oplus \Gamma_\varphi^\infty(M, VF) \oplus \dots \oplus \wedge^i(\Gamma_\varphi^\infty(M, VF))).$$

To take this limit in cdgAlg , we must at first verify that $\overline{C}_*^{\text{diff}} / \sim_{\pi_i}$ still has the structure of a commutative differential graded algebra. But the multiplication of $\overline{C}_*^{\text{diff}}$ is defined in such a way that $\mu \cdot \lambda$, when evaluated at $x_1 \wedge \dots \wedge x_j$ only depends on the values of μ and λ at terms of the form $x_{i_1} \wedge \dots \wedge x_{i_h}$, where $(x_{i_1}, \dots, x_{i_h})$ is a subset of $(x_1, \dots, x_j)$. Hence multiplication on $\overline{C}_*^{\text{diff}}$ induces a multiplication on $\overline{C}_*^{\text{diff}} / \sim_{\pi_i}$.

That the limit indeed agrees with $\overline{C}_*^{\text{diff}}$ holds as every element of $C_*(\mathcal{L}_1)$ in each degree is a sum of only finitely many non-zero elements. This implies that the action on $C_m(\mathcal{L}_1)$ that is given by an element of the limit of the $\overline{C}_*^{\text{diff}} / \sim_{\pi_i}$ in degree m is well-defined and even smooth. So every element in the limit gives us an element in $\overline{C}_*^{\text{diff}}$, which in turn gives can be naturally projected to the $\overline{C}_*^{\text{diff}} / \sim_{\pi_i}$ and hence to the limit, where both constructions are inverse to each other.

By Proposition 3.2.10, we know that any element of $\overline{C}_m^{\text{diff}} / \sim_{\pi_k}$ can be represented by an element of $\mathfrak{T}_m^1(\mathfrak{F})$. It remains to be shown that the relation imposed on $\mathfrak{T}^1(\mathfrak{F})$ arising from the projection to $\overline{C}^{\text{diff}} / \sim_{\pi_k}$ is the same relation as described in the proposition.

At first, let us show that the projection from $\mathfrak{T}_m^1(\mathfrak{F}) / \sim_k$ to $\overline{C}_m^{\text{diff}} / \sim_{\pi_k}$ is still well-defined. Assume that we are given two elements $A, B \in \mathfrak{T}^1(\mathfrak{F})$ that agree under the given relation $\sim_k$. Then we especially have that for any $\psi_1, \dots, \psi_l$ with $l \leq k$ we can consider the affine map of vector spaces

$$\begin{aligned} \mathbb{R}^l &\rightarrow \mathfrak{F} \\ e_i &\mapsto \varphi + \psi_i. \end{aligned}$$

Here we have that differentiating a function in direction x_i on $\mathbb{R}^l$ is the same as differentiating it in direction ψ_i on $\mathfrak{F}$ at the given point. So knowing that $A \sim_k B$ under the given relation implies that

$$\frac{\partial}{\partial \psi_1} \dots \frac{\partial}{\partial \psi_l} A(y_1 \wedge \dots \wedge y_m)|_{\varphi} = \frac{\partial}{\partial \psi_1} \dots \frac{\partial}{\partial \psi_l} B(y_1 \wedge \dots \wedge y_m)|_{\varphi},$$

as we can compute both as

$$\frac{\partial}{\partial x_1} \dots \frac{\partial}{\partial x_l} A(y_1 \wedge \dots \wedge y_m)|_0 = \frac{\partial}{\partial x_1} \dots \frac{\partial}{\partial x_l} B(y_1 \wedge \dots \wedge y_m)|_0.$$

So we have that their images λ_A and λ_B also agree under the relation $\sim_{\pi_k}$.

So the projection is well-defined as a map, but we have also claimed that $\mathfrak{T}^1(\mathfrak{F}) / \sim_k$ is a commutative differential graded algebra and would hope for the induced morphism to be an algebras morphism. Therefore, we would like to show that the algebra structure of $\mathfrak{T}^1(\mathfrak{F})$ induces an algebra structure on $\mathfrak{T}^1(\mathfrak{F}) / \sim_k$. Then the fact that the projection is an algebra morphism between $\mathfrak{T}^1(\mathfrak{F})$ and $\overline{C}_*^{\text{diff}}$ would immediately imply that the morphism it induces is one as well.

But when we look at multiplication on $\mathfrak{T}^1(\mathfrak{F})$, we can analyze the evaluation of $A \cdot B$ at any smooth $\omega \in \Gamma^\infty(\mathfrak{F}, \wedge^l(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty}^* \varphi J_m^\infty F)))$. By definition, it arises as a sum of terms of the form $A(\omega_1) \cdot B(\omega_2)$, for some ω_1 and ω_2 with $\omega_1 \wedge \omega_2 = \omega$. The value and derivatives at 0 up to order k that we obtain through some $d : \mathbb{R}^l \rightarrow \mathfrak{F}$ can thus be determined by the product rule. They only depend on the value and derivatives at 0 up to

order k of A and B , when evaluated at ω_1 and ω_2 respectively. So we have that $\mathfrak{T}^1(\mathfrak{F})/\sim_k$ actually possesses an algebra structure.

This means that indeed the projection induces a morphism of commutative differential graded algebras between $\mathfrak{T}_m^1(\mathfrak{F})/\sim_k$ and $\overline{C}_m^{\text{diff}}/\sim_{\pi_k}$. We still need to show that the image of $A \in \mathfrak{T}_m^1(\mathfrak{F})/\sim_k$ under the projection to $\overline{C}_m^{\text{diff}}/\sim_{\pi_k}$ is unique.

So assume we are given two elements A and B such that $\lambda_A \sim_{\pi_k} \lambda_B$ and a smooth map $d : \mathbb{R}^l \rightarrow \mathfrak{F}$ with $f(0) = \varphi$. Because of the properties of partial derivatives, it is sufficient to prove here that when evaluated at $\omega \in \Gamma^\infty(\mathfrak{F}, \wedge^m(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^\infty \varphi}^* J_m^\infty F)))$, we have that

$$\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_k}} A(\omega) = \frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_k}} B(\omega).$$

We will now present an argument why this holds in the case that $A = f$ and $B = g$ with $f, g \in \mathfrak{T}_0^1(\mathfrak{F}) = C^\infty(\mathfrak{F})$. The argument for the more general case follows afterwards. It depends on decomposing the term above such that instead of ω and $\frac{\partial}{\partial x_{i_j}}$ the expressions contain the terms of the form $\frac{\partial}{\partial \rho_j}$ for some constant vector fields ρ_j and ω' , which is constant with respect to the connection on $T^* J_m^\infty F$.

At every point $z \in \mathbb{R}^l$ we get a map

$$\begin{aligned} T_z \mathbb{R} &\rightarrow T_{d(z)} \mathfrak{F} \\ \frac{\partial}{\partial x_j} &\mapsto \psi_j(z). \end{aligned}$$

By analyzing the evaluations of the tangent vectors at all points $m \in M$ and using the identification of all $T_{d(z)} \mathfrak{F}$ with $T_\varphi \mathfrak{F}$, we can then write

$$\psi_j(z) = \psi_j^{(0, \dots, 0)} + z_1 \cdot \psi_j^{(1, 0, \dots, 0)} + z_2 \cdot \psi_j^{(0, 1, 0, \dots, 0)} + \dots + z_l^k \cdot \psi_j^{(0, \dots, 0, k)} + R,$$

where the sum runs over all multi-indices of combined degree up to k and the $\psi_j^{(i_1, \dots, i_l)}$ are constant with respect to the identification. The term R is of the form

$$R = \sum_{|(i_1, \dots, i_l)|=k+1} z_1^{i_1} \cdot z_2^{i_2} \cdot \dots \cdot \psi_j^{(i_1, \dots, i_l)}(z),$$

where the components $\psi_j^{(i_1, \dots, i_l)}(z)$ now explicitly depend on z .

This helps us in so far, as for any function $f \in C^\infty(\mathfrak{F})$ we have that

$$\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_{(k-1)}}} \frac{\partial}{\partial x_j} f|_0 = \frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_{(k-1)}}} \frac{\partial}{\partial (\psi_j - R)} f|_0$$

as

$$\frac{\partial}{\partial x_{i_1}} \cdots \frac{\partial}{\partial x_{i_{(k-1)}}} \frac{\partial}{\partial R} f|_0$$

by the product rule.

When applying the product rule on the summands of $\psi_j - R$ we are then able to rephrase the term even further, as we can directly compute the derivatives of the polynomials in the coordinates. The term we find depends on the values and derivatives up to order $k - 1$ at 0 of all terms of the form

$$\frac{\partial}{\partial \psi_j^{(m_1, \dots, m_k)}} f$$

for any multi-index $(m_1, \dots, m_k)$ with $|(m_1, \dots, m_k)| \leq k$.

By iterating, we can hence express

$$\frac{\partial}{\partial x_{i_1}} \dots \frac{\partial}{\partial x_{i_k}} f|_0$$

through terms of the form

$$\left(\frac{\partial}{\partial \rho_1} \dots \frac{\partial}{\partial \rho_k} f \right) (0) = \frac{\partial}{\partial \rho_1} \dots \frac{\partial}{\partial \rho_k} f|_\varphi,$$

where the ρ are constant vector fields with respect to the identification of tangent spaces. So we already know for $f, g \in C^\infty(\mathfrak{F})$ that if their images $\lambda_f, \lambda_g \in \overline{C}_0^{\text{diff}} / \sim_{\pi_k}$ agree, already $f \sim_k g$.

For general $A, B \in \mathfrak{T}^1(\mathfrak{F})$, we can in a similar fashion also analyze the section $\omega \in \Gamma^\infty(\mathfrak{F}, \wedge^m(\Gamma^\infty(M, \wedge^n(T^*M) \otimes T_{j_m^*}^* J_m^\infty F)))$ we are evaluating at. Along the vectors ρ_j that are relevant for every expression, we can express ω as

$$\omega(\varphi + \sum t_i \cdot \rho_i) = \omega^{(0, \dots, 0)} + t_1 \cdot \omega^{(1, 0, \dots, 0)} + t_2 \cdot \omega^{(0, 1, 0, \dots, 0)} + \dots + t_l^k \cdot \omega^{(0, \dots, 0, k)} + R'$$

for coefficients that are constant sections. We find by using the point-wise linearity of A and B that the derivatives up to order k of $A(\omega)$ and $B(\omega)$ do not depend on R' .

Now we have reduced

$$\frac{\partial}{\partial x_{i_1}} \dots \frac{\partial}{\partial x_{i_k}} A(w)|_0$$

to an expression that is only dependent on values and derivatives with respect to constant vector fields of $A(\omega')$ or $B(\omega')$ for sections ω' that are constant (with respect to the identification of tangent spaces given by the vector bundle structure of $\mathfrak{F}$). But we can recover these values at φ and all derivatives up to order k from $\lambda_A, \lambda_B \in \overline{C}_m^{\text{diff}} / \sim_{\pi_k}$, where we assumed they agreed.

So we indeed find that it is sufficient that $\lambda_A \sim_{\pi_k} \lambda_B$ to conclude $A \sim_k B$. This means that the projection between $\mathfrak{T}^1(\mathfrak{F}) / \sim_k$ and $\lambda_A, \lambda_B \in \overline{C}_*^{\text{diff}} / \sim_{\pi_k}$ is well-defined, injective and surjective. That the limit over the diagrams agrees then follows from the fact these are the isomorphisms in our chosen category. $\square$

Outlook. This result is helpful to our cause, as it tells us that the complex is independent of the vector bundle structure of $F \rightarrow M$, even while $\mathcal{L}$ is not. With this result, we can even hope to generalize the construction to ordinary fiber bundles.

As the tangent space of a field theory is a local construction on M and $P\delta L$ is a local functional, we may restrict any fiber bundle to an open around $\Gamma(\varphi)$ that has fibers isomorphic to $\mathbb{R}^k$. If we can endow this bundle with any kind of smooth coordinates on the fiber that have linear transition functions, we have found a vector bundle structure with respect to which we can compute the perturbative observables.

The question when this is possible exceeds the scope of this thesis, but I suspect that it will be in many, maybe even all cases.

Another question that arises is if it might be possible to use a similar construction for example on the chain complex $\mathfrak{T}^2(\mathfrak{F})$ and if it can be expressed as an L_∞ -algebra as well. Following the same arguments that we have already seen in this section, it should be closely related to the complex $\dots \longrightarrow 0 \longrightarrow \Gamma_\varphi^\infty(M, VF) \longrightarrow (\Gamma_{c,\varphi}^\infty(M, VF))^* \longrightarrow 0 \longrightarrow \dots$, with L_∞ -algebra structure and pairing of degree 3 given similar to 3.2.4.

This is an object that differs from the previously defined $\mathcal{L}_1$ in the sense that the complex does not arise any more as the sections of a bundle over M . At first, this seems like a grave difference that disfavors this L_∞ -algebra as a generalization of Definition 3.2.1. However, it turns out that Costello and Gwilliam themselves sometimes consider chain complexes not arising as sections but distributional sections, just as the diffeological dual occurring here (cf. [5] p.9, p.71).

So the studies, how these (and possibly other) L_∞ -algebras can be used to express physical observables, is far from complete. As with general models of physics, time will determine how much application this notion of “observables” will find, especially in quantisation, where Costello and Gwilliam ([5]) apply it to.

4 Hamiltonian Classical Observables

4.1 The Poincaré-Cartan form and its Hamiltonian forms

Aside from the constructions discussed in the previous sections, there is also another concept known as “observables”. This notion arises from symplectic geometry and comes with a construction of an L_∞ -algebra that is known as “the L_∞ -algebra of observables”. There is, however, no direct relation to the L_∞ algebras in the previous chapter. Nonetheless, I find it important to review this concept as well. This allows a more complete picture of the studies of physical observables in mathematics. (Also, I would like to highlight the need for distinct nomenclature of these concepts.)

The idea behind this further notion is that we usually consider a time-dependent physical systems on a manifold N . This can be expressed a fiber bundle $F \rightarrow M$ which is of the form

$$E \times \mathbb{R} \rightarrow N \times \mathbb{R}.$$

The observables we can study with this method are expressed through a “density” on N for each point in time. Examples of this are energy, that is computed as the integral of an energy density over N , or an electric charge being determined by an electric charge density.

To define this notion, we need to look a little into multi-symplectic geometry first. We will follow the lines of [6] here.

Definition 4.1.1. A pre- k -symplectic form on a manifold M is an element $\omega \in \Omega^{(k+1)}(M)$ that is closed, i.e. $d\omega = 0$.

Definition 4.1.2. Let ω be a pre- k -symplectic form on M . A pair (v, α) , where $v \in \mathfrak{X}^1(M)$ and $\alpha \in \Omega^{(k-1)}(M)$ is called a Hamiltonian pair if

$$\iota_v \omega = d\alpha.$$

v is also called a Hamiltonian vector field, while α is called Hamiltonian differential form.

There is a well-known constructions of an L_∞ -algebra from the Hamiltonian differential forms, attributed to Rogers (cf. [16], Theorem 3.14). As shown in [6], we will see later that these constructions generalize to the diffeological setting we are interested in.

Definition 4.1.3. Let ω be a pre- m -symplectic form on a manifold M . Consider the graded vector space defined by

$$L_n = \begin{cases} \Omega_{\text{ham}}^{m-1}(M) & \text{if } n = 1 \\ \Omega^{m-n}(M) & \text{if } 2 \leq n \leq m \\ \{0\} & \text{else.} \end{cases}$$

The L_∞ -algebra of Hamiltonian forms is given by these vector spaces together with the maps

$$l_n(\alpha_1, \dots, \alpha_n) = \begin{cases} \iota_{v_1} \iota_{v_2} \dots \iota_{v_n} \omega & \text{if } \iota_{v_i} \omega = d\alpha_i \text{ for } 1 \leq i \leq n \\ 0 & \text{else.} \end{cases}$$

Now going back to our consideration of a field theory $F \rightarrow M$, we find the following: We have that every vector field $\chi \in \mathfrak{X}^1(J^\infty F)$ decomposes into $\chi = \xi + X$, called the vertical and horizontal component. At the point $j_m^\infty \varphi$, we have that X arises from first projecting χ to $T_m M$ and then mapping it back along the jet prolongation $j^\infty \varphi : M \rightarrow J^\infty F$ (where X turns out to be independent of the choice of φ).

This decomposition of $\mathfrak{X}^1(J^\infty F)$ induces a decomposition of $\Omega^1(J^\infty F)$. This in turn gives us a notion of homogeneous elements, not only on $\Omega^1(J^\infty F)$ but also on the higher differential forms $\Omega^n(J^\infty F)$. They come with a bigrading, depending on which order they depend on vertical and horizontal tangent vectors respectively. The arising bicomplex is called the variational bicomplex. It is equipped with a vertical and a horizontal differential, δ and d_M , which arise from the differentials on $\mathfrak{F}$ and M respectively. This construction and its properties are for example covered in [1] or [2].

Let us return to our setting of a Lagrangian field theory: Let M be an n -dimensional manifold and $F \rightarrow M$ a fiber bundle. We have that the Lagrangian naturally defines an element in $\Omega^{(0,n)}(J^\infty F)$, where $(0,n)$ denotes the bigrading with respect to $\mathfrak{F} \times M$. Both δL and $P\delta L = EL$ are objects of bigrading $(1,n)$.

By the properties of the operator P , the term

$$\delta L - EL \in \Omega_{\text{loc}}^{(1,n)}(\mathfrak{F} \times M)$$

can be expressed by $-d_M \gamma$ for some local form $\gamma \in \Omega^{(1,n-1)}(\mathfrak{F} \times M)$ (cf. [1] Theorem 2.6, [2] Theorem 2.12). By a theorem of Taken, we even additionally have that this already defines γ up to some term $d_M \eta$ with $\eta \in \Omega_{\text{loc}}^{(1,n-2)}(\mathfrak{F} \times M)$ (cf. [19], Theorem 4.6).

Definition 4.1.4. The Poincaré-Cartan form ω is given by

$$\omega = EL + \delta \gamma.$$

Remark. This form arises naturally for example through the Legendre transformation as described in [18] (cf. Prop. 1.2.285) and is local, i.e. can be described as a form on $J^\infty F$.

Example 4.1.5. Consider a particle in $\mathbb{R}^3$ that moves depending on the time $t \in \mathbb{R}$. The fiber bundle to describe this theory is given by

$$F = \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R} = M$$

and the space of fields by $\mathfrak{F} = \Gamma^\infty(M, F) = C^\infty(\mathbb{R}, \mathbb{R}^3)$. The Lagrangian for some $\varphi \in \mathfrak{F}$ is given by

$$L(\varphi, t) = \frac{1}{2} \left((\dot{q}^1(\varphi, t))^2 + (\dot{q}^2(\varphi, t))^2 + (\dot{q}^3(\varphi, t))^2 \right) d_M t,$$

where we denote $\dot{q}^i(\varphi, t) = \frac{\partial}{\partial s} x^i(\varphi(s))|_t$ for the coordinate functions x^i that we have on $\mathbb{R}^3$.

Now we can simply compute δL as

$$\delta L = \sum_{i=1}^3 \dot{q}^i(\varphi, t) \delta \dot{q}^i \wedge d_M t,$$

and $\text{EL} = P\delta L$ as

$$P\delta L = \sum_{i=1}^3 -\ddot{q}^i(\varphi, t) \delta q^i \wedge d_M t,$$

where we have $\ddot{q}^i(\varphi, t) = \frac{\partial^2}{\partial s^2} x^i(\varphi(s))|_t$ for the coordinate functions x^i .

We can explicitly check that

$$\gamma = \sum_{i=1}^3 \dot{q}^i(\varphi, t) \delta q^i$$

satisfies

$$-d_M \gamma = \delta L - \text{EL} = \sum_{i=1}^3 \dot{q}^i(\varphi, t) \delta \dot{q}^i \wedge d_M t + \ddot{q}^i(\varphi, t) \delta q^i \wedge d_M t,$$

such that

$$\omega = \text{EL} + \delta \gamma = \sum_{i=1}^3 \delta \dot{q}^i \wedge \delta q^i - \ddot{q}^i(\varphi, t) \delta q^i \wedge d_M t.$$

Now let us start to study the properties of ω , so we can later analyze properties of the arising structures.

Lemma 4.1.6. (cf. [6], p.125) *The Poincaré-Cartan form ω is closed in $\Omega^{\dim(M)+1}(\mathfrak{F} \times M)$.*

Proof. We have that

$$d\omega = d_M \omega + \delta \omega.$$

From the formula

$$\delta L = \text{EL} - d_M \gamma$$

we can obtain that

$$\delta \text{EL} = \delta d_M \gamma,$$

therefore

$$\delta\omega = \delta EL + \delta^2\gamma = \delta d_M\gamma.$$

On the other hand, we have that for dimension reasons

$$d_M EL = 0,$$

such that

$$d_M\omega = d_M\delta\gamma = -\delta d_M\gamma.$$

This combines such that indeed

$$d\omega = 0.$$

□

As we have defined Hamiltonian differential forms and Hamiltonian vector fields for manifolds, we can now just do the same for our diffeological space. To retain the notion of locality, however, that we have on the variational bicomplex, we can also consider the jet bundle instead.

Definition 4.1.7. A Hamiltonian observable α is a Hamiltonian differential form with respect to the Poincaré-Cartan form ω , i.e. $\alpha \in \Omega^{\dim(M)-1}(\mathfrak{F} \times M)$ is a Hamiltonian observable if there exists a $\chi \in \mathfrak{X}^1(\mathfrak{F} \times M)$ such that

$$d\alpha = \iota_\chi\omega.$$

Definition 4.1.8. A local Hamiltonian observable α is a local Hamiltonian differential form with respect to the Poincaré-Cartan form ω , i.e. $\alpha \in \Omega^{\dim(M)-1}(J^\infty F)$ is a Hamiltonian observable if there exists a $\chi \in \mathfrak{X}^1(J^\infty F)$ such that

$$d\alpha = \iota_\chi\omega.$$

This allows us to adapt our previous construction

Definition 4.1.9. Consider the graded vector space defined by

$$L_n = \begin{cases} \Omega_{\text{ham}}^{m-1}(\mathfrak{F} \times M) & \text{if } n = 1 \\ \Omega^{m-n}(\mathfrak{F} \times M) & \text{if } 2 \leq n \leq m \\ \{0\} & \text{else.} \end{cases}$$

The L_∞ -algebra of Hamiltonian observables is given by these vector spaces together with the maps

$$l_n(\alpha_1, \dots, \alpha_n) = \begin{cases} \iota_{\chi_1}\iota_{\chi_2}\dots\iota_{\chi_n}\omega & \text{if } \iota_{\chi_i}\omega = d\alpha_i \text{ for } 1 \leq i \leq n \\ 0 & \text{else.} \end{cases}$$

Definition 4.1.10. The L_∞ -algebra of local Hamiltonian observables is defined as the sub-algebra of the L_∞ -algebra of Hamiltonian observables given only by the local forms on $\mathfrak{F} \times M$.

The question naturally poses itself, how the Hamiltonian observables are related to the observables as we have seen them in the previous chapters, i.e. as functions in $C^\infty(\mathfrak{F})$.

Definition 4.1.11. Consider a manifold M with a designated co-dimension 1 Cauchy hyper-surface S . The associated observable to a (local) Hamiltonian observable α is the function $f_\alpha \in C^\infty(\mathfrak{F})$ given by

$$f_\alpha(\varphi) = \int_S \iota_{S \hookrightarrow \{\varphi\} \times M}^* (\alpha),$$

where $\iota_{S \hookrightarrow \{\varphi\} \times M}^*$ denotes the pull-back of differential forms on $\mathfrak{F} \times M$ to differential forms on S .

We want to analyze this term for a local observable: The integral above does only depend on the values of α under the very specific pull-back of the definition. So we would like to know how the pull-back looks like along a map $S \rightarrow \mathfrak{F} \times M$ that is constant on $\mathfrak{F}$. Let us decompose α into

$$\alpha = \sum_{i=0}^{\dim(M)-1} \alpha_i$$

for $\alpha_i \in \Omega_{\text{loc}}^{(i, \dim(M)-1-i)}(\mathfrak{F} \times M)$. It turns out that pull-back depends on α_0 , the component of α with the bigrading $\alpha_0 \in \Omega_{\text{loc}}^{(0, \dim(M)-1)}(\mathfrak{F} \times M)$. (The reason behind this is, that TS is entirely mapped to the space of horizontal vector fields). This raises the question how this component looks like for an Hamiltonian vector field.

Proposition 4.1.12. (cf. [6], Proposition 1.1.7) *Let α be a local Hamiltonian observable arising from*

$$d\alpha = \iota_\chi \omega$$

for some $\chi = \zeta + X$. Then the following equations are satisfied (and actually sufficient for the existence of α): If we write $\alpha = \sum_{i=0}^{\dim(M)-1} \alpha_i$ for $\alpha_i \in \Omega_{\text{loc}}^{(i, \dim(M)-1-i)}(\mathfrak{F} \times M)$ we have that

$$\begin{aligned} \iota_\xi \text{EL} &= d_M \alpha_0 \\ \iota_\xi \delta \gamma &= \delta \alpha_0 + d_M \alpha_1 - \iota_X \text{EL} \\ \iota_X \delta \gamma &= \delta \alpha_1 + d_M \alpha_2 \\ 0 &= \delta \alpha_i + d_M \alpha_{i+1}, \end{aligned}$$

where the last equation holds for $i \geq 2$.

Example 4.1.13. Assume that we are given a manifold $M = N \times \mathbb{R}$ with N compact. Further, let us consider a fiber bundle $E \times \rightarrow N$, which gives rise to a fiber bundle $E \times \mathbb{R} = F \rightarrow M = N \times \mathbb{R}$. Then for the Cauchy hyper-surfaces $S_t \subset M$ of the form $N \times \{t\}$ we can compute

$$\begin{aligned}
f_{\alpha,s}(\varphi) - f_{\alpha,t}(\varphi) &= \int_{S_s} \iota_{S_s \hookrightarrow \{\varphi\} \times M}^* (\alpha) - \int_{S_t} \iota_{S_t \hookrightarrow \{\varphi\} \times M}^* (\alpha) \\
&= \int_{S_s} \iota_{S_s \hookrightarrow \{\varphi\} \times M}^* (\alpha_0) - \int_{S_t} \iota_{S_t \hookrightarrow \{\varphi\} \times M}^* (\alpha_0) \\
&= \int_{N \times (s,t)} d \iota_{S_t \hookrightarrow \{\varphi\} \times M}^* (\alpha_0) \\
&= \int_{N \times (s,t)} \iota_{S_t \hookrightarrow \{\varphi\} \times M}^* (d_M \alpha_0) \\
&= \int_{N \times (s,t)} \iota_{S_t \hookrightarrow \{\varphi\} \times M}^* (\iota_{\xi} \text{EL}),
\end{aligned}$$

where we have repeatedly applied that the pull-back along a map constant in direction of $\mathfrak{F}$ only depends on the terms of bigrading $(0, k)$.

The last line of our computation still depends on EL. But if we assume that φ is a solution of the field theory, i.e. that EL vanishes at φ , we have that the given term equals 0. So we have found that the associated observable f_{α} is independent of t on solutions of the field equation. This can be interpreted as a measurement on N that is conserved over time throughout the experiment.

Remark. As explained in [6] (cf. p. 188), this type of statement can be generalized to any co-dimension 1 Cauchy hyper-surface $S \subset M$, where the value only depends on the homology class of S .

In general, however, there is no obvious reason why the assignment from Hamiltonian observables to observables should be a one-to-one correspondence. For example, it is even in the case of 4.1.13 not a priori possible to express a function that just depends on the value of a field at $(n, t) \in N \times \mathbb{R} = M$ as an integral over all of $N \times \{t\}$.

So in the end, it always depends on the application if the Hamiltonian observables or the observables in the sense of functions on $\mathfrak{F}$ are the objects that one would like to consider in any context that arises from field theories.

References

- [1] Ian M. Anderson. “Introduction to the variational bicomplex”. In: *Mathematical aspects of classical field theory (Seattle, WA, 1991)*. Vol. 132. Contemp. Math. Amer. Math. Soc., Providence, RI, 1992, pp. 51–73. DOI: 10.1090/conm/132/1188434. URL: <https://doi.org/10.1090/conm/132/1188434>.
- [2] Ian M. Anderson. *The Variational Bicomplex*. Book draft, available at <http://differentialgeometry.org/papers> (24.8.2020).
- [3] Christian Blohmann. “Lagrangian Field Theory: Lecture Notes for Advanced Geometry I”. Version: 11.02.2020. Not publicly available, please contact blohmann@mpim-bonn.mpg.de.
- [4] J. Daniel Christensen and Enxin Wu. “Tangent spaces and tangent bundles for diffeological spaces”. In: *Cah. Topol. Géom. Différ. Catég.* 57.1 (2016), pp. 3–50.
- [5] Kevin Costello and Owen Gwilliam. “Factorisation Algebras in Quantum Field Theory, Volume 2”. Not published yet, draft from 28.04.2016, available at <https://people.math.umass.edu/~gwilliam/vol2may8.pdf> (7.4.2020).
- [6] Néstor León Delgado. “Lagrangian Field Theories: Ind/Pro-Approach and L_∞ -algebra of Local Observables”. Available at <http://hss.ulb.uni-bonn.de/2018/5025/5025.pdf> (8.4.2020). PhD thesis. Rheinische Friedrich-Wilhelms Universität Bonn, 2018.
- [7] Gerald B. Folland. *A guide to advanced real analysis*. 1st ed. Vol. 37. The Dolciani Mathematical Expositions. MAA Guides, 2. Mathematical Association of America, Washington, DC, 2009, pp. x+107. ISBN: 978-0-88385-343-6.
- [8] Owen Gwilliam. “Factorisation Algebras and Free Field Theories”. Available at <https://people.math.umass.edu/~gwilliam/thesis.pdf> (02.10.2019). PhD thesis.
- [9] Igor Kříž and J. P. May. “Operads, algebras, modules and motives”. In: *Astérisque* 233 (1995), iv+145pp. ISSN: 0303-1179.
- [10] Tom Lada and Jim Stasheff. “Introduction to SH Lie algebras for physicists”. In: *Internat. J. Theoret. Phys.* 32.7 (1993), pp. 1087–1103. ISSN: 0020-7748. DOI: 10.1007/BF00671791. URL: <https://doi.org/10.1007/BF00671791>.
- [11] Jacob Lurie. *Derived Algebraic Geometry X: Formal Moduli Problems*. Available at <http://people.math.harvard.edu/~lurie/papers/DAG-X.pdf> (8.4.2020). 2011.

- [12] Katsumi Nomizu and Hideki Ozeki. “The existence of complete Riemannian metrics”. In: *Proc. Amer. Math. Soc.* 12 (1961), pp. 889–891. ISSN: 0002-9939. DOI: 10.2307/2034383. URL: <https://doi.org/10.2307/2034383>.
- [13] Joost Nuiten. *Lie Algebroids in Derived Differential Topology*. Thesis manuscript, available at <https://imag.umontpellier.fr/~nuiten/Writing/LieAlgdDerDiffTopScreen.pdf> (10.8.2020). 2018.
- [14] Mauro Porta. “Derived Formal Moduli Problems”. Available at <http://algant.eu/documents/theses/porta.pdf> (8.4.2020). MA thesis. Università degli studi di Milano, Université de Paris-Sud XI, 2013.
- [15] J. P. Pridham. *An outline of shifted Poisson structures and deformation quantisation in derived differential geometry*. 2018. arXiv: 1804.07622v2 [math.DG].
- [16] Christopher L. Rogers. “Higher Symplectic Geometry”. PhD thesis. University of California Riverside, 2011. arXiv: 1106.4068v1 [math-ph].
- [17] D. J. Saunders. *The geometry of jet bundles*. Vol. 142. London Mathematical Society Lecture Note Series. Cambridge University Press, Cambridge, 1989, pp. viii+293. ISBN: 0-521-36948-7. DOI: 10.1017/CB09780511526411. URL: <https://doi.org/10.1017/CB09780511526411>.
- [18] Urs Schreiber. *Differential cohomology in a cohesive infinity-topos*. 2013. arXiv: 1310.7930v1 [math-ph].
- [19] Floris Takens. “A global version of the inverse problem of the calculus of variations”. In: *J. Differential Geometry* 14.4 (1979), 543–562 (1981). ISSN: 0022-040X. URL: <http://projecteuclid.org/euclid.jdg/1214435235>.
- [20] Terence Tao. Lecture notes (Math 131BH), available at <https://www.math.ucla.edu/~tao/resource/general/131bh.1.03s/week45.pdf> (20.8.2020). 2003.
- [21] *The Stacks Project*. Section 15.57, available at <https://stacks.math.columbia.edu/tag/06XY> (21.4.2020).
- [22] Bertrand Toën. *Derived Algebraic Geometry*. 2014. arXiv: 1401.1044v2 [math.AG].
- [23] Martin Vincent. “Diffeological Differential Geometry”. Available at https://pdfs.semanticscholar.org/718b/dae69ed8029a35e5de54bf74f0c22abf0b.pdf?_ga=2.90241436.707467964.1598790553-876133039.1587984269 (30.8.2020). MA thesis. University of Copenhagen, 2008.